TCS CORPORATE WEBSITE

Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
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Last validated	2026-11-08
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Classification	INTERNAL — classification to be confirmed

Generated from the Markdown sources of docs/handover (branch migrated-page-content-edam); do not edit this PDF.

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Developer Guide

Attribute	Value
Document ID	DEV-000
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	Entry point for developers: setup, workflow, standards, loading sequence, shared code, validation and where each topic is detailed
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	AGENTS.md, package.json, scripts/scripts.js
Related documents	—
Related AEM paths	—

1. Prerequisites

- Node.js and npm (the versions of package.json); Git; access to the GitHub repository sradha-tcs/tcs-corp.
- Universal Editor access to the AEM Sites author for authoring checks (not required for code changes).

2. Workflow in short

1. Read the knowledge base README and the topic pages for the area (knowledge base), then AGENTS.md.
2. Branch from `main` (`feature/`, `fix/`, `docs/`).
3. Change code, models (`npm run build:json`), imports (`npm run build:modules`), block docs (`npm run build:docs`).
4. Run `npm run lint` and `npm test`; report the real results.
5. Open a PR into `main` with a feature preview URL; checks green; a human merges.
6. **Update this documentation set** (docs/handover on migrated-page-content-edam) in the same change (see the documentation governance).

3. Commands (results of 8 November 2026)

Command	Purpose	Result in this program
`npm run lint`	ESLint (Airbnb, json, xwalk model rules) and Stylelint	Passed (exit 0)
`npm test`	Unit tests (Node test runner)	387 of 389 passed, 2 failed: the 2 Form Variations motion tests (known before this program)

Command	Purpose	Result in this program
`npm run build:json`	Aggregate the Universal Editor models	Passed; no change to the committed files
`npm run build:docs`	Generated sections of the block READMEs	Passed; no change to the committed files
`node docs/handover/knowledge-base/validate-knowledge-base.mjs`	Knowledge base structure and pointers	Passed (34 pages, nothing missing or failing)
`node capture.mjs (documentation program)`	Automated checks and screenshots of 200 reference pages at 4 widths	Completed: 200 pages, 800 screenshots
`node analyze-links.mjs (documentation program)`	Links of the published pages of the roots	Completed for the three roots
`perf.mjs (documentation program)`	Load timing and query index requests of a page	Before PR #203: index done after 18.4 s (24 requests, 1.37 MB); after: 2.9 s (9 parallel requests, 0.87 MB) on /language-masters/en, 1.1 s (1 request) on /design-block-master/en

4. Main loading sequence

5. Runtime architecture

Three-phase loading

`loadPage()` in `scripts/scripts.js` runs `loadEager`, then `loadLazy`, then `loadDelayed`.

Phase	Steps (in code order)
Eager (`loadEager`)	`decorateHead()` (SEO head tags and `<html lang>`, `decorators/seo-decorator.js`); `trackInteractions()` (page entry and capture listeners, `decorators/data-layer-decorator.js`); `decorateTemplateAndTheme()`; `decoratePageType()` (Page Type body class); on the 404 page `loadErrorPage(main)`, else `loadTemplate(main)`; `decorateMain(main)`; `reserveHeaderSpace()`; `body.appear`; `preloadBlockModules(first section)`; `loadSection(first)` with `prepareMotion` and `waitForFirstImage` (LCP); `fonts.css` only at 900px and wider or when the session flag `fonts-loaded` is set
Lazy (`loadLazy`)	`autolinkModals` (links containing `/modals/` open the Modal block); `preloadBlockModules(main)` and of the chrome blocks (`header`, `footer`, `fragment`, `tcs-header`, `conversation-canvas`); `prefetchData(main)`; `loadHeader`; `loadFooter` (hidden until the sections have loaded); `loadMainSections` (each section with `prepareMotion`, RUM enhancement after the first); footer shown; `trackBlocks(main)`; `addErrorBackButton` on 404 pages; scroll to the URL hash; `lazy-styles.css`; `loadFonts()`
Delayed (`loadDelayed`)	After 3 seconds: `trackBlocks(document)` (header, footer, late blocks such as modals) and `import('./delayed.js')`. `scripts/delayed.js` contains only a comment (`// add delayed functionality here`)

`decorateMain(main)` runs `decorateIcons` → `buildAutoBlocks` (an empty placeholder) → `decorateSectionsWithOptions` (section options, then the container sections Gen Carousel, Tabs Container and Columns Container wrapped in synthetic blocks) → `decorateBlocks` → `applyBlockOptions` (Additional tab) → `decorateButtons`.

`prefetchData(main)` starts the query index (`getPages`) and the taxonomy (`loadTaxonomy`) at the lazy phase only when the page has a `.list`, `.filter-list` or `.conversation-canvas`, and the redirects sheet (`loadRedirects`) only with a `.list`. It isn't run in the eager phase so it doesn't compete with the LCP image (code comment). See indexing-and-search.md.

Block loading

From `scripts/aem.js`:

- `decorateBlock(block)` adds `.block`, `data-block-name` (the first class), `data-block-status="initialized"`, wraps text nodes and adds `{name}-wrapper` to the parent.
- `loadBlock(block)` sets `loading`, loads `/blocks/{name}/{name}.css` and imports `/blocks/{name}/{name}.js` in parallel, awaits the default export, catches and logs errors (`failed to load module for {name}`, `failed to load block {name}`) and always ends with `data-block-status="loaded"`.
- `loadSection(section, callback)` loads the section's blocks one after another, runs the callback (here `prepareMotion`), then shows the section.

Module preloading

There is no bundler, so a module's imports are only discovered after it loads (AGENTS.md "Module preloading"). Two generated lists remove these round trips (`npm run build:modules`, `tools/modules/generate-module-graph.mjs`):

- `head.html` has `modulepreload` links for every module `scripts.js` imports (between `<!-- modulepreload -->` markers).
- `scripts/module-graph.js` lists, per block, the modules it imports beyond `scripts.js` and whether it has a stylesheet; `preloadBlockModules()` adds `modulepreload` links (head) and stylesheet `preload` links (body, because `loadCSS` skips a stylesheet already linked in the head).

`npm test` includes `test/modules/module-graph.test.mjs`; AGENTS.md states the tests fail when the lists are out of date.

Templates

`loadTemplate(main)` reads the Page Type (`page-type` metadata, else `template`) and, when it is one of `TEMPLATES = ['home', 'article', 'blog', 'press-release']`, imports `templates/{name}/{name}.js` and loads its CSS, then calls the default export with `main` before decoration. Failures are logged (`failed to load template {name}`) and the page continues. What each template adds: templates/README.md and content-architecture.md.

6. Guides

- Local development set-up
- Repository structure
- Coding standards
- Design style guide
- Block development
- Curated design page development
- Content modeling
- Universal Editor models
- Decorators and utilities
- Analytics data layer
- Accessibility
- Performance
- SEO
- Unit testing
- Debugging and troubleshooting
- Pull request review checklist
- Release and deployment
- Block inventory, variation matrix, dependency matrix, known limitations
- Curated page inventory

7. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

8. Known limitations

- None recorded for this document.

9. Open items

- None.

10. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Accessibility Development Guide

Attribute	Value
Document ID	DEV-010
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	WCAG 2.1 AA implementation rules and evidence
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/accessibility.md, decorators/a11y-decorator.js
Related documents	—
Related AEM paths	—

1. Target and verified standard

Item	Value
Knowledge-base target	**WCAG 2.2 Level AA**
Project rule (verified)	WCAG **2.1** AA: AGENTS.md "Accessibility" ("Blocks meet WCAG 2.1 AA") and "Testing & Quality Assurance" ("Follow WCAG 2.1 AA guidelines"); handover 1.7.2 "WCAG 2.1 AA conformance (target)"; `blocks/README.md` "Shared Conventions"
Difference	WCAG 2.2 adds, at A / AA, 2.4.11 Focus Not Obscured (Minimum), 2.5.7 Dragging Movements, 2.5.8 Target Size (Minimum), 3.2.6 Consistent Help, 3.3.7 Redundant Entry and 3.3.8 Accessible Authentication (Minimum), and removes 4.1.1 Parsing. The automated audits already ran axe with the WCAG 2.2 AA rules (`wcag22aa` tag in `docs/audit/tools/harness.mjs`); the project rule text still says 2.1. Raising the rule to 2.2 AA is a decision for the product owner: **Not recorded**
Compliance claim	None. Evidence is automated axe runs and code review; see Evidence

How the WCAG 2.2 additions stand in the code:

WCAG 2.2 criterion	Implementation	Status
2.4.11 Focus Not Obscured	Conversation Canvas: `scroll-padding-bottom` keeps focused content clear of the floating dock; Anchor List Sticky: open ("[To be confirmed …]" in the handover, listed in `docs/blocks/reports/anchor-list.md`)	Requires validation
2.5.7 Dragging Movements	Scrollers that can be dragged also have previous / next buttons and keyboard steps (`decorators/scroller-decorator.js`)	Implemented; not separately tested

WCAG 2.2 criterion	Implementation	Status
2.5.8 Target Size (Minimum, 24px)	Buttons `--button-height` 3.125rem (50px), icon buttons `--icon-button-size` 2.75rem (44px), Canvas voice button 2rem (32px)	Requires validation (no systematic measurement)
3.2.6 Consistent Help	Header and footer come from shared reference pages per locale	Not recorded (no help mechanism defined)
3.3.7 Redundant Entry	Forms are single-step (Form Variations)	Not applicable as far as identified
3.3.8 Accessible Authentication	No visitor login exists	Not applicable

2. Semantic HTML, headings and landmarks

- **Landmarks**: `header` (with `nav#nav`), `main` and `footer` come from the boilerplate page; the Conversation Canvas search form has `role="search"`; tables with a scrollable width become labelled `role="region"` elements. No skip link was identified (bypass relies on the landmarks): see Recommendations.
- **Native elements**: Accordion is built on `details` / `summary`; Modal and the video pop-up use `dialog` with `showModal()`; the Separator renders `hr`; tables keep header cells.
- **Headings**: blocks with a title have an authored level (Title Type / Title heading level, `decorators/heading-decorator.js` `HEADINGS` h1–h6); the level renders the element and its size follows it, so "the outline matches the look" (dialog options report, section 10). The Page Header renders the page H1 from metadata. A hidden List Item title stays in the DOM, visually hidden. Authors are responsible for a logical order.
- **Language**: `decorators/seo-decorator.js` sets `<html lang>` from the locale (or `language` metadata); content loaded from another locale (the `/en` 404 fallback) is marked with its `lang` (`loadErrorPage` in `scripts/scripts.js`).

3. Keyboard, focus order and focus visibility

Area	Behaviour (code)
Global focus ring	`a:focus-visible, button:focus-visible`: 2px (`--border-width-200`) solid `--color-border-brand`, offset 0.125rem; brand 300 inside dark backgrounds (theme contract)
Focus order	DOM order; visual reordering (Teaser image Top / Bottom / Left / Right) uses CSS `order` and doesn't change the DOM (dialog options report, section 10)
Header menu (tablet / mobile)	Buttons with `aria-expanded` / `aria-controls`; while the menu is open `main` and `footer` are `inert`; Escape closes it and returns focus to the menu button; focus moving to an element outside the nav closes it (F-71)
Tabs	`role="tablist"` / `tab` / `tabpanel`, `aria-selected`, Arrow keys, Home and End; `tabs:reveal` opens the tab holding a deep-link target
Gen Carousel	Inactive slides are `inert`; Manual modes: arrow buttons and ArrowLeft / ArrowRight; Auto modes: pause / play button
Scrollers (List, Cards, Filter List)	Keyboard-focusable region while it scrolls, previous / next buttons, keyboard steps
Modal, video pop-up	Native `dialog` (Escape closes); focus returns to the opener
Conversation Canvas	Disclosure buttons with `aria-expanded` / `aria-controls` / `aria-pressed`, Escape and focus return; a hidden dock row is `inert`
Hidden blocks	Hide block (`hideblock`) is `display: none` on the page; nothing focusable is left in it
Skeletons	No focusable element; focus doesn't move when the skeleton is replaced

4. Accessible names and UI text

- All UI text (button names, status messages, hints) comes from the locale's placeholders with an English fallback: `t('nextSlide', 'Next slide')` (`utils/placeholders.js`); AGENTS.md forbids hard-coded user-facing strings.
- Icon-only buttons get their name from a visually hidden label (`hideVisually`, `createHiddenText` in `decorators/a11y-decorator.js`; `.visually-hidden` in `styles/styles.css`).
- Examples: Gen Carousel `aria-label` "Carousel", slides "{n} of {total}", dots "Show slide {n} of {total}", toggle "Pause slides" / "Play slides"; Embed Play button "Play video: {title}" when the placeholder image has alt text; a hidden List Item title names the card's call to action ("Read more: The agentic bank").

5. Images, alt text and decorative images

- Image fields have alt text fields; an empty alt marks the image as decorative (`alt=""`). Section backgrounds have their own alt field (`backgroundimagealt`) and default to decorative (`buildSectionBackground` in `scripts/scripts.js`).
- Authored icons are `img` with an empty alt (`decorateIcons` in `scripts/aem.js`); button arrows and list arrows are CSS masks, not content.
- Lottie animations: `role="img"` with the label, or `aria-hidden` without one; a non-image reference's link text is its Alt (`animationLabel`, `utils/media.js`; AGENTS.md "AEM delivery rules").
- Background videos are `aria-hidden` and their iframe has `tabindex="-1"`.
- Text over images (Hero, Teaser Background image / Card) relies on overlays; axe can't judge text over images, so the re-audit tested them in targeted browser checks (re-audit, section 2).

6. Links and buttons

- Actions are `<button>` elements; navigation is `<a href>`. Every button and link button goes through the button decorators (`decorators/button-decorator.js`, `icon-button-decorator.js`, `content-buttons-decorator.js`).
- **New tab**: `setLinkTarget` sets `target="_blank"` and `rel="noopener noreferrer"`; `announceNewTab(a, hint)` adds visually hidden text, or extends an existing `aria-label`, with the `opensInNewTab` placeholder (WCAG 3.2.5 practice). Used by the button decorator, card lists, Teaser, Page Header and Footer; social links add the hint to their hidden label. **Open**: the article author bio's "View LinkedIn profile" link opens a new tab without the hint (`decorators/author-bio-decorator.js`; handover F-52).
- Links in running text are underlined on dark backgrounds so they are not told apart by colour alone (theme contract).
- Disabled buttons use `aria-disabled="true"` or `:disabled`.

7. Forms

Form Variations (`blocks/form-variations/form-variations.js`):

- Each field has a visible label; required fields are validated on submit; the first invalid field receives focus.
- Errors are shown next to their field and linked with `aria-invalid="true"` and `aria-describedby` (removed again once the field is valid).
- `autocomplete` is set from the field's name or type (WCAG 1.3.5 Identify Input Purpose).
- A textarea's character counter is linked with `aria-describedby`.
- The submit outcome is a `role="status"` message that receives focus. Integrations report their outcome through `form-variations:submit` and `detail.respondWith(promise)` (AGENTS.md "Shared events and helpers"); no submit endpoint is configured in the repository (handover F-01).

8. Interactive components

Component	Pattern	README
Accordion	Native `details` / `summary`	accordion
Tabs Container / Tabs	ARIA tabs (tablist, tab, tabpanel, `aria-selected`), arrows, Home / End	tabs
Gen Carousel	Named region, slide labels, `inert` inactive slides, `aria-live="polite"` track (off while playing), pause / play for Auto, arrows for Manual; autoplay starts paused for reduced motion	gen-carousel
Modal	Native `dialog`, named, focus return	modal
Header navigation	Disclosure buttons, `inert` page while the menu is open, Escape, focus loss closes; `header:menu` event	header
Anchor List	Links to section anchors; the current link is marked; Sticky pins the row	anchor-list
Table	Header cells; a scrollable table is a focusable, labelled region	table

9. Search and filters

- **Conversation Canvas** search: `role="search"` form with an `input type="search"`, a hidden label, and a `role="status"` region; the Canvas search pushes a redacted term to the data layer (privacy, `utils/privacy.js`). See the Canvas README.
- **Filter List**: filter pills are buttons (`tag-filter`, active state with `aria-current` styling); the result count is a live region whose `role="status"` is set after the first count, so the initial render isn't announced (audit D-A-25); the count can be visually hidden but stays announced; the empty state shows "No resources found." and "Try selecting another filter." from placeholders.

10. Video and media

Media	Control	Reference
Lottie (Hero, Teaser)	Pause / play button (`pauseAnimation` / `playAnimation` placeholders, WCAG 2.2.2); paused off screen; first frame only for reduced motion	`decorators/animation-decorator.js`
Video (DAM, YouTube, Vimeo)	Named player and Play button; pause / stop toggle for background and autoplay (WCAG 2.2.2); captions `track` (Captions, language, label fields); transcript link; never autoplays for reduced motion	video
Embed	A placeholder with a named Play button when a placeholder image is authored; privacy-enhanced players (youtube-nocookie, Vimeo dnt)	embed
Gen Carousel Auto	Pause / play button	above

Open: the video pause button's brand outline may have low contrast on dark posters ("[To be confirmed: contrast measurement]", handover F-56).

11. Dynamic announcements, loading, error and empty states

State	Implementation
Live regions	`role="status"`: Form Variations outcome, Filter List count, Conversation Canvas search, article meta "Link copied" (`decorators/article-meta-decorator.js`); `aria-live="polite"` on the Gen Carousel track

State	Implementation
Loading (skeletons)	Data later than 150ms shows a skeleton: the skeleton is `aria-hidden="true"`, the block `aria-busy="true"` until the content is rendered (`decorators/skeleton-decorator.js`); no focus move; the pulse stops for reduced motion
Empty	List: the Empty message (default "There are no items to show.") or hidden, per the List's empty behaviour; Filter List: "No resources found." with a hint; Canvas menu: "No pages found"
Error page	The authored 404 page per locale, else `/en`, else built-in content; "Go back" is added for referrers from the site (`scripts/scripts.js`)
Editor-only	Items without content show a hint in Universal Editor only (`itemEditorHint`)

12. Contrast

- **Tokens**: `--color-text-secondary` (`--slate-700`) 6.9:1 on white and `--color-text-brand` / `--color-bg-brand` (`--brand-400`) 5.4:1, replacing `--grey-400` (2.84:1) and `--brand-500` (4.43:1) after the first audit found 3,737 axe colour-contrast nodes; this change still needs design sign-off (audit, sections 1, 2 and 13).
- **Dark backgrounds**: the theme contract (see design-system.md) was measured on "Sections: colours with blocks": 43 axe violations before (Grey Teaser in a dark Section 1.09:1, inline links 1.72:1 to the text, List image cards 1.05–1.48:1), 0 after; focus ring brand 300 11.34:1 on Dark and 7.23:1 on Dark Green; Grey islands brand 500 4.07:1; inverted Primary button 19.6:1 (dialog options report, section 10; handover 2.34, F-105).
- **Register entries not yet updated**: handover F-27 (video caption `#999`) and F-28 (secondary text `#999`, white on `#4673db`) still read as open, but the code now uses `--color-text-secondary` for video captions and the darker tokens; non-text contrast of the video pause outline (F-56) and the Teaser's own Dark Primary buttons (F-49) remain open.
- **Forced colours**: a `forced-colors: active` query exists only in `blocks/separator/separator.css` (the handover's "present in the CSS" covers only that block).

13. Reflow, zoom and touch targets

Topic	What is implemented	Evidence
Reflow (1.4.10)	Mobile-first layouts, rem sizes, `overflow-wrap: break-word` on links, sideways motion clipped (`motion-clip`)	Audit harness: 0 loads with horizontal overflow at 1440 / 1024 / 768 / 600 / 390px; **320px not tested**
Zoom / resize text (1.4.4)	`<meta name="viewport" content="width=device-width, initial-scale=1">` (zoom not disabled); rem-based type	No zoom test recorded
Text spacing (1.4.12)	No fixed heights on text known to clip	Not recorded
Touch targets	Button 3.125rem, icon button 2.75rem, Page Header actions 2.75rem, Canvas menu button 59px, nav items 48px, voice button 32px	Values from CSS and READMEs; no systematic measurement
Print	`@media print` exists only in `blocks/separator/separator.css`	—

14. Reduced motion

`utils/environment.js` `prefersReducedMotion()` and `scrollBehavior()` (`auto` instead of `smooth`) are shared. With `prefers-reduced-motion: reduce`: the Additional tab animations aren't prepared (content shows at once), skeletons don't pulse, Lottie stays on its first frame, videos don't autoplay, the Gen Carousel starts paused, smooth scrolling is off. Nothing animates in Universal Editor edit mode. Background animation on the first hero delays the LCP image (F-94; author guidance). See animation report section H.

15. Personalized content

No personalization runtime exists (no Adobe Target, AJO or edge personalization): every visitor gets the same authored content, so there is no personalized-content accessibility case to test. Blocked until a personalization design exists (audit, section 8).

16. Responsibilities

Role	Responsibilities
Content authors	Write meaningful alt text, or leave it empty only for decorative images; choose heading levels in a logical order (one H1 per page, from the Page Header or the first Title); write link and button text that makes sense out of context; add captions / transcripts to videos; don't put information only in an animation, image or colour; check pages in the block library's language; keep placeholders translated
Developers	Use semantic elements and the shared decorators (buttons, links, headings, `announceNewTab`, `hideVisually`); keep all UI text in placeholders with an English fallback; keyboard support (Tab, Shift+Tab, Enter, Space, Escape, arrows, Home / End where the pattern needs it); visible focus; pause controls for moving content; reduced motion; semantic tokens for text colours so the theme contract applies; update the block README's Accessibility section
Designers	Provide colours that pass 4.5:1 (text) and 3:1 (non-text, focus) on every background, including the dark bands; sign off the contrast token change (audit); define focus, hover, disabled and error states; keep targets at least 1.5rem (24px, WCAG 2.5.8); design dark and Grey island combinations
QA	Run the axe harness at 1440 and 390px on the block library and curated pages; keyboard-test each changed block; check reduced motion, 200% zoom and 320px reflow; screen-reader runs when available; record results with the test vocabulary (Passed, Failed, Corrected and passed, Blocked, Not applicable)
Site administrators	Keep the placeholders (`/{locale}/placeholders.json`) complete per locale, including the accessibility strings (`opensInNewTab`, `pauseAnimation`, `playAnimation`, slide labels); publish error pages per locale; keep `<html lang>`-relevant metadata (language) correct
AI agents	Follow AGENTS.md "Accessibility" and this page; never claim a test that wasn't run; use `announceNewTab`, placeholders and the button decorators instead of re-implementing them; check new blocks inside the dark theme contract; record gaps as "Not recorded" or "Blocked"

17. Definition of done and evidence

A block or page change is accessibility-done when:

1. Semantics: native elements or the ARIA pattern of the component; headings at the authored level; landmarks unchanged.
2. Keyboard: every action reachable and operable; Escape closes popups and returns focus; no keyboard trap; focus visible on light and dark backgrounds.
3. Names: every control has a name from visible text or placeholders; new-tab links announce it.
4. Media and motion: pause / play for anything that moves for more than 5 seconds; nothing animates for reduced motion or in edit mode.
5. Contrast: axe colour-contrast clean on the block library page, including a dark Section variant; text over images checked manually.
6. Automated run: the audit harness (axe WCAG 2.0 / 2.1 / 2.2 A and AA at 1440 and 390px) reports 0 violations for the affected pages.
7. Documentation: the block README's Accessibility section is updated.

Recorded evidence:

Evidence	Scope	Result	Status
First audit (19 October 2026), axe at 1440 and 390px	188 pages (block library and curated design pages), 940 loads	3,745 violation nodes before (colour-contrast 3,737, scrollable-region-focusable 8), 0 after	Corrected and passed

Evidence	Scope	Result	Status
Re-audit (20 October 2026)	Same 188 pages × 5 widths	0 violation nodes before and after the re-audit fixes; new defects found by code review and targeted browser tests	Passed
Theme contract (2.34)	"Sections: colours with blocks" at 1440 / 390px	43 → 0 violations; live check after package 1.0.13: 0 violations	Corrected and passed
Performance regression run (2.36)	955 loads	0 axe violations	Passed
Code review per block	Semantics, headings, keyboard, focus, names, alt, reduced motion, pause controls, live regions, duplicate ids	23 accessibility defects corrected in the first audit (D-A-2, D-A-4, D-A-25, D-B-1, …)	Corrected and passed
Screen reader (NVDA, VoiceOver)	Published pages	Not executed: no screen reader in the environment	**Blocked**
Universal Editor dialogs (authoring accessibility)	Real dialogs	Needs AEM author access	**Blocked**
Reflow at 320px, 200% / 400% zoom, text spacing	—	Not recorded	Not recorded

Tools: `docs/audit/tools/harness.mjs` (all pages × widths with axe and data-layer checks) and `compare.mjs` (published vs local). The scripts use paths of the delivery workspace (the drafts folder, a local `axe-core` and Playwright install), so they need adapting before re-use: Requires validation.

18. Gaps and recommendations

Gap	Status	Proposed next step (not an approved decision)
Project rule says WCAG 2.1 AA; this knowledge base targets 2.2 AA	Requires validation	Product owner decides; update AGENTS.md and handover 1.7.2 together
Screen-reader testing	Blocked: NVDA / VoiceOver on the published pages	Run on the curated pages and each interactive block
No skip link	Not recorded as a decision	Consider a "Skip to content" link before the header
Anchor List Sticky and Focus Not Obscured (2.4.11)	Requires validation	Test tabbing under the pinned list
Author bio LinkedIn link has no new-tab hint (F-52)	Open	Use `announceNewTab`
Video pause outline contrast (F-56), Teaser Dark Primary buttons (F-49)	Open	Design and development fix
Contrast token change needs design sign-off	Requires validation	Design sign-off
`forced-colors` and print styles only in the Separator	Not recorded	Review in Windows High Contrast
Register F-27 / F-28 read as open though the code changed	Documentation	Update the handover register

19. Related

- design-system.md: tokens, theme contract, focus states, motion
- performance.md: skeletons and loading behaviour

- Other knowledge base pages: testing-and-quality.md, blocks.md, authoring.md, development-standards.md, personalization.md
- AGENTS.md "Accessibility" (Testing) and "Accessibility" in Blocks
- `blocks/README.md` and each block README's "Accessibility" section
- `decorators/README.md` (a11y, link, button, skeleton, animation decorators)
- Block ecosystem audit, re-audit, dialog options report, animation report
- Handover: Part 1 — 1.7.2 Accessibility, Part 16 — Findings

20. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

21. Known limitations

- None recorded for this document.

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Analytics Data Layer Guide

Attribute	Value
Document ID	DEV-009
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	The Adobe Client Data Layer implementation and the rules for developers
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	decorators/data-layer-decorator.js, utils/analytics.js, docs/handover/knowledge-base/data-layer-and-analytics.md
Related documents	—
Related AEM paths	—

1. Status at a glance

Capability	Status	Where
Data layer array `window.adobeDataLayer` (created on the first push, `window.adobeDataLayer || []`)	Active	`utils/analytics.js` `pushToDataLayer`
Page entry (once per page)	Active	`trackPage`
Component entry per block (once per block)	Active	`trackBlocks`
Events `cmp:click`, `cmp:show`, `cmp:hide`, `cmp:media`, `cmp:search`, `cmp:submit`, `cmp:change`	Active	`trackInteractions` (document listeners) and Tabs, Gen Carousel, Modal
Richer state for List, Cards, Separator, Tabs panels	Active	`decorators/card-list-decorator.js` `trackCardList`, `blocks/separator/separator.js`, `blocks/tabs/tabs.js`
Search term redaction	Active (unit tested)	`utils/privacy.js` `redactSearchTerm`, `test/privacy/privacy.test.mjs`
Adobe Client Data Layer library (`getState()`, `addEventListener()`)	Not loaded: the array stays a plain JavaScript array	—
Adobe Tags (Data Collection), AEP Web SDK, Edge Network, datastreams, XDM schema	Blocked: measurement plan (not configured)	`scripts/delayed.js` holds only a comment
Adobe Analytics, Customer Journey Analytics	Blocked: measurement plan (not configured)	—
Consent management, visitor identity	Not implemented	—

Capability	Status	Where
Personalized impressions	None (no personalization runtime)	Personalization
Real User Monitoring (RUM, including the 404 checkpoint)	Active (platform)	`scripts/aem.js` `sampleRUM`, `404.html`

Nothing is sent to an analytics server by project code today: the data layer lives only in the visitor's browser until a tag manager processes it. RUM (`sampleRUM`, part of the Edge Delivery library) is the only measurement that leaves the page.

2. Architecture



3. Modules

Module	Exports	Role
`utils/analytics.js`	`pushToDataLayer(entry)`, `pushComponents(components)`, `pushComponentEvent(event, id, info)`	Low-level pushes; every push is wrapped in `try / catch` so analytics never breaks rendering or links
`decorators/data-layer-decorator.js`	`trackPage()`, `trackBlocks(container)`, `trackInteractions()`, `componentIdOf(block)`	Page and component state, document-wide listeners (capture phase), component ids
`decorators/card-list-decorator.js`	`trackCardList(block, options)`	List and Cards state with one entry per card
`blocks/separator/separator.js`, `separator-config.js`	`analyticsEntry(config, id)`	Separator state (opt-in Analytics field)
`blocks/tabs/tabs.js`, `blocks/gen-carousel/gen-carousel.js`, `blocks/modal/modal.js`	—	`cmp:show` for tab panels, slides and dialogs; Tabs also pushes its panels as components
`decorators/skeleton-decorator.js`	`showSkeleton`	Marks a block `data-analytics="pending"` while its skeleton shows and tracks it once rendered
`utils/privacy.js`	`redactSearchTerm(value)`	Data minimisation of search terms
`utils/viewport.js`	`currentBreakpoint()`	`breakpoint` of clicks: `mobile` (< 600px), `tablet` (600–1199px), `desktop` (>= 1200px)
`utils/metadata.js`	`getPageMetadata()`, `getPageLanguage()`, `getPageTags()`	Page entry values

4. When entries are pushed

Phase (`scripts/scripts.js`)	Call	What is pushed
Eager, start of `loadEager`	`trackInteractions()`	The page entry (first entry); the document listeners start
Eager and lazy, while blocks decorate	`trackCardList`, Separator, Tabs	List / Cards (and their cards), Separator (when its Analytics is on), tab panels

Phase (scripts/scripts.js)	Call	What is pushed
Lazy, after `loadMainSections(main)`	`trackBlocks(main)`	One entry for every block in `main` not yet tracked
Skeleton removed (List, Filter List)	`trackBlocks(block.parentElement)`	The block once it has rendered, not its skeleton
Delayed, 3 seconds after load (`loadDelayed`)	`trackBlocks(document)`, then `import('./delayed.js')`	Header, footer and blocks added later (for example modals); `delayed.js` adds nothing today
Any time	document listeners, Tabs, Gen Carousel, Modal	`cmp:*` events

The same code runs in Universal Editor, so entries are also pushed while authors edit (nothing leaves the browser).

5. Page entry

`trackPage()` pushes one entry: `{ page: { 'page-{hash}': { ... } } }`. The id is `page-` plus a djb2-style hash of `window.location.pathname` in base 36, so the same path always gets the same id.

Field	Value
`@type`	`tcs-corp/components/page`
`dc:title`	`og:title` metadata, else `document.title` (read before the SEO decorator applies `meta-title`)
`xdm:language`	The `language` metadata when it is a language code, else the locale's language tag (`utils/site.js`)
`xdm:template`	The Page Type (`page-type` metadata, else `template`); omitted when empty
`xdm:tags`	`article:tag` meta values and the `cq-tags` meta, else the comma-separated `tags` meta; de-duplicated
`repo:path`	`window.location.pathname` (the published path)

Region, country, persona, content type, author and campaign id are page metadata (`utils/metadata.js`) but are not in the page entry. There is no separate page-view event: the page entry is the page-view signal a future Tags rule would use.

6. Component entries, naming and ids

Generic entries

`trackBlocks(container)` adds every decorated `.block` once: `{ component: { '{id}': { ... } } }`.

Field	Value
(key)	The component id (`componentIdOf`, below)
`@type`	`tcs-corp/components/{block}`: the block's `data-block-name`, else its first class (for example `teaser`, `hero`, `header`, `footer`, `columns-container`)
`parentId`	The id of the nearest enclosing block (a Hero in a Gen Carousel, a block in a Fragment), else the page id
`dc:title`	The first heading (h1–h6) of the block: its `aria-label` or text, whitespace collapsed, at most 100 characters; omitted without a heading

A block is skipped when it already has `data-cmp-tracked`, or when it or an ancestor carries `data-analytics` (`off`, `custom` or `pending`). Blocks hidden with the Additional tab's Hide block are still added. The Tab and Column Break markers set `data-analytics="off"` so they aren't registered as components.

Component ids

`componentIdOf(block)` stores the id as `data-cmp-id`:

1. An existing `data-cmp-id`.
2. Else the block's `id`: the Anchor ID of the Additional tab (`decorators/block-options-decorator.js`), the Component ID of List, Cards and Separator, or an id the block sets itself.
3. Else `{block}-{n}`: a counter per block name in the order ids are first requested (normally page order), for example `teaser-1`, `teaser-2`.

`{block}-{n}` ids shift when blocks are added, removed or reordered; give blocks that matter for reporting a stable Anchor ID (or Component ID).

Item ids and positions

Block	Child component ids	Position
List	`{list id}-item-{hash}` per card (`@type tcs-corp/components/list/item`)	`xdm:position` in the card entry; `data-cmp-position` on the card `li` (the feature card is position 1)
Cards	`{cards id}-item-{n}` per card (`tcs-corp/components/cards/item`)	As List
Tabs (block and Tabs Container)	Each panel id (`tcs-corp/components/tabs/item`, `parentId` the Tabs block); columns built in a panel get `{panel id}-columns`	—
Gen Carousel	`data-cmp-id` per slide (`{carousel}-slide-{n}`); slides aren't component entries	`position` in `cmp:show`

Card entries carry `dc:title` (the item's Analytics label, else its title, else the start of its description), `xdm:linkURL`, `xdm:position` and `repo:modifyDate`. The List entry adds `xdm:listSource` (`parent`, `fixed`, `tags`) and `xdm:variant`; the Separator entry adds its configured options (see `blocks/separator/README.md`).

Variations are not recorded as a generic field: only List / Cards (`xdm:variant`, the card style or `links`) and Separator push a variant. Block classes (Additional tab, Style) are not in the entries.

7. Events

All events use the envelope pushed by `pushComponentEvent`:

```
{ "event": "cmp:click", "eventInfo": { "path": "component.{id}", "...": "..." } }
```

`eventInfo.path` is `component.{id}`, or `page.page-{hash}` for clicks outside any block (links in default content). The component of an event (`componentOf`) is the nearest `data-cmp-id` inside the same block (a card, a slide), else the block, else the page. The document listeners run once, in the capture phase.

Event	Fires when	Extra `eventInfo` fields	Not pushed for
`cmp:click`	Click on `a[href]`, `button`, `[role="tab"]`, `[role="button"]` in a block, or a link in `main` default content	`interaction` (`data-cmp-interaction`, else `link` / `button`), `dc:title` (label, 100 characters), `xdm:linkURL`, `xdm:linkType` (`download`, `exit`, `other`), `value` (`data-cmp-value`), `position` (`data-cmp-position` of the element or its item), `breakpoint`	`summary` elements; elements in `data-analytics="off"`; the submit button of a search form; the click ending a scroller drag (`data-dragged`); `aria-disabled="true"` controls; elements outside blocks and outside `main` links

Event	Fires when	Extra eventInfo fields	Not pushed for
cmp:show	An accordion <details> opens (toggle); the user selects another tab; the active Gen Carousel slide changes (not the first slide on load; autoplay changes included); a Modal opens	dc:title (item or tab label), position (slide), interaction: 'dialog' (Modal)	The initial tab and the first slide
cmp:hide	An accordion item closes; any <dialog> closes (close button, Escape, backdrop)	dc:title (item), interaction: 'dialog'	—
cmp:media	play, pause, ended of a <video> / <audio> in a block	interaction (the event type), xdm:mediaURL (currentSrc), position (seconds)	Background videos (no controls, loop, muted); play / pause made by the page (data-cmp-auto, set by the Video block's off-screen pause / resume); YouTube and Vimeo iframes
cmp:search	Submit of a form in a block with role="search" or an input[type="search"] (today the Conversation Canvas search)	xdm:searchTerm: the redacted term	Empty terms (after redaction)
cmp:submit	Submit of any other form in a block (today Form Variations) that passes the browser's constraint validation (novalidate forms are checked with checkValidity())	dc:title (the form's aria-label, else its aria-labelledby heading text, else its id, else the block name), fields (unique field names, never values)	Invalid submissions
cmp:change	change of an input / select with data-cmp-interaction (today Filter List options)	interaction, value (data-cmp-value, else the control value: the tag id), checked (checkboxes)	Controls without data-cmp-interaction (all Form Variations fields)

xdm:linkType is download when the link has a download attribute or its URL ends in .pdf, .doc(x), .xls(x), .ppt(x), .zip, .csv, .txt, .mp3 or .mp4; exit when it goes to another origin (also mailto: and tel: links); else other.

Pushes outside the listeners: Tabs cmp:show (blocks/tabs/tabs.js, also the DOM event tabs:change), Gen Carousel cmp:show, Modal cmp:show (blocks/modal/modal.js). Tab buttons also push a cmp:click (interaction button). Other DOM events (form-variations:submit, conversation-canvas:search, header:menu, tabs:reveal) are not in the data layer.

8. data-cmp-* attributes

Attribute	Set by	Effect
data-cmp-id	componentIdOf on blocks; blocks on child components (cards, slides)	The component of events inside it
data-cmp-interaction	Blocks and decorators on links, buttons, inputs	interaction of cmp:click; makes an input / select push cmp:change
data-cmp-value	Blocks on links, buttons, inputs	value of cmp:click / cmp:change (for example a filter tag id or all)
data-cmp-position	decorators/card-list-decorator.js on card items	position of cmp:click (1-based)

Attribute	Set by	Effect
`data-cmp-tracked`	`trackBlocks`	Prevents a second state entry
`data-cmp-auto`	`blocks/video/video.js` before an automatic pause / play	The next media event is not pushed
`data-analytics="off"`	List / Cards / Separator with Analytics off, Tab, Column Break	No state and no events inside the block
`data-analytics="custom"`	`trackCardList`, Separator with Analytics on	The block pushes its own state; its events are still tracked
`data-analytics="pending"`	`decorators/skeleton-decorator.js`	Not tracked until rendered
`data-analytics-label`	Card links (List, Cards)	Informational; the Analytics label becomes the card entry's `dc:title`

`data-cmp-interaction` values in the code:

Value	Where
`card`, `cta`, `previous`, `next`	`decorators/card-list-decorator.js` (List, Cards)
`filter`, `filter-option`, `load-more`	`blocks/filter-list/filter-list.js`
`filter` (tag filters)	`blocks/conversation-canvas/conversation-canvas.js`
`previous`, `next`, `pause` / `play` (the toggle's current action)	`blocks/gen-carousel/gen-carousel.js`
`play`	`blocks/embed/embed.js` (placeholder play button)
`playback`	`blocks/video/video.js` (playback toggle)
`listen`, `download`, `share`	`decorators/article-meta-decorator.js` (Page Header)
`animation`	`decorators/animation-decorator.js` (Lottie pause / play)

Author fields that affect analytics: the Anchor ID (Additional tab) becomes the component id; List / Cards Component ID, Analytics (`analyticsEnabled`, default on) and the items' Analytics label; Separator Component ID, Analytics (default off) and Analytics label. See the block READMEs (List, Cards, Separator).

9. What is measured today

Measurement	Status	How (in the data layer)
Page views	Active (data layer only)	The page entry, once per page load; no page-view event
Component impressions	Active (state only)	One component entry per block and per List / Cards card when the block is decorated; no visibility-based impression events
Shown content	Active	`cmp:show` for accordion items, tabs, carousel slides, modals
Clicks (links, buttons, CTAs, cards, navigation)	Active	`cmp:click` with `interaction`, `xdm:linkType`, `position`, `breakpoint`; header and footer links are attributed to the header / footer blocks
Downloads	Active	`cmp:click` with `xdm:linkType download`; the Page Header download (print) action is `interaction download`
Exit links	Active	`cmp:click` with `xdm:linkType exit`
Internal search	Active	`cmp:search` with the redacted term (Conversation Canvas); no search results page code in the repository
Filters	Active	Filter List pills (`cmp:click filter`), advanced options (`cmp:change filter-option`), Load more; Conversation Canvas tag filters (`cmp:click filter`)

Measurement	Status	How (in the data layer)
Forms	Active (attempts)	`cmp:submit` for valid submissions, field names only; the outcome (sent / failed) is not pushed (blocked: measurement plan)
Video	Active (file videos)	`cmp:media` play / pause / ended for <video> files; YouTube / Vimeo iframes are not tracked beyond the Embed play button and the Video playback toggle (`cmp:click`)
Errors (404)	Active (RUM only)	`404.html` calls `sampleRUM('404', { source: document.referrer })`; the data layer pushes the regular page entry of the 404 document, with no error event
Personalized impressions	None	No personalization runtime (Personalization)

RUM (`sampleRUM` in `scripts/aem.js`, never modified) samples page views (by default one in 100, overridable with `?rum=` / `?optel=`), sends checkpoints of sampled page views with `navigator.sendBeacon` to the RUM collector and loads the RUM enhancer for them (`sampleRUM.enhance()` after the first section in `loadMainSections`). It is independent of the data layer.

10. Duplicate-event prevention

Rule	Code
One page entry per page	`trackPage` keeps the page id
Listeners registered once	`trackInteractions` (`listening` flag)
One state entry per block	`data-cmp-tracked`; blocks with `data-analytics` (custom state, off, pending) are skipped
A search is one `cmp:search`, not also a click	`onClick` ignores the submit button of a search form
A drag isn't a click	`decorators/scroller-decorator.js` marks `data-dragged="true"` during a drag
A disabled control isn't a click	`aria-disabled="true"` ignored
Accordion summaries push show / hide, not a click	`summary` ignored by `onClick`
Automatic media actions aren't the visitor's	`data-cmp-auto` (Video block off-screen pause / resume); background loops ignored
No empty searches, no invalid submits	`redactSearchTerm` returns `''`; `checkValidity()` for `novalidate` forms

The audit harness confirms it: in the re-audit, 758 of 758 test clicks pushed exactly one `cmp:click`, and no page had duplicate component ids (`docs/audit/REAUDIT-2026-10-20.md`).

11. Privacy and data minimisation

Rule	Code
No form values	`cmp:submit` carries field names only; Form Variations fields have no `data-cmp-interaction`, so typing pushes nothing
Search terms redacted	`redactSearchTerm`: e-mail addresses become `[email]`, phone-like numbers (7+ digits) `[number]`, trimmed, at most 100 characters; unit tested
Labels truncated	`dc:title` of clicks and generic entries at most 100 characters
Opt-out per block	`data-analytics="off"` (List, Cards, Separator Analytics fields)
No storage	The data layer code writes no cookies or web storage

Known residual risks (not corrected in code): `mailto:` and `tel:` links are pushed in full as `xdm:linkURL` (handover finding F-05); query strings of link URLs are pushed as they are; the Conversation Canvas Search page URL carries the raw query (`?q=`), which a future Web SDK would collect as the page URL (re-audit R-D-12, blocked until Tags exists). See Security, privacy and consent.

12. Adobe Data Collection, Web SDK, Analytics and CJA: not configured

Blocked: measurement plan. No Adobe Tags property, Web SDK (`alloy`), Edge Network datastream, XDM schema, Analytics report suite or CJA data view exists in the repository, and no ids are recorded here. The handover Part 11 (section 11.7, "Integrating Adobe Tags and the Web SDK") describes the planned flow as a recommendation only:



Planned points from the handover (not implemented): the Tags embed loads from `scripts/delayed.js` (delayed phase, so `strict-dynamic` in the CSP trusts it), never from `head.html` and never in Universal Editor or on AEM author; staging library on `.aem.page`, production on `.aem.live` and the production domain; the ACDL extension processes the entries already in the array; `xdm:linkType` values already match the XDM `web.webInteraction.type` enumeration; a page-view rule, link / exit / download rules and rules per `cmp:*` event. The measurement plan, the Tags property, the datastreams, the XDM mapping and the consent platform are open items (handover Part 16, O-07).

13. Consent and identity

No consent banner, consent management platform or consent check exists, and no visitor identity (ECID, login, CRM id) is read or set. The data layer is filled without a consent check because it stays in the page; consent must govern any future sending (handover 11.7.6; Security, privacy and consent). The consent categories and the identity strategy are blocked dependencies (re-audit / audit section 9).

14. Debugging and validation

Console checks on preview or live (not in Universal Editor edit mode, where clicks select components):

```
window.adobeDataLayer;                                  // all entries
adobeDataLayer.find((e) => e.page);                     // the page entry
adobeDataLayer.filter((e) => e.component)
  .flatMap((e) => Object.entries(e.component))
  .map(([id, c]) => [id, c['@type'], c.parentId]);      // component ids and types
adobeDataLayer.filter((e) => e.event);                  // events only
adobeDataLayer.filter((e) => e.event).at(-1);           // the last event
```

`adobeDataLayer.getState()` and `addEventListener()` exist only once an ACDL library is loaded. Enable "Preserve log" (or Ctrl / Cmd + click) when testing links that navigate. `?debug=list` and `?debug=separator` log the List / Separator configuration to the console (not data layer entries).

Validation	Status	Source
Audit harness: data layer present, page entry, component count, duplicate component ids, one `cmp:click` per test click (local preview, 188 pages × 5 widths)	Passed (re-audit, 20 October 2026)	`docs/audit/tools/harness.mjs`, `docs/audit/REAUDIT-2026-10-20.md`
Search redaction unit tests (`npm test`)	Passed (5 of 5, run locally on 21 October 2026; CI runs lint only, not `npm test`)	`test/privacy/privacy.test.mjs`
Unit tests of the data layer decorator	Not applicable (none exist)	—
Adobe Tags, Web SDK, Analytics / CJA, duplicate-event checks in the AEP Debugger	Blocked: Tags property, datastream and measurement plan	Audit section 9

Manual test cases (page entry, 3-second component entries, link / download / exit clicks, cards, accordion, tabs, carousel, modal, video, search, form, filter, Analytics off, breakpoint) are listed in the handover 11.8; troubleshooting is handover Part 13 (13.20 Data Layer Events Missing).

15. Rules for developers

- Reuse the data layer code: new blocks get state, clicks, accordion toggles, dialogs, media and forms automatically. Describe interactions with `data-cmp-interaction`, `data-cmp-value`, `data-cmp-position` and `data-cmp-id`; don't push clicks from blocks (`AGENTS.md`, Data layer).
- Push only state the central code can't know, with `pushComponentEvent` / `pushComponents` from `utils/analytics.js` (for example a tab or slide shown); for richer state set `data-analytics="custom"` (see `trackCardList`).
- Don't add parallel tracking libraries, a second data layer or a tag script in `head.html`; a future Tags embed belongs in `scripts/delayed.js`.
- Never push form values or other personal data; run free text through `utils/privacy.js`.
- Document new interactions in `decorators/README.md` and the block README.

16. Related

- Architecture, Blocks, Indexing and search, Decision log
- Personalization
- Security, privacy and consent
- `decorators/README.md`, `utils/README.md`
- `AGENTS.md` (Data layer, Three-Phase Page Loading)
- Handover Part 11 — Analytics & Data Layer, Part 13 — Global Troubleshooting Guide, Part 16 — Handover Findings and Open Items Register (`TCS-CORP-HANDOVER.md`)
- `docs/audit/BLOCK-ECOSYSTEM-AUDIT.md` (sections 6, 7 and 9), `docs/audit/REAUDIT-2026-10-20.md`
- Block READMEs with data layer sections: `blocks/README.md`

17. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

18. Known limitations

- None recorded for this document.

19. Open items

- None.

20. Change history

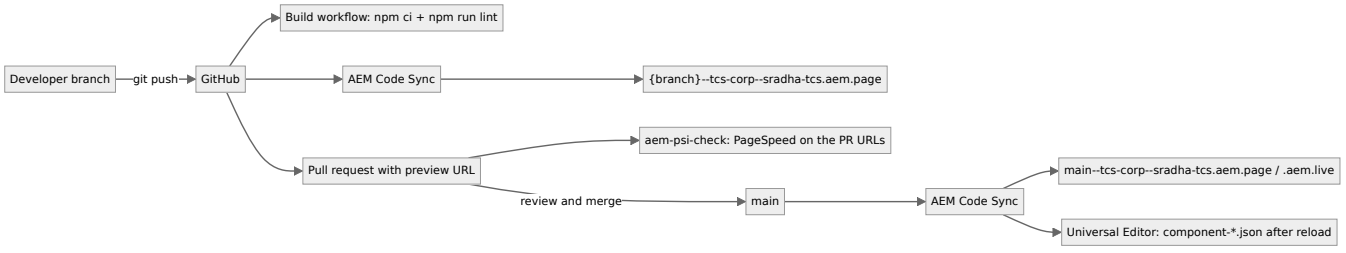
Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Coding Standards

Attribute	Value
Document ID	DEV-003
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	JavaScript, CSS, HTML, model, shared-code, error handling, logging, responsive, accessibility, performance and review standards
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/development-standards.md, .eslintrc.js, .stylelintrc.json
Related documents	—
Related AEM paths	—

1. Repository and Branching Workflow

The repository is `sradha-tcs/tcs-corp` on GitHub, based on `adobe-rnd/aem-boilerplate-xwalk`. There is no build step: the committed JavaScript and CSS are what the site serves. The model is trunk-based: short-lived branches merged into `main` through pull requests with merge commits (`Merge pull request #NN from sradha-tcs/<branch>`).



Steps

1. Branch from an up-to-date `main`.
2. `npm install` once per clone: it also installs the Husky pre-commit hook (`prepare`).
3. Run the local server: `npx -y @adobe/aem-cli up --no-open --forward-browser-logs` (or `aem up`), at `http://localhost:3000`. Add `--html-folder drafts` for static test pages in `drafts/`.
4. Implement the change following AGENTS.md.
5. Run the generators the change needs (see Definition of Done): `npm run build:json`, `npm run build:modules`, `npm run build:docs`, `npm run build:paths`.
6. `npm run lint` and `npm test`.

7. Push the branch and check the feature preview `https://{branch}--tcs-corp--sradha-tcs.aem.page/{path}` (slashes in branch names become hyphens in the host).
8. Run PageSpeed Insights on the feature preview URL; the target is 100 (AGENTS.md).
9. Open the pull request to `main` with a link to the feature preview page that shows the change: AGENTS.md states that a PR without it is rejected.
10. `gh pr checks`: Build, AEM Code Sync and the PageSpeed check must be green; a reviewer inspects the URL and merges.
11. After the merge, Code Sync updates `main`; authors reload Universal Editor to get changed models.

Branch naming

Prefix (from `git log --merges`)	Merges	Used for	Examples
`feature/`	85 (74 of them `feature/tcs-…`)	New features (the earlier history)	`feature/tcs-cards-manual-list`, `feature/columns-container`
`fix/`	30	Defects and audits	`fix/block-reaudit`, `fix/section-options-delivery`
`docs/`	6	Handover and documentation	`docs/handover-2-35`, `docs/block-readmes`
`feat/`	2	Features (recent)	`feat/list-item-display`, `feat/section-bg-colors`
`perf/`	2	Performance	`perf/module-preload`, `perf/font-fallback`

Keep names short, lower case and valid in a host name (branch, repository and owner form the `aem.page` sub-domain). The handover (12.3) recommends `feature/tcs-<topic>`; the recent history uses `<type>/<topic>` without `tcs-`. No naming rule is enforced by tooling.

Automated checks

Check	Defined in	Runs	Verifies
Husky pre-commit	.husky/pre-commit.mjs	Every local commit	If a model partial (`_*.json`) is staged, runs `npm run build:json` and stages `component-models.json`, `component-definition.json`, `component-filters.json`
Build	.github/workflows/main.yaml	Every push, every branch	`npm ci`, `npm run lint` (Node 24)
AEM Code Sync	GitHub app (not in the repository)	Every push	Syncs the branch to the code bus
PageSpeed (aem-psi-check)	AEM GitHub app (not in the repository)	Pull requests	PageSpeed Insights on the URLs in the PR description
Renovate	.renovaterc.json	Scheduled	Dependency PRs; devDependencies labelled `ignore-psi-check` and auto-merged when tests pass; ESLint pinned to `^8.0.0`
Unit tests	`package.json` `test`	**Manual only**	Not run by CI: run `npm test` before every PR and state it in the PR

Merging, tags and releases

- Never force-push or rewrite `main`; undo a merge with a revert PR (handover 12.6 Rollback).
- Branch protection rules and required approvals on `main`: Not recorded (handover 12.3 lists them as to be confirmed).
- Git tags: the repository has `home-content-package_<date>` tags that mark the content-package downloads branch before a handover PDF update; production code releases are not tagged.

- `package.json` keeps the boilerplate name and version (`@adobe/aem-boilerplate`, `1.3.0`); there is no changelog file: PR titles, the handover revision history and each block README's Change History record changes.

2. Naming Conventions

Item	Convention	Source
Block folder and class	kebab-case; the folder name is the block class (`blocks/teaser`, `.teaser`)	AGENTS.md "Blocks"
Block files	`{block}.js`, `{block}.css`, `_{block}.json`, `README.md`; helper modules beside them (`list-data.js`, `cards-data.js`, `separator-config.js`, `canvas-journey.js`)	`blocks/`
Component template name	AEM renders a block with the class of its **template name** (class-name form), not its component id: the template name must name a block folder ("Column Break", "Button Block")	test/models/block-names.test.mjs, AGENTS.md "Columns Container", "Button"
Model field names	Don't end a field that must be read in Text, Title, Type, Alt or MimeType (it collapses into its base field: `buttonVariant`, `titleHidden`); no grouped `prefix_name` fields in key-value blocks (`poster`, not `placeholder_image`)	AGENTS.md "AEM delivery rules"
Additional tab fields	`classes_anchor`, `classes_hideblock`, `classes_padding*`, `classes_margin*`, `classes_animation*`, `classes_bganimation` on blocks; `container*` on container sections (sections don't get `classes_*`)	AGENTS.md "Additional tab", "AEM delivery rules"
Anchor ids	`anchor-` prefix, validated `^anchor-[a-z0-9][a-z0-9-]*$`; duplicates get `-2`, `-3`	decorators/block-options-decorator.js
CSS classes	Scoped to the block (`.{block} .item-list`); never `{block}-container` or `{block}-wrapper` (used by sections)	AGENTS.md "CSS"
Placeholder keys	camelCase in code with an English fallback: `t('nextSlide', 'Next slide')`; the sheet's Key may use any casing	utils/README.md
DOM events	`{block or module}:{event}` custom events: `tabs:reveal`, `tabs:change`, `tabs:tab-update`, `form-variations:submit`, `header:menu`, `conversation-canvas:search`, `animation:failed`	AGENTS.md "Shared events and helpers", block READMEs, `grep "new CustomEvent"`
Data layer types	`tcs-corp/components/{block}`; ids from the block id (Anchor ID), else `{block}-{n}`	decorators/data-layer-decorator.js
Tests	`test/<area>/<topic>.test.mjs` (Node test runner)	package.json
Shared modules	`decorators/{purpose}-decorator.js`, `utils/{purpose}.js`, one purpose per file	decorators/README.md, utils/README.md

3. Block Folder Structure

```
blocks/{block}/
├── {block}.js        # export default async function decorate(block)
├── {block}.css       # block-scoped styles (optional; preloaded when listed in scripts/module-graph.js)
├── _{block}.json     # Universal Editor definitions, models and filters (aggregated by npm run
build:json)
├── README.md         # written sections + sections generated by npm run build:docs
└── {block}-*.js      # optional pure helpers, unit tested in test/{block}/
```

- `decorate(block)` loads dependencies, reads the configuration, transforms the DOM and adds listeners; it must tolerate missing or extra fields (authors may omit any field).
- Decide the authored structure (the content contract) before writing code, and treat it as an interface: existing pages depend on it.
- Register a new block in the section / container filters, the module graph (`npm run build:modules`), the docs inventory (`tools/docs/blocks.json`, `npm run build:docs`), and, if animated, `CONTENT_TARGETS` /

`BACKGROUNDS` in decorators/motion-decorator.js.
- Inventory, categories, containers and deprecated blocks: blocks/README.md.

4. JavaScript Conventions

- Vanilla ES modules (ES6+), no transpiling, no bundler, no runtime npm dependencies (vendored libraries: `scripts/dompurify.min.js` for Universal Editor updates, `scripts/lib/lottie/` for Lottie animations).
- ESLint: `airbnb-base`, `plugin:json/recommended`, `plugin:xwalk/recommended`, with `import/extensions` requiring `.js` in imports, Unix line endings, and `no-param-reassign` allowing property changes (.eslintrc.js). Tests (`test/**/*.mjs`) run with Node globals and also require `.mjs` extensions.
- scripts/aem.js is the core library: **never modify it**. Page logic goes in scripts/scripts.js; post-load third-party code in scripts/delayed.js (empty today).
- Scripts must pass the Content Security Policy of `head.html` / `404.html` (`'nonce-aem' 'strict-dynamic'`): load third-party scripts from code (`loadScript`), never as inline tags without the nonce.
- Universal Editor: `scripts/editor-support.js` re-renders the changed block from the editor's update and reloads the page when it can't apply it; a block must be decoratable again from fresh markup.
- After adding, removing or changing an `import`, run `npm run build:modules` (preload lists in `head.html` and `scripts/module-graph.js`); `npm test` fails when they are stale.

5. CSS Conventions

- Stylelint `stylelint-config-standard` (.stylelintrc.json) on `blocks/**/*.css` and `styles/*.css`. `templates/*/*.css` is not covered by `npm run lint:css`.
- Mobile first; `min-width` media queries at 600px, 900px and 1200px (written as `@media (width >= 600px)`); the desktop header starts at 1200px (`blocks/header/header.js` `matchMedia('(min-width: 1200px)')`).
- All sizes in `rem` (1rem = 16px, at most 4 decimals); only breakpoints stay in px. Letter-spacing derived from the rule's font size and reset where a rule resizes a heading, button or tag.
- Every selector scoped to its block; no `{block}-container` / `{block}-wrapper` classes.
- Use the design tokens of styles/styles.css (not primitives) for text, links and focus so blocks follow the dark-background theme contract (AGENTS.md "Backgrounds and theme").
- `styles/styles.css` is eager (LCP), `styles/lazy-styles.css` loads in the lazy phase, `styles/motion.css` only when an animation is used.

6. Model and Definition Conventions

Models live in the partials (`blocks/{block}/_{block}.json`, `models/_*.json`); the aggregated `component-definition.json`, `component-models.json` and `component-filters.json` are generated by `npm run build:json` (the pre-commit hook does it for staged partials) and never edited by hand. The rules that matter most:

- At most three short dialog tabs, the last one **Additional** (every block except header and footer); in item models each tab is an empty cell, so keep `ITEM_LAYOUT` / `CARD_LAYOUT` and their tests in step.
- Repeating content is **item children** added with Add, not composite multifields.
- No sample content in templates or model defaults (they are stored on every new component).
- `xwalk/max-cells`, `no-duplicate-fields`, `invalid-field-name`, `no-orphan-collapsible-fields` and `no-custom-resource-types` run in `npm run lint`; per-model `max-cells` limits are in `.eslintrc.js`. Prefer field collapsing and element grouping to raising a limit.

Details, delivery rules per field type and the editor behaviour: universal-editor.md and AGENTS.md "Content Modeling", "Dialog tabs", "AEM delivery rules", "Item children".

7. Shared Code: Decorators and Utilities

Blocks don't re-implement shared behaviour: `decorators/` create or decorate elements (buttons, links, headings, images, media, data layer, motion, skeletons), `utils/` build no UI (properties, cells, links, dates, tags, URLs, ids, environment, viewport, media, metadata, placeholders, taxonomy, site, query index). Decorators may import utils; **utils never import decorators**; one purpose per file; every button and link button goes through the button decorators. Before adding a module, search both READMEs for an existing export, then list the new module in its README. See decorators-and-utilities.md, decorators/README.md and utils/README.md.

8. Error Handling

Verified patterns to follow:

Layer	Behaviour in code
Block loading	`loadBlock` (`scripts/aem.js`) catches import and `decorate()` errors, logs `failed to load module for <block>` and still marks the block `data-block-status="loaded"`: one block never stops the page
Templates, auto blocks	`loadTemplate` logs `failed to load template <name>`; `buildAutoBlocks` logs `Auto Blocking failed`
Shared data	Placeholders, taxonomy, data sheets, the query index and redirects resolve to empty values on a failed request (`.catch(() => new Map())`, `[]`), never throw; UI text falls back to the English text in code
Analytics	Every data layer push is wrapped in `try / catch` (`utils/analytics.js`): analytics never breaks rendering or links
Motion, skeletons, Lottie	`decorators/motion-decorator.js` and `skeleton-decorator.js` catch and log (`motion`, `skeleton`); a failed Lottie logs `failed to play the Lottie animation` and dispatches `animation:failed`
Fragments	`loadFragment(path, from)` returns `null` for paths not starting with `/` (or starting with `//`), for the page itself and for recursion along the including chain; in the editor the Fragment shows a hint
Authoring gaps	Missing or invalid fields fall back to defaults; in Universal Editor blocks show placeholder hints (List setup message, empty item placeholder `itemEditorHint`) instead of failing
Fallback chains	Header / footer fall back to the locale's `references/nav` / `references/footer`; the 404 page falls back locale → `/en` → built-in content

Rules: validate authored values against allow-lists (unknown values never become classes), keep decoration idempotent, and never let a block's failure hide the rest of the page.

9. Logging

- No `console.log` in site code. Errors that need support attention use `console.error` with a stable message (listed in handover 14.3 Logs); each needs `// eslint-disable-next-line no-console`.
- Debug output only behind a URL switch: `?debug=list` (`[list] …`) and `?debug=separator` (`[separator] …`) via `console.debug` (enable the console's Verbose level).
- RUM (`sampleRUM` in `scripts/aem.js`; `?rum=on` forces sampling) is the only telemetry; never log personal data, form values or search terms.

10. Responsive Design

Mobile-first CSS with the three breakpoints above; `utils/viewport.js` names the breakpoint (`mobile` < 600px, `tablet` 600–1199px, `desktop` ≥ 1200px), which the data layer also reports. Responsive block options (List grid / scroll per breakpoint, Columns Container "Side by side from") use the same breakpoints. Check new work at the five harness widths (1440, 1024, 768, 600, 390px): no horizontal overflow. See testing-and-quality.md.

11. Accessibility First

WCAG 2.1 AA is the target (AGENTS.md "Accessibility"): real `<button>`s with `aria-expanded` / `aria-controls` for disclosures, Escape closes popups and returns focus, pause / play for moving content, `prefers-reduced-motion` honoured (nothing animates for reduced motion or in edit mode), new-tab links announced (`announceNewTab`), form errors next to the field and linked with `aria-describedby`, one `h1` per page, alt text fields on images, and **all UI text from placeholders**. Use the tokens so contrast holds on dark backgrounds.

12. Performance First

- Three-phase loading (eager: first section and LCP; lazy: other sections, header, footer, `lazy-styles.css`, fonts; delayed: after 3 seconds, late block tracking and `delayed.js`): AGENTS.md.
- **Module preloading**: no build step, so each import level is a round trip; keep `head.html` and `scripts/module-graph.js` current with `npm run build:modules`. Stylesheet preloads go in the body because `loadCSS` skips any stylesheet whose href is already linked in the head, whatever its `rel`.
- **No forced layout while decorating**: measure in one shared animation frame, reads before writes (`decorators/scroller-decorator.js`); write `<html>` custom properties only when they change.
- Keep the fallback font faces (`inter-fallback`, `-roboto`, `-dejavu`) when touching fonts (CLS).
- Data blocks render through `renderWhenLoaded` (skeleton after 150ms); shared data loads once per page.
- No third-party script on load; nothing render-blocking added to `head.html`.
- Verify with PageSpeed Insights / Lighthouse on the feature preview (target 100).

13. Data Layer Integration

Every block is in the Adobe Client Data Layer (`window.adobeDataLayer`, Core Components schema), filled centrally by decorators/data-layer-decorator.js through utils/analytics.js: the page entry once, one component entry per block, and document-level listeners for click, toggle, close, media, submit and change.

- Blocks **don't push clicks**. They describe them with `data-cmp-interaction`, `data-cmp-value`, `data-cmp-position` and `data-cmp-id` (child components such as cards); the central capture pushes exactly one `cmp:click`.
- Blocks push only state the central code can't know, with `pushComponentEvent` (a tab or slide shown).
- `data-analytics="off"` stops state and events for a block (List, Cards, Separator Analytics fields); blocks inside `[data-analytics]` are skipped by the generic tracking and push their own state.
- Never push form values (field names only), raw search terms (`redactSearchTerm`, `utils/privacy.js`) or other personal data.
- Adobe Tags and the Web SDK are **not configured** (blocked: measurement plan), so nothing leaves the browser; `adobeDataLayer` stays a plain array (no `getState`) until the ACDL library is loaded.

Contract and events: handover Part 11 — Analytics & Data Layer (TCS-CORP-HANDOVER.md) and decorators/README.md.

14. Personalization

None. No Adobe Target, AJO, CJA or audience integration exists in the repository; every block renders its authored default content, which Edge Delivery caches. The only personalization option (a Teaser marker nothing read) was removed in the block audit (D-B-3). Don't add personalization hooks without an approved personalization design (blocked dependency in docs/audit/BLOCK-ECOSYSTEM-AUDIT.md).

15. Tests

- Pure logic goes in pure modules (no DOM) so it can be unit tested with Node's test runner: `npm test` runs `test/**/*.test.mjs` (14 files, 311 tests at commit 2d06316).
- Tests also guard generated files: the module graph (`npm run build:modules`) and the block documentation (`npm run build:docs`) fail `npm test` when stale; component template names must map to block folders.
- When an item model changes, update its layout constant and the cell tests (`test/model-layout.mjs` derives the expected cells from the model).
- Browser behaviour is checked on the local preview and the feature preview (and with the audit harness); see testing-and-quality.md.

16. Documentation

Update in the same change (AGENTS.md):

Change	Update
A block's model, fields, files or library usage	`npm run build:docs` (generated sections of `blocks/README.md` and the block README); the written sections by hand; `tools/docs/blocks.json` for status, type, purpose and relationships
A block's behaviour	Its README (written sections, Change History)
A decorator or util export	decorators/README.md / utils/README.md
A project rule, delivery rule, shared event or convention	AGENTS.md
Significant changes	The handover TCS-CORP-HANDOVER.md: component chapter, Part 16 findings, revision history; then regenerate the PDF (docs/handover/README.md)
Anything this knowledge base summarizes	The knowledge base page (see Knowledge Base Governance)
A design proposal	docs/proposals/

17. Backward Compatibility and Migration

Content authored before a change must keep rendering. Code and the local preview must assume what AEM really delivers (AGENTS.md "AEM delivery rules", verified on aem.page): empty rows for each dialog tab in non key-value blocks, `<node name>`, `<classes>` first cells for items with a `classes` field, collapsed `*Text` / `*Title` / `*Type` / `*Alt` / `*MimeType` fields, no grouped fields in key-value blocks, no `classes_*` on sections, item names in class-name form, template values stored on new components, non-image DAM files served only once published.

Legacy values the code still reads (verified in code):

Legacy value	Still handled by	Current replacement
Composite multifields (CTA List, links, stats, filter tags, footer links)	The block reads them before the item children	Item children (`hero-cta`, `teaser-link`, `footer-link`, …)
Hero Motion Type `motion-fade` / `motion-slide` / `motion-zoom`; Table List `motion`	`LEGACY_MOTIONS` in `blocks/hero/hero.js`; `addLegacyMotion` in `utils/motion.js`	Additional tab Animation Type / Background animation
Video `placeholder_image` / `placeholder_imageAlt`	`blocks/video/video.js` (where delivered; the grouped field mostly wasn't, so posters need re-selection)	`poster` / `posterAlt`
Column items named "Column" (class `column`)	`scripts.js` replaces `column` with `column-break`	Column Break
Built-in Columns and Button components	Definitions kept (`button-basic` model); not offered in filters	Columns Container / Column Breaks; Button Block

Legacy value	Still handled by	Current replacement
Section Style `highlight`	`main .section.highlight` in `styles/styles.css`	Style Grey (`bg-grey`)
Teaser `titleSize`	No longer delivered; the title takes its level's size	Title Type
List Item rows of 16 cells (before the Display toggles)	`itemLayoutFor` in `blocks/list/list-data.js`	22-cell rows
Form item `name` (lower-cased by AEM)	Keys still work in lower case	`fieldName`
Index columns `featureCard` / `featuredPriority`	`utils/query-index.js` fallbacks	`featured` / `priority`
Separator Width Auto, Variant Custom; List / Cards Component ID	Still read	The widths, Default, Anchor ID

Migration rules: keep reading a removed value while AEM still delivers it; migrate the library content (the delivery workspace's migration scripts are not in the repository); record the migration in the block README, the handover and the audit register (BLOCK-ECOSYSTEM-AUDIT.md, DIALOG-OPTIONS-2026-10-20.md). Production content migration of the items above is an open content action (re-audit section 5).

18. Code Review

- Template: .github/pull_request_template.md asks for the GitHub issue (`Fix #<id>`) and Before / After test URLs. It still contains the boilerplate `{repo}` / `{owner}` placeholders: write `tcs-corp` / `sradha-tcs` in the URLs.
- The PR description must link `https://{branch}--tcs-corp--sradha-tcs.aem.page/{path}` showing the change (AGENTS.md "Publishing Process"); the PageSpeed check reads the URLs from it.
- Reviewer checks: Build, Code Sync and PageSpeed green; `npm test` passed (not in CI); aggregated `component-*.json` included when partials changed; generated files current; scoped CSS in rem; placeholders for UI text; no clicks pushed by blocks; legacy content still renders; docs updated.
- Required reviewers and branch protection: Not recorded.

19. Definition of Done

A change is done when all of these hold:

1. `npm run lint` passes (ESLint with xwalk model rules, Stylelint).
2. `npm test` passes; new pure logic has unit tests.
3. Generators run and their output committed where relevant: `npm run build:json` (models), `npm run build:modules` (imports), `npm run build:docs` (models, new blocks), `npm run build:paths` (`LOCALES`).
4. Verified on the local preview and the feature preview at mobile, tablet and desktop widths; no console errors, no overflow; authored content from before still renders.
5. Keyboard, focus, names and contrast checked on changed blocks; reduced motion honoured.
6. Data layer: one `cmp:click` per click, no duplicate component ids, no personal data.
7. PageSpeed Insights on the feature preview (target 100); no new render-blocking resources.
8. Documentation updated in the same change (table above).
9. PR with the preview URL; checks green; reviewed and merged with a merge commit.
10. Items that need an author, a measurement plan or another environment are reported as **Blocked**, not as passed.

20. Do and Don't

Do	Don't
Search `decorators/`, `utils/` and similar blocks before writing code	Re-implement buttons, links, images, ids or placeholders in a block

Do	Don't
Read rows by `data-aue-prop` in the editor and skip empty tab rows on the page (`itemRows`)	Assume "the first row" or a fixed cell count
Use names that survive delivery (`buttonVariant`, `titleHidden`, `container*` on sections)	End a readable field in Text, Title, Type, Alt or MimeType; group fields in key-value blocks
Use placeholders with English fallbacks for every UI string	Hard-code user-facing text
Scope CSS to the block, in rem, mobile first, with tokens	Style `{block}-container` / `-wrapper` or use px sizes
Describe interactions with `data-cmp-*`	Push `cmp:click` from a block
Keep reading legacy values; migrate content	Rename or remove a field without checking existing content
Run `build:json`, `build:modules`, `build:docs` when relevant	Edit `component-*.json`, `scripts/module-graph.js` or generated README sections by hand
Load third-party scripts from `delayed.js`	Add inline scripts or edit `scripts/aem.js`
Report what couldn't be tested as Blocked	Claim exhaustive or unexecuted tests as passed

21. The Five Rules

1. **Search for reusable implementations first**: decorators, utils, shared card-list code (`decorators/card-list-decorator.js`, `utils/card-list-config.js`, `models/_card-list.json`), existing blocks and their READMEs.
2. **Avoid duplicates**: one option per purpose (the audits removed duplicate dialog options such as Teaser Title size vs Title Type, Section Highlight vs Grey); one module per purpose.
3. **Validate existing content before removal or renaming**: check usage (block library, the audit inventories in docs/audit/data/, the published pages, and production content through an author); production usage needs AEM author access (Blocked for the agent).
4. **Preserve backward compatibility or provide migration**: keep reading values AEM still delivers, map legacy values in code, migrate content, and document the mapping.
5. **Update tests and documentation in the same change**: unit tests, item layouts, generated sections, READMEs, AGENTS.md, the handover and this knowledge base.

22. Knowledge Base Governance

Rules for `docs/handover/knowledge-base/`:

- **Update the knowledge base in the same change** when a change alters a fact it states: the workflow or checks (`.github/`, `.husky/`, `package.json` scripts), lint or style rules, model or delivery rules, decorators / utils exports, the data layer contract, loading and performance mechanisms, test suites, locales and site data, troubleshooting behaviour (console messages, fallbacks), or an integration's status (Adobe Tags, consent, personalization).
- Summarize and link; the detailed sources (AGENTS.md, block READMEs, decorators / utils READMEs, the handover, the audit reports) stay authoritative.
- Verify every statement in the repository (code first); write "Not recorded" or `Blocked: <what is needed>` rather than inventing.
- **Freshness**: set "Last reviewed" only with evidence (the commit the page was checked against and the date); otherwise "Not recorded". A page with unverified sections has Status "Requires validation".
- No secrets or confidential identifiers (program / environment ids, author hosts, tokens, personal data).

23. Related

- testing-and-quality.md, troubleshooting.md

- universal-editor.md, decorators-and-utilities.md, architecture.md, repository-structure.md, deployment-and-release.md, data-layer-and-analytics.md, accessibility.md, decision-log.md
- AGENTS.md, blocks/README.md, decorators/README.md, utils/README.md, templates/README.md
- Handover (TCS-CORP-HANDOVER.md): Part 12 — Publishing, Deployment & Release; 15.4 Content Modeling Standards; Part 16 — Handover Findings and Open Items Register
- docs/audit/README.md, docs/blocks/BLOCK-DOCUMENTATION-REPORT.md, docs/proposals/README.md

24. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

25. Known limitations

- None recorded for this document.

26. Open items

- None.

27. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Content Modeling Guide

Attribute	Value
Document ID	DEV-006
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How blocks are modelled for authors and what AEM really delivers
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	AGENTS.md, docs/handover/knowledge-base/universal-editor.md, docs/handover/knowledge-base/blocks.md
Related documents	—
Related AEM paths	—

1. AEM delivery rules

The rules below are verified on aem.page (block audit of 19 October 2026) and in AGENTS.md; code and the local preview must assume what AEM really delivers.

AEM delivery rules (verified on aem.page, block audit of 19 October 2026, docs/audit/ on the branch migrated-page-content-edam). Code and the local preview must assume what AEM really delivers:

- A block that isn't key-value gets an **empty row for each dialog tab** of its model (Form Variations, Table, Accordion, Hero, Embed, Fragment, Modal): read rows by `data-aue-prop` in the editor, else skip empty tab rows, never "the first row".
- An item row whose model has a `classes` field starts with a cell **"<node name>, <classes>"** (e.g. `email, field-text`); a `name` field is the item node's name, not a cell.
- In key-value blocks, **grouped fields** (`prefix_name`) aren't delivered: use plain names (Video `poster` / `posterAlt`).
- Fields ending in Text, Title, Type, Alt or MimeType **collapse** into their base field (`linkType` becomes bold / italic around the link): use another name (`buttonVariant`) for a value that must be read.
- The built-in Columns component delivers no `classes_*` (no Additional tab); the built-in Button, Title, Image and Text deliver only their fixed fields.
- **Sections don't get their `classes_*` fields delivered** (only `style`, as classes, and the other fields as data attributes): the container sections' Additional tab fields are `container*` (`containerAnchor`, `containerHide`, `containerAnimation`, `containerAnimationSpeed`, `containerAnimationTrigger`, `containerPadding*`, `containerMargin*`), which `scripts.js` `applySectionBlockOptions` turns into the synthetic block's classes.
- New components get their **definition template values** stored (a sample text in a template appears on every new item): no sample content in templates or model defaults.
- A non-image reference (a Lottie `.json`, a PDF) is delivered as a link whose **text is its Alt** (no title): read it with `animationLabel` (`utils/media.js`).

- An item's `name` property is delivered in **class-name form** (`firstName` -> `firstname`): a value whose case matters needs its own field (Form Variations `fieldName`).
- **DAM files other than images** (Lottie JSON, videos, captions) are served only once published: images travel with the page, other assets don't.
- `decorateBlock` wraps the text of a block's cells (`wrapTextNodes`): decorate a synthetic block while it is empty, and don't test cells for bare text after decoration. The library preview (`.home-work/render-lib.mjs`) reproduces these rules; compare aem.page with the local preview after model changes.

2. Field and Property Naming

The naming rules follow what AEM delivers, verified on aem.page in the block audit of 19 October 2026 (AGENTS.md, "AEM delivery rules"):

Rule	Effect	Convention	Example
Field collapse	Fields ending in `Text`, `Title`, `Type`, `Alt` or `MimeType` fold into their base field (`linkText` is the link's text, `imageAlt` the image's alt, `linkType` becomes bold / italic around the link)	Use the suffix only to fold; for a value that must be read, use another name	`buttonVariant` (not `buttonType`); `posterAlt` folds into `poster`
Element grouping	`<group>_<field>` fields render in one cell in row-per-field blocks; **in key-value blocks grouped fields aren't delivered**	Plain names in key-value blocks	Video `poster` / `posterAlt` (was `placeholder_image`, lost: audit L-2); Hero `content_*` (row-per-field block)
Block options	A field named `classes` or `classes_<name>` arrives as block classes, never as a content row	Presentation options and the Additional tab	`classes_anchor`, `classes_paddingtop`, Teaser `classes`
Section fields	**Sections don't get their `classes_*` fields delivered**: only `style` (as classes) and other fields as data attributes	Section containers use `container*` names, turned into classes by `applySectionBlockOptions` (`scripts/scripts.js`)	`containerAnchor`, `containerHide`, `containerAnimation`, `containerPaddingTop` (re-audit R-L-1)
Item node name	An item's `name` property is delivered in class-name form, and an item model with `classes` starts with a cell `<node name>, <classes>`	A value whose case matters gets its own field	Form Variations `fieldName` (re-audit R-D-1); first cell `email, field-text`
Non-image references	A Lottie `.json` or PDF reference is delivered as a link whose text is its Alt	Read it with `animationLabel` (`utils/media.js`)	Hero / Teaser Lottie alt (re-audit R-B-3)
Built-in components	Built-in Button, Title, Image and Text deliver only their fixed fields; built-in Columns no `classes_*`	Don't add fields to built-in models	Button (basic), Columns Additional tab removed
Field name syntax	Letters, digits, `_`, `-`, `:`; starts with a letter; at most one `_` and one `:`	`xwalk/invalid-field-name`	`classes_animationspeed`, `jcr:title`

Property naming habits in the models: camelCase names (`showEyebrow`, `desktopColumns`); `show*` booleans for optional parts; `*Alt` next to each image; `titleType` next to `title`; Additional tab names are lower case after the prefix (`classes_paddingleft`), container names camelCase (`containerPaddingLeft`).

3. Defaults and the Template-Value Rule

- **Template values are stored.** New components get their definition template values written to the content, so a sample text in a template appears on every new item: no sample content in templates or model defaults. Use them only for settings (`buttonView: button`, `titleType: h2`, `desktopMode: grid`).

- Corrected cases (re-audit): Table List "Explore" (R-A-6), Tab title "Tab" (R-C-10), Accordion Item Lorem Ipsum (R-C-11), Table rows "Column1" (R-D-9). Retained exceptions: the Submit Button `label` "Send" (template value; localised by translation only, re-audit review D, low) and the TCS Header logo alt defaults ("Tata Consultancy Services", "TATA Group").
- **Model `value`** (672 fields) is the dialog default; keep it equal to the code's default so an empty value and the default render the same (Hero CTA Arrow default corrected to Auto in the audit).
- UI text that isn't authored content comes from placeholders (`utils/placeholders.js`), never from defaults.

4. Multifields and Item Children

Approach	Status	Rule
Item children (components added with Add, own model, in the container's filter)	Standard	All repeating content: List Items, Cards, CTAs, Teaser Links and Stats, Testimonial Stats, Anchor Links, Filter List tags / categories / pages, Footer Links, form fields, Table rows, Tabs items, Canvas buttons
Composite multifields (`container` with `multi: true`)	Compatibility-only	They didn't render from content packages; the 12 remaining ones are labelled "authored before" and still read (`readMultifieldItems` in `utils/multifield.js`, `readLinkItems` in `utils/links.js`, `buildStats` in `decorators/stats-decorator.js`)
`multi: true` on simple fields	Used	`aem-tag` (tags) and Page Keywords; Universal Editor doesn't save `multi: true` on `aem-content` fields
Non-multi `container`	Used	Collapsible groups in the Conversation Canvas dialog (`menu`, `searchBar`)

The audit recommends removing the legacy multifields after confirming that no production content uses them (deferral D-B-14, blocked: production content check).

5. Container-Child Models

- An item model is a `core/franklin/components/block/v1/block/item` definition with its own model, listed in the container's filter; it has no folder and is documented in its container's README.
- AEM renders each item as a row after the block's own rows, one cell per field (empty and tab cells included); an item model with a `classes` field starts with a `<node name>, <classes>` cell (form items: `classes` is a read-only Field type).
- `itemRows(block, models, isItem)` in `utils/cells.js` finds the item rows: by `data-aue-model` in Universal Editor, else by the cell count (`KEY_VALUE_ITEM` for key-value blocks: 1 or 3+ cells, so a two-field item starts with a tab; `FIELD_ROW_ITEM` for row-per-field blocks: 2+ cells), tested in `test/items/item-rows.test.mjs`.
- An item without content yet (just added) is a placeholder in the editor (`decorators/editor-item-decorator.js`, placeholder key `itemEditorHint`) and is left out on the page.
- `xwalk/max-cells` limits the cells of non key-value models; the item models that need more cells have their own limits in `.eslintrc.js`.

6. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

7. Known limitations

- None recorded for this document.

8. Open items

- None.

9. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Debugging and Troubleshooting Guide

Attribute	Value
Document ID	DEV-014
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	Debugging blocks, delivery, data layer and indexing
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/troubleshooting.md
Related documents	—
Related AEM paths	—

1. Triage First

1. **Where does it happen?** Universal Editor (author content, `main` code), preview (`*.aem.page`), live (`*.aem.live`) or local (`localhost:3000`, local code on previewed content). This separates authoring state, publication state, caching and code.
2. **Which locale?** Most site data is per locale (nav, footer, placeholders, taxonomy, sheets, 404).
3. **How wide?** One page points to content; one block type to block code or its content; every page to code, configuration or the platform.
4. **What changed just before?** A merge, a publication, a package install or a configuration change.
5. **Which breakpoint?** Mobile < 600px, tablet 600–1199px, desktop ≥ 1200px (the header's desktop layout also starts at 1200px).

Diagnostic toolkit

Tool	Use
Console	`failed to load module for <block>`, `failed to load block <block>`, `failed to load template <name>`, `Auto Blocking failed`, `failed to play the Lottie animation`, `motion` / `skeleton` errors, CSP violations
Network	Block JS / CSS, `*.plain.html` (fragments, nav, footer, error pages), `/{locale}/placeholders.json`, `/{locale}/taxonomy.json`, `/{locale}/data/*.json`, `/query-index.json?offset=0&limit=500`
Delivered markup	`curl https://main--tcs-corp--sradha-tcs.aem.page/{path}.plain.html` shows the rows and cells a block receives; `curl -I` shows status and cache headers
Debug switches	`?debug=list`, `?debug=separator` (`console.debug`, enable Verbose); `?rum=on` forces RUM sampling
Data layer	`window.adobeDataLayer`, `adobeDataLayer.filter((e) => e.event)`
Journey state	`document.documentElement.dataset.canvasJourneyState` (Conversation Canvas)

Tool	Use
Admin API	`GET https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}` (preview, live, source times; authenticated)
Local server	`aem up` to tell code from content

2. Runbook

Development and code

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Local dev server doesn't start or pages 404 on `localhost:3000`	AEM CLI not installed or the port is taken. The page isn't previewed: the local server serves previewed content. Node older than 20 (README: Node 20+; CI uses 24)	`node -v`; read the CLI output; open the page on `main--tcs-corp--sradha-tcs.aem.page`. For test pages, check the file is in `drafts/`	`npx -y @adobe/aem-cli up --no-open --forward-browser-logs`; free the port; preview the page in AEM, or serve static pages with `--html-folder drafts`	The page loads on `localhost:3000` with the local code
Pre-commit doesn't rebuild `component-*.json`	`npm install` never ran (Husky is installed by `prepare`), or the commit used `--no-verify`	Check `.husky/pre-commit` runs; `npm run build:json` then `git diff`	`npm install`; `npm run build:json` and commit the three aggregated files	No diff after `npm run build:json`
`npm test` fails in `module-graph` or `block-docs`	An import or model changed without regenerating	Read the failing assertion	`npm run build:modules` and / or `npm run build:docs`; commit the output	`npm test` passes
Build check red on a PR	ESLint (Airbnb, `.js` extensions, line endings), xwalk model rules or Stylelint	Actions log; `npm run lint` locally	`npm run lint:fix`; for `xwalk/max-cells` use field collapsing / element grouping before raising the limit in `.eslintrc.js`	Build green on the latest commit
Block shows raw rows or nothing; `failed to load module for <block>`	JS / CSS 404, syntax error, `decorate()` threw on unexpected content, block not on `main` yet, or a renamed import	Console stack; Network status of `/blocks/{name}/{name}.js`; `.plain.html` of the block vs its `_{name}.json`; reproduce with `aem up`	Fix the code (or revert the merge), or fix the content; merge the branch that adds the block	No load error in the console. Note: `data-block-status="loaded"` is set **even when loading failed**, so it doesn't prove success
Decoration error only on the published page, not in the editor or local preview	The code assumes a structure AEM doesn't deliver: an empty row for each dialog tab, item first cell `<node name>`, `<classes>`, collapsed `*Text` / `*Title` / `*Type` / `*Alt` fields, lower-cased item names	Compare `.plain.html` on aem.page with the local markup; count the item's cells	Read rows by `data-aue-prop` in the editor and skip tab rows on the page (`itemRows`, `utils/cells.js`); never assume "the first row"	The block renders on aem.page; unit tests for the cell layout pass
Block CSS not applied although the CSS file exists	`loadCSS` (`scripts/aem.js`) skips a stylesheet when a `head > link` with the same href exists, **whatever its rel**: a `preload` link added to the head suppresses the stylesheet. Other causes: CSS 404; selector not scoped or overridden; style only in `lazy-styles.css`	Elements panel: is there a `rel="stylesheet"` link for `/blocks/{name}/{name}.css`? Is there a head link with the same href? Network status	Put stylesheet preloads in the body (as `preload` in `scripts.js` does); fix the path or the selector	Computed styles show the block rules; no 404

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Universal Editor reloads the page on every edit, or shows stale rendering	`editor-support.js` couldn't match the update to an element and called `window.location.reload()`. Container sections re-render as a whole. Blocks that read the index or sheets can't show published data	Console during the update; which block	Reload the editor; make the block re-decoratable from fresh markup; report with the block and page path	Edits of simple blocks apply in place
Code change not visible after merge	Code Sync didn't process the push; browser cache; the file is ignored by `.hlxignore` (`_*`, `docs/*`, `test/*`, `*.md`)	Code Sync check on the merge commit; request the file on `main--tcs-corp--sradha-tcs.aem.page`	Re-trigger the sync (Admin API code operation or an empty commit); hard reload	The served file matches the commit
Third-party script blocked (CSP)	`head.html` / `404.html` CSP `'nonce-aem' 'strict-dynamic'`: tags without the nonce are blocked	Console CSP message (blocked URI, directive)	Load the script with `loadScript()` from `scripts/delayed.js`; change the policy only in both files with a security review	No CSP violation; the script runs

Authoring and delivery

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Dialog field, tab or block missing in Universal Editor	Aggregated `component-*.json` not rebuilt; change only on a branch (the editor uses `main`); block not in the section or container filter; editor not reloaded; field hidden by its visibility condition	Search `component-models.json` on `main`; `npm run build:json` and diff; check `component-filters.json`; read the block README's field table (conditions)	Rebuild, merge, reload the editor; add the block to the filter; set the controlling field	The field or block appears after a full reload
A dialog value doesn't persist or changes nothing on the page	Delivery rules: the field name ends in Text / Title / Type / Alt / MimeType and collapses; grouped `prefix_name` field in a key-value block; `classes_*` on a section (not delivered; use `container*`); `multi: true` on `aem-content` (not saved); built-in Button / Columns options (not delivered)	`.plain.html` on aem.page: is the value there? For sections, look for `data-container*` attributes	Rename the field (`buttonVariant`, `titleHidden`, `poster`), use item children or `container*` fields; migrate content; use the Button Block / Columns Container	The value appears in the delivered markup and the block reflects it
Hidden block, or a faded block with a dashed outline	Additional tab Hide block (`hideblock`): hidden outside edit mode, dimmed in edit mode, still in the HTML	Select the block; check Hide block	Clear Hide block and republish (never use it for confidential content)	Visible in Preview and on the site
`#anchor` link doesn't jump	Anchor ID includes its `anchor-` prefix in the id; duplicates get `-2`; sections use Anchor name	Inspect the element `id`	Link `#anchor-…`; use unique ids	`page#anchor-…` scrolls to the block
Fragment, Modal or Tab content empty	Fragment page not published; path not starting with `/` (full URL) or `//`; the fragment references the page itself or its including chain (recursion guard)	Network: `{path}.plain.html` status; the Reference value	Publish the fragment; pick it with the content picker	Content renders on reload

Layout, media and links

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Layout wrong at one width, or horizontal scrolling	Rule outside the mobile-first breakpoints (600 / 900 / 1200px); fixed px sizes; an element wider than the viewport; the editor canvas is narrower than the window (e.g. Teaser image Right applies from 600px)	Reproduce at 1440, 1024, 768, 600 and 390px; in the console compare `document.documentElement.scrollWidth` with `clientWidth`; find the element extending past the right edge	Fix the block CSS (rem, `min-width` queries, scoped selectors); test the editor at its canvas width	No overflow at the five widths; the harness overflow check is 0
Header layout unexpected between 900 and 1199px	The desktop header (logo bar) starts at 1200px (`header.js` `matchMedia('(min-width: 1200px)')`); some handover pages still say 900px	Check the viewport width	Expected behaviour; test at ≥ 1200px for desktop	Logo bar at ≥ 1200px, menu below
Lottie, video or captions 404 on aem.page / aem.live while images work	Non-image DAM files (Lottie JSON, MP4, WebVTT) are served only once previewed / published; images travel with the page	Network: the asset URL status	Publish the assets from AEM Assets (publish with references)	The file returns 200; the animation or video plays
Lottie shows a still frame, or video waits for a click	Reduced motion (by design: paused animations, click-to-play videos); `failed to play the Lottie animation`; Image type not Lotto	OS reduced-motion setting; console; asset status	Accept reduced motion; publish the JSON; set the Image type	Plays with reduced motion off
Image not optimised or large	Typed / external URLs (only same-origin `/media_` images are optimised by `createOptimizedPicture` / `decorators/responsive-image-decorator.js`); large committed SVGs or Lottie files	Network (Img): `/media_…?width=…&format=webply` or external	Use AEM Assets references; compress committed files	`<picture>` with width variants; PageSpeed no longer flags it
Broken image in a fragment or card	Asset unpublished, not a `/media_` path, or a typed URL	Inspect `img src`	Re-select the image with the picker and republish	Image loads
Link opens wrong target or isn't announced	List cards link to the redirect target of `/{locale}/redirects.json` (`utils/redirects.js`, up to 5 hops); a new-tab link without `announceNewTab`; unsafe schemes are rejected (`isSafeUrl`)	Check the sheet; inspect the link's text / `aria-label`	Correct the redirects sheet; use the link decorators	Card URL as expected; new-tab hint present
Old URL returns 404 instead of redirecting; `/` returns 404	The platform applies only a root `/redirects.json`; per-locale sheets aren't applied; `/content/tcs-corp` isn't published	`curl -I` the old URL / the root	Configure the redirect mechanism (e.g. CDN rules); decision owned by the site administrator	301 / 302 with the expected `location`

Index, lists and search

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Page missing from `/query-index.json`	Not published to live; under an excluded path (`/*/references/**`, `/*/error-pages/**`, `/*/data/**`, `/*/nav`, `/*/footer`, block libraries, `site-global-templates`, `/**/fragments/**`, `/**/modals/**`); not reindexed after a `helix-query.yaml` change	Search the path in `/query-index.json?limit=500` (and further `offset` pages); check the excludes	Publish; move the page out of excluded folders; reindex	The row exists with the expected columns

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Index row lacks columns (card, eyebrow, hide-in-nav, featured)	The index was built before the `helix-query.yaml` change (finding F-10)	Inspect the row's keys	Reindex the affected paths	Columns present
List / Filter List empty in Universal Editor	Expected: no query index on AEM author; the hint says the pages show on the published page, or names missing settings	Read the hint	Complete the named settings; review on aem.page	Cards on the preview
List / Filter List incomplete on the site	Pages not published or indexed; Parent page / depth, Search root, tags (any / all), Root path, Page types / Content types don't match; `hide-in-nav = yes`; Maximum items; rows filtered to the locale root (`indexRoot`)	`?debug=list` (configuration, warnings, cards); compare index rows with the settings	Fix settings or metadata; publish the pages	Expected cards and result count
Filter List "No resources found." or pills with derived labels	Filters too narrow; tags missing from `/{locale}/taxonomy.json`; values that aren't tag IDs	Index rows below the Root path; the taxonomy JSON	Fix Root path / types; publish the taxonomy; pick tags with the picker	Cards and taxonomy titles
Canvas search does nothing visible	No Search page and no integration listening to `conversation-canvas:search`	Check the Search page field; listen for the event in the console	Set the Search page or connect an integration	A query opens results
Search term missing or altered in the data layer	By design `cmp:search` carries the redacted term (e-mail addresses and phone numbers replaced, cut at 100 characters; `utils/privacy.js`); an empty search pushes nothing	`adobeDataLayer.filter((e) => e.event === 'cmp:search')`	None (privacy rule)	Redacted term present

Data layer, analytics and Adobe integrations

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
`adobeDataLayer.getState` / `addEventListener` undefined	Expected: the ACDL library isn't loaded (normally by the Tags ACDL extension), so `adobeDataLayer` is a plain array	`Array.isArray(window.adobeDataLayer)`	None until Adobe Tags is configured; read the array	Entries present in the array
Block or component entries missing	Blocks are added once decorated (main content in the lazy phase; header, footer and late blocks after 3 seconds in `loadDelayed`); `data-analytics="off"` on the block; blocks inside `[data-analytics]` push their own state	Wait 3 s; check `data-cmp-id`, `data-analytics` on the element	Turn the block's Analytics option on (List, Cards, Separator); report missing entries for tracked blocks	Component entry with `@type tcs-corp/components/{block}`
Clicks missing	Clicks on `<summary>` push `cmp:show` / `cmp:hide`, not clicks; the element is inside `data-analytics="off"`; a handler stopped the event before the capture listener	`adobeDataLayer.filter((e) => e.event === 'cmp:click')` with Preserve log on	Describe the interaction with `data-cmp-*`; don't stop propagation in capture	One `cmp:click` per click
Clicks duplicated, or duplicate component ids	A block pushes its own `cmp:click` in addition to the central capture; ids reused (`data-cmp-id` from duplicate block ids)	Count events per click; list component ids	Remove block-level click pushes (blocks only push state via `pushComponentEvent`); keep ids unique (Anchor IDs are de-duplicated with `-2`)	Harness click test: exactly one `cmp:click`; no duplicate ids

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
No data in Adobe Analytics / CJA	Adobe Tags (Data Collection) and the Web SDK are **not configured**; `scripts/delayed.js` is empty	Check `delayed.js`; no Tags / `alloy` requests in Network	Blocked: Tags property, datastream and the measurement plan (handover 11.7)	After integration: AEP Debugger shows `sendEvent` calls
Target / AJO activity not shown	Not implemented: no Target, AJO or audience integration exists	—	Not applicable until a personalization design is approved	—
Consent banner missing, data before consent, identity questions	Not implemented: no consent mechanism or identity service; nothing is sent to third parties; the data layer stays in the page	—	Blocked: the consent model and CMP	—

Accessibility and performance

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
axe contrast violations on a dark Section / Columns Container	Block uses colour primitives instead of the theme tokens; a block with a fixed light colour inside a dark background	axe at 1440 and 390px; the "Sections: colours with blocks" library page	Use the tokens (`--color-text-default`, `--link-color`, `--color-border-brand`, ...); reset tokens inside Grey islands	0 axe violations on the combination page
Keyboard trap, focus lost, Escape doesn't close	Disclosure without `<button>` / `aria-expanded` / `aria-controls`; focus not returned on close	Keyboard pass (Tab, Shift+Tab, Enter, Space, Escape, arrows)	Follow AGENTS.md "Accessibility"; return focus to the opener	Keyboard-only pass succeeds
English UI labels on another locale	Placeholder key missing or the sheet unpublished; built-in `404.html` texts aren't placeholders	`/{locale}/placeholders.json`; find `t('key', 'fallback')`	Add the key to the locale sheet and publish	Translated label on reload
LCP, TBT or CLS regression	Stale preload lists; forced layout during decoration; heavy or lazy-image first section; scripts outside `delayed.js`; fallback font faces removed; content arriving without reserved space	PageSpeed / Lighthouse diagnostics; Performance panel; `npm test` (module graph)	`npm run build:modules`; batch reads / writes in one animation frame; keep the LCP image eager in the first block; keep `inter-fallback*` faces	PageSpeed back to target; RUM CLS ≤ 0.1, LCP < 2.5 s

Packages, publishing and reindexing

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
After installing a content package, pages don't appear on aem.page / aem.live	Installing in AEM doesn't preview or publish; sheets (placeholders, metadata) are separate	Admin API status per path; Sites console publication status	Publish sheets first, then fragments, then pages (handover 12.4); or Admin API `preview` then `live` per path; retry paths that time out (one did in 1.0.12)	Every page previewed and published (e.g. 315 / 315 for 1.0.13)
Packaged pages render without CTAs, stats, filters or footer links	Content written as composite multifields (`ctas/item0`) doesn't render (finding F-87)	`.plain.html` : empty item rows	Use item children (`hero-cta`, `teaser-link`, `footer-link`, ...)	Items render on the published page

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Package procedure itself (build, install) unclear	Not recorded in the repository: the packages are built in the delivery workspace	—	Blocked: package build and install procedure from the delivery team	—
New pages missing from lists after publishing	Not yet indexed, or the index predates a `helix-query.yaml` change	Search `/query-index.json`	Reindex: POST `https://admin.hlx.page/index/sradha-tcs/tcs-corp/main/{path}` per path (authenticated); publish sheets before pages so rows carry their metadata	Row present (e.g. 312 / 312 reindexed for 1.0.10); block library pages stay out by design
Live page still old after publish	Published to preview only; a reference (fragment, nav, footer, asset, sheet) not published; caching; inherited Navigation / Footer / 404 values cached per session (`sessionStorage` `tcs-inherited-metadata`)	Compare aem.page / aem.live in a private window; Admin API status	Republish with references; purge the cache; new browser session	Fresh `last-modified`; change visible

3. Further Site-Wide Issues

Each of these has symptoms, root cause, diagnosis, resolution, validation and escalation in the handover's Part 13 — Global Troubleshooting Guide (TCS-CORP-HANDOVER.md):

Issue	Handover section
Pages published at the wrong path, locale missing	13.4 (`paths.json` and `public.json` paths must match)
Wrong header or footer	13.6 (inherited Navigation / Footer, `<meta name="nav">`)
404 in the wrong locale or the built-in English 404	13.7 (`loadErrorPage` fallback chain)
Tags showing IDs	13.14 (`/{locale}/taxonomy.json`)
Bulk metadata not applied	13.17 (site configuration `metadata.source`)
Form submit sends nothing	13.21 (`form-variations:submit`; see also the Form Variations README)
Embeds not loading	13.24
Sitemap host	13.25 (`helix-sitemap.yaml` `origin`)
Template (Page Type) not applied	13.26 (TEMPLATES in `scripts.js`: home, article, blog, press-release)
Page Header author details missing	13.27 (Authors sheet)

Block-specific runbooks: the "Troubleshooting" section of each block README, for example List, Filter List, Header, Fragment, Video, Hero, Teaser, Form Variations, Conversation Canvas.

4. Evidence to Collect

Collect:

- the page path (AEM and published) and the environment (UE / aem.page / aem.live / local);
- the locale, viewport width, browser and OS (including the reduced-motion setting);
- the time in UTC and the related merge, publication or package;
- console messages with stacks;
- Network status codes of the block, sheet, fragment and index requests;
- the block's `.plain.html` excerpt and outer HTML;

- the block's dialog settings (screenshot), the `?debug=list` / `?debug=separator` output, the relevant `/query-index.json` rows;
- `copy(JSON.stringify(adobeDataLayer, null, 2))` with `mailto:` / `tel:` link URLs removed.

Don't collect:

- cookies, tokens or Authorization headers;
- author host names;
- AEM program or environment ids;
- form values;
- visitors' personal data;
- unredacted search terms;
- screenshots showing personal details.

5. Related

- development-standards.md, testing-and-quality.md, universal-editor.md, decorators-and-utilities.md, architecture.md, indexing-and-search.md, data-layer-and-analytics.md, security-privacy-consent.md, personalization.md, administration-and-operations.md, deployment-and-release.md
- Handover (TCS-CORP-HANDOVER.md): Part 13 — Global Troubleshooting Guide, 14.3 Logs, 14.4 Common Errors, 14.5 Escalation Matrix, 12.6 Rollback
- AGENTS.md "AEM delivery rules", "Module preloading"; blocks/README.md; docs/audit/READUIT-2026-10-20.md (operations and content actions)

6. Debugging and validation

Console checks on preview or live (not in Universal Editor edit mode, where clicks select components):

```
window.adobeDataLayer;                                  // all entries
adobeDataLayer.find((e) => e.page);                     // the page entry
adobeDataLayer.filter((e) => e.component)
  .flatMap((e) => Object.entries(e.component))
  .map(([id, c]) => [id, c['@type'], c.parentId]);      // component ids and types
adobeDataLayer.filter((e) => e.event);                  // events only
adobeDataLayer.filter((e) => e.event).at(-1);           // the last event
```

`adobeDataLayer.getState()` and `addEventListener()` exist only once an ACDL library is loaded. Enable "Preserve log" (or Ctrl / Cmd + click) when testing links that navigate. `?debug=list` and `?debug=separator` log the List / Separator configuration to the console (not data layer entries).

Validation	Status	Source
Audit harness: data layer present, page entry, component count, duplicate component ids, one `cmp:click` per test click (local preview, 188 pages × 5 widths)	Passed (re-audit, 20 October 2026)	`docs/audit/tools/harness.mjs`, `docs/audit/READUIT-2026-10-20.md`
Search redaction unit tests (`npm test`)	Passed (5 of 5, run locally on 21 October 2026; CI runs lint only, not `npm test`)	`test/privacy/privacy.test.mjs`
Unit tests of the data layer decorator	Not applicable (none exist)	—
Adobe Tags, Web SDK, Analytics / CJA, duplicate-event checks in the AEP Debugger	Blocked: Tags property, datastream and measurement plan	Audit section 9

Manual test cases (page entry, 3-second component entries, link / download / exit clicks, cards, accordion, tabs, carousel, modal, video, search, form, filter, Analytics off, breakpoint) are listed in the handover 11.8; troubleshooting is handover Part 13 (13.20 Data Layer Events Missing).

7. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

8. Known limitations

- None recorded for this document.

9. Open items

- None.

10. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

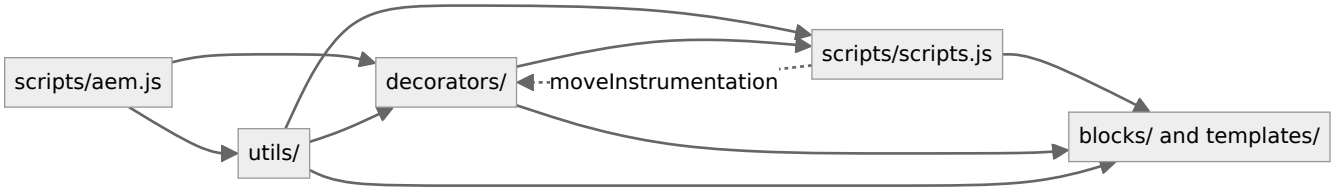
Decorators and Utilities Guide

Attribute	Value
Document ID	DEV-008
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	The shared decorators and utilities, their layering and rules
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	decorators/README.md, utils/README.md
Related documents	—
Related AEM paths	—

1. Layering

Layer	Builds UI	May import	Verified
`scripts/aem.js`	Core decoration (Edge Delivery library)	nothing project-specific	Never modified (AGENTS.md)
`utils/` (26 modules)	No	other utils, `scripts/aem.js` (`toClassName`, `getMetadata`, `loadScript`)	**No util imports a decorator** (grep of every `import` in `utils/` at commit 2d06316; the only mentions of `decorators/` are in comments)
`decorators/` (28 modules)	Yes (elements, behaviour)	utils, other decorators, `scripts/aem.js`, `scripts/scripts.js` (`moveInstrumentation`)	Yes
`scripts/scripts.js`	Page decoration and loading	decorators, utils, `aem.js`, `module-graph.js`	Yes
`blocks/`, `templates/`	Blocks and templates	all of the above	Yes

`scripts/scripts.js` imports seven decorators and some decorators import `moveInstrumentation` from `scripts/scripts.js` (`content-decorator`, `card-list-decorator`, `editor-item-decorator`, `heading-decorator`, `image-decorator`, `stats-decorator`): these are ES module cycles resolved at call time. `responsive-image-decorator.js` is kept free of `scripts.js` imports so `scripts.js` can use it.



2. Decorators Inventory

Status: every decorator is Active (imported by at least one block, template or script). Tests: decorators touch the DOM and have no unit tests; they were validated by the browser runs of the audits (188 pages at 5 widths).

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
a11y	decorators/a11y-decorator.js	Visually hidden content for assistive technology	teaser, testimonial, conversation-canvas; link, social-links, card-list decorators	`hideVisually(el)`, `createHiddenText(text)` return the element	Adds `.visually-hidden`	None	Active
animation	decorators/animation-decorator.js	Lottie animations (the "Lotto" image type): loaded near the viewport, paused off screen and for reduced motion, pause / play button	hero, teaser	`createAnimation(path, options)` returns a container with `animationToggle`; `playAnimation(container, path)`	Loads the vendored Lottie player once (`utils/lottie.js`); observers on the container	None	Active
article-meta	decorators/article-meta-decorator.js	Published date, reading time, listen / download / share actions, author lockup (Figma "L3 Header")	page-header, teaser, filter-list; card-list decorator	`buildPublished`, `buildReadingTime`, `buildPageActions`, `buildAuthorLockup` return elements	Share button handler (Web Share API, else copies the link to the clipboard)	None	Active
author-bio	decorators/author-bio-decorator.js	"About the author" card from the Authors sheet	templates/article (also used by blog through article)	`buildAuthorBio(author, labels)` returns an element	None	None	Active
block-options	decorators/block-options-decorator.js	Additional tab: an `anchor-…` class becomes the block id (unique, `-2`, `-3`)	`scripts.js` (`decorateMain`), `editor-support.js`	`applyBlockOptions(container)`	Sets `id` on blocks	None	Active
button	decorators/button-decorator.js	The single place for button views, types, alignment and states	13 blocks (button-block, hero, teaser, header, footer, modal, video, ...), `scripts.js`, 5 decorators	`decorateButton(a, options)`, `createButton`, `setButtonDisabled`, `setButtonAlignment`, `removeButtonDecoration`	Classes and ARIA on the element; new-tab hint	None	Active

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
card-list	decorators/card-list-decorator.js	Cards of the TCS DS List variations: header, image / panel / report / stat / feature cards, rows, grid or scroll per breakpoint, skeleton, empty state, data layer state	list, cards	`renderCardList(block, config)` and the card renderers, `applyCardListLayout`, `buildCardListSkeleton`, `renderCardListEmpty`, `renderEditorItems`, `cardListId`, `trackCardList`	Replaces block content; scroller; pushes component state	Indirect: `test/cards`, `test/list` (data side), Passed	Active
contact-list	decorators/contact-list-decorator.js	Groups of contacts with mail / phone links (Media Contacts sheet)	blocks/media-contacts	`buildContactGroups(groups, { headingLevel, labels })` returns a list, or null without groups	None	Links carry `data-cmp-interaction` email / phone and the group as `data-cmp-value`	Active
content-buttons	decorators/content-buttons-decorator.js	Standalone authored links in content become buttons, icon buttons or link buttons	`scripts.js` (re-exported for `editor-support.js`)	`decorateButtons(container)`	Changes link classes in the container	None	Active
content	decorators/content-decorator.js	Moves a property cell's content into a new element, keeping editor instrumentation	teaser, quote, form-variations, testimonial, video, footer	`fromCell(cell, tagName, className)` returns the element	Moves `data-aue-*` attributes	None	Active
data-layer	decorators/data-layer-decorator.js	Page, blocks and interactions in the Adobe Client Data Layer (Core Components schema)	`scripts.js`, tabs, modal; skeleton, card-list decorators	`trackPage()`, `trackBlocks(container)`, `trackInteractions()`, `componentIdOf(block)`	Pushes to `window.adobeDataLayer`; document-level capture listeners (click, toggle, close, play, pause, ended, submit, change), registered once; `data-cmp-id`, `data-cmp-tracked` on blocks	Indirect: `test/privacy` (search redaction, contact links), Passed	Active
date	decorators/date-decorator.js	<time> for authored or page-property dates	filter-list; article-meta decorator	`buildTime(value, format, className)` returns `HTMLTimeElement` or null	None	Indirect: `formatDate` in `test/cards`, Passed	Active
editor-item	decorators/editor-item-decorator.js	In Universal Editor, an item the block doesn't render yet (just added) as a selectable placeholder	hero, teaser, testimonial, page-header, anchor-list, accordion, filter-list, footer; stats decorator	`editorItemHint(rows)` (placeholder `itemEditorHint`, loaded only in the editor), `buildEditorItems(rows, hint)`	None on the published page	None	Active
heading	decorators/heading-decorator.js	Heading from a title property and its Title Type	teaser, quote, form-variations, testimonial; card-list decorator	`buildHeading(cell, typeCell, className, fallback)`, `HEADINGS`	Moves instrumentation	None	Active

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
icon-button	decorators/icon-button-decorator.js	Icon-only <button>s on the button decorator	conversation-canvas, gen-carousel, modal, embed, video, filter-list; card-list, animation, article-meta decorators	`createIconButton({ label, icon, ... })` returns a button	None	None	Active
image	decorators/image-decorator.js	The picture of an image property with its alt field	tcs-header, teaser, quote, testimonial, video	`pictureOf(cell, altCell)` returns <picture> or null	Sets `alt` (empty = decorative)	None	Active
link	decorators/link-decorator.js	New-tab links and their announcement	teaser, page-header, footer; button, card-list, social-links, author-bio decorators	`setLinkTarget(a, newWindow)`, `announceNewTab(a, hint)`	`target`, `rel`, hidden hint text	None	Active
masonry	decorators/masonry-decorator.js	Packs grid items of different heights into columns in reading order; native masonry where supported	filter-list	`layoutMasonry(grid)` returns `{ disconnect }`	ResizeObserver and MutationObserver until `disconnect()`	None	Active
media	decorators/media-decorator.js	YouTube, Vimeo, iframe and video players built with DOM APIs (no URL in HTML strings)	embed, video	`createFrame`, `createYoutubeFrame`, `createVimeoFrame`, `createVideoElement` return elements	None until inserted	Indirect: `test/media/vimeo-url.test.mjs` (URLs), Passed	Active
motion	decorators/motion-decorator.js	Content and Background animations of the Additional tab, prepared before a section shows, run by one IntersectionObserver	`scripts.js`, fragment, header, footer; skeleton decorator	`prepareMotion(container)`, `releaseMotion(block)`	Loads `styles/motion.css` when needed; one page-wide observer; classes and inline styles removed at the end (safety timeout 4 s); nothing in edit mode or for reduced motion	Indirect: `test/motion/motion.test.mjs` (`utils/motion.js`), Passed	Active
responsive-image	decorators/responsive-image-decorator.js	Image from a URL; site media optimized as <picture>	conversation-canvas, page-header, filter-list, `scripts.js`; social-links, card-list, author-bio decorators	`buildImage({ src, alt }, breakpoints)` returns an element	None	None	Active
scroller	decorators/scroller-decorator.js	Previous / next, indicator, drag and keyboard for a CSS horizontal scroller, active only while it scrolls	card-list decorator	`createScroller(viewport, options)` returns `{ update, scrollToItem, destroy }`	Listeners and a ResizeObserver until `destroy()`; measurements batched in one animation frame (reads, then writes)	None	Active

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
seo	decorators/seo-decorator.js	Bulk metadata SEO / social columns to the standard head tags, `<html lang>`	`scripts.js` (`loadEager`)	`decorateHead()`	Writes `<title>`, meta and link tags, `lang`	None	Active
skeleton	decorators/skeleton-decorator.js	Block-shaped placeholders while data loads; render directly when data is there within 150 ms	list, filter-list, conversation-canvas; card-list decorator	`renderWhenLoaded(block, { data, render, skeleton })`, `skeletonWhile`, `showSkeleton`, `skeletonElement`, `bone`	Timers (150 ms delay, 250 ms minimum); skeleton removed on success and failure	None	Active
social-links	decorators/social-links-decorator.js	Social Links sheet as an icon list	footer	`buildSocialLinks(links, className, labelOf, newTabHint)` returns a list	None	None	Active
stats	decorators/stats-decorator.js	Stats from Stat items and the legacy Stats multifield, as a description list	teaser, testimonial	`buildStats(cell, prefix, rows, hint)` returns `<dl>`	Moves instrumentation	None	Active
tag	decorators/tag-decorator.js	A list of tag labels	teaser, page-header	`buildTagList(tags, className)` returns `<ul>` or null	None	None	Active
template	decorators/template-decorator.js	Structure a Site Global Template adds before decoration (Page Header, loaded sections)	templates/article, templates/press-release (blog through article)	`ensurePageHeader(main)`, `appendLoadedSection(main, className, build)`	Adds sections to `main`	None	Active

3. Utilities Inventory

Status: every util is Active. "Passed" means the listed test file passed in `npm test` (311 tests) at commit 2d06316.

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
analytics	utils/analytics.js	Adobe Client Data Layer pushes (Core Components schema); never throws	tabs, separator, gen-carousel, modal; data-layer, card-list decorators	`pushToDataLayer(entry)` returns a boolean; `pushComponents(map)`, `pushComponentEvent(event, id, info)`	Creates / pushes to `window.adobeDataLayer`	None	Active
buttons	utils/buttons.js	Button vocabulary and reading button options from authored link markup	button-block, conversation-canvas, tabs, hero, table-list; button, content-buttons, card-list decorators	`BUTTON_VIEWS`, `BUTTON_TYPES`, `BUTTON_ALIGNMENTS`, `pick`, `readButtonOptions(a)`	None (pure)	None	Active

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
card-list-config	utils/card-list-config.js	Presentation settings of List and Cards without the DOM: defaults, validation, card style / colour precedence	list (`list.js`, `list-data.js`), cards (`cards.js`, `cards-data.js`), filter-list; card-list decorator	`normalizePresentation(read)`, `kindOf`, `themeOf`, option lists, `isSafeUrl`, `isExternal`, `toISODate`, `toSafeId`, `hash`, `normalizePath`	None (pure)	`test/cards/cards-data.test.mjs`, `test/list/*`, Passed	Active
cells	utils/cells.js	Item rows of a block, tab cells, cells matched to model fields	11 blocks (list, cards, tabs, hero, teaser, testimonial, page-header, anchor-list, filter-list, footer, table-list)	`itemRows(block, models, isItem)`, `KEY_VALUE_ITEM`, `FIELD_ROW_ITEM`, `withoutTabCells(cells, layout)`, `matchCells(cells, fields, matchers)`	None (pure)	`test/items/item-rows.test.mjs`, `test/list/list-data.test.mjs`, Passed	Active
data	utils/data.js	The locale's data sheets (authors, locations, partners, social links, media contacts), loaded once per page	list, page-header, filter-list, footer, media-contacts; templates/article	`getAuthor(id)`, `getAuthors`, `getSocialLinks`, `getMediaContacts`, `getLocations`, `getPartners`, `inRegion`, `loadDataSheet` return typed rows (empty lists for missing sheets)	Fetches `/{locale}/data/*.json`; module cache	None	Active
dates	utils/dates.js	Formats authored dates without time zone shifts	list-data, cards-data; date decorator	`formatDate(value, format)` returns a string	None (pure)	`test/cards/cards-data.test.mjs`, Passed	Active
environment	utils/environment.js	Reduced motion and Universal Editor checks	13 blocks; motion, animation, scroller decorators	`prefersReducedMotion()`, `scrollBehavior()`, `isAuthor()` (path starts with `/content/`), `isEditing()` (`adobe-ue-edit`), `isInEditor(el)`	None (reads `matchMedia`, location, classes)	None	Active
header-space	utils/header-space.js	Reserves the header's remembered height before it loads (no layout shift); the header overlay mode	header, tcs-header, `scripts.js`	`reserveHeaderSpace()`, `rememberHeaderSpace(header)`, `setHeaderOverlay(mode)`	Reads / writes `localStorage` keys `header-height-v2-{breakpoint}` and `header-overlay-v1`; sets `--header-reserved` and `data-header-overlay` on `<html>`	None	Active
ids	utils/ids.js	Unique element ids per block instance; authored ids kept unless taken	9 blocks, `scripts.js`; block-options, article-meta, card-list decorators	`uniqueId(name)`, `nextInstance(name)`, `createIdFactory(prefix)`, `availableId(id, self)`	Per-page counters (module state)	None	Active
inherited-metadata	utils/inherited-metadata.js	Navigation, Footer and 404 Page inherited from the nearest parent page up to Home	header, footer, `scripts.js` (404)	`getInheritedMetadata(name)` returns a promise of the value; `INHERITED_METADATA`	Fetches parent page heads in parallel; caches values in `sessionStorage` for the session	None	Active

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
links	utils/links.js	Authored links and their Link Type / Open in New Window values	10 blocks (hero, teaser, page-header, footer, list, button-block, conversation-canvas, form-variations, video, tcs-header)	`linkOf(cell)`, `linkPathsOf`, `readLinkItems`, `readLinkRows`, `NEW_WINDOW`	None (pure)	None	Active
lottie	utils/lottie.js	Loads the vendored Lottie player once (`scripts/lib/lottie/lottie_light.min.js`)	animation decorator	`loadLottie()` returns a promise of the player	Adds a script once	None	Active
media	utils/media.js	Video providers and player URLs; Lottie link label	teaser, hero, embed, video; media decorator	`isYoutube`, `isVimeo`, `isAnimation`, `youtubeId`, `youtubeEmbedUrl`, `vimeoEmbedUrl`, `animationLabel`, `videoType`	None (pure)	`test/media/vimeo-url.test.mjs`, Passed	Active
metadata	utils/metadata.js	Page metadata from page properties and the bulk metadata sheet	conversation-canvas, page-header; seo, data-layer decorators; templates	`getPageMetadata()`, `getPageTags()`, `getPageLanguage()`	None (reads `<meta>`)	None	Active
motion	utils/motion.js	Additional tab animation values read from block classes; legacy Motion Type mapping	hero, table-list; motion decorator	`contentMotionOf`, `backgroundMotionOf`, `speedOf`, `triggerOf`, `isSeen`, `staggerIndex`, `isHorizontal`, `isVisibleColor`, `addLegacyMotion`, option lists	None (pure; unknown values ignored)	`test/motion/motion.test.mjs`, Passed	Active
multifield	utils/multifield.js	Reads composite multifields authored before item children	anchor-list, filter-list; stats decorator; links util	`readMultifieldItems(cell)` returns item element runs	None (pure)	None	Active (compatibility path)
placeholders	utils/placeholders.js	UI text from the locale's placeholders sheet, with English fallbacks	24 blocks, `scripts.js`, templates; editor-item, animation decorators	`getPlaceholders()` returns `t(key, fallback, values)`; `fillPlaceholder`, `loadPlaceholders`	Fetches `/{locale}/placeholders.json` once (empty map on failure)	None	Active
privacy	utils/privacy.js	Data minimisation of values that leave the page	data-layer decorator	`redactSearchTerm(value)`: e-mail addresses and phone numbers replaced, trimmed, cut at 100 characters; `contactLinkKind(href)`: `mailto:` / `tel:` links as email / phone (the data layer's clicks without the address)	None (pure)	`test/privacy/privacy.test.mjs`, Passed	Active
properties	utils/properties.js	Reads key-value block properties	18 blocks; image, content, card-list, heading decorators; tags util	`readProperties(block)`, `textOf`, `isOn`, `isOnByDefault`, `optionOf`, `normalizeKey`, `textSource`	None (pure)	Indirect through `test/cards`, `test/list`, Passed	Active

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
query-index	utils/query-index.js	Published pages of the locale from `/query-index.json` (or the language master under review), loaded once	conversation-canvas, list, list-data, filter-list, `scripts.js` (prefetch)	`getPages()`, `getPagesAt(paths)` (pages read from their HTML, e.g. on author), `tagsOf`, `columnOf`, `listOf`, `rowOfDocument`	Fetches the index in pages of 500; module cache	`test/list/query-index-tags.test.mjs`, Passed	Active
redirects	utils/redirects.js	Redirect targets from the locale's redirects sheet (not applied by the platform)	list, `scripts.js` (prefetch)	`loadRedirects()` (a map), `resolveRedirect(map, href)` (up to 5 hops), `normalizeRedirectPath`	Fetches `/{locale}/redirects.json` once	`test/list/list-data.test.mjs`, Passed	Active
site	utils/site.js	The 13 locale sites, AEM and published paths, sheets, parent pages; temporary language master review	6 blocks, `scripts.js`, 8 utils, `tools/paths/generate-paths.mjs`	`LOCALES`, `DEFAULT_LOCALE`, `getLocale`, `localePagePath`, `resolvePagePath`, `localeSheetPath`, `toPublishedPath`, `toAemPath`, `getParentPages`, `pageLink`, `indexRoot`, `sheetRows`, `REVIEW_*`	None (reads location)	`test/site/index-root.test.mjs`, Passed	Active
tags	utils/tags.js	Tag ids to ids and labels (fallback when the taxonomy lacks them)	teaser, list, cards, filter-list; taxonomy util	`parseTag(id)`, `readTagValues(cell)`, `readTags`	None (pure)	None	Active
taxonomy	utils/taxonomy.js	Tag labels from the locale's taxonomy (language sheet when translated)	teaser, conversation-canvas, list, cards, page-header, filter-list, `scripts.js` (prefetch)	`loadTaxonomy()`, `resolveTags(values)`	Fetches `/{locale}/taxonomy.json` once	None	Active
urls	utils/urls.js	AEM content paths to published paths	conversation-canvas, table-list	`toSitePath(path)`	None (pure)	None	Active
viewport	utils/viewport.js	Defers work until content is near the viewport; breakpoints	gen-carousel, embed, video, filter-list; card-list, animation, data-layer decorators; header-space util	`onVisible(el, callback)` (once, disconnects), `watchVisibility(el, callback)`, `currentBreakpoint()`	IntersectionObserver per call	None	Active

Exports without a consumer outside their module

Found by searching every exported name across `blocks/`, `decorators/`, `utils/`, `scripts/`, `templates/`, `test/` and `tools/`. Most are used inside their own module or kept as part of a small API; `getLocations` and `getPartners` (`utils/data.js`) are used nowhere and are **Unused candidates** (their sheets, `data/locations` and `data/partners`, are published per locale; remove them only after checking planned use).

Module	Exports
decorators	`playAnimation`, `setButtonDisabled`, `cardHeadingLevel`, `renderRow`, `renderFeature`, `SKELETON_DELAY_MS`, `SKELETON_MIN_MS`, `showSkeleton`

Module	Exports
utils	`STYLE_OPTIONS`, `loadDataSheet`, `getLocations`, `getPartners`, `nextInstance`, `INHERITED_METADATA`, `MOTION_TRIGGERS`, `loadPlaceholders`, `rowOfDocument`, `normalizeRedirectPath`, `REVIEW_SHEETS`, `REVIEW_PAGES`, `readTags` (`LOCALES` and `toAemPath` are used by `tools/paths/generate-paths.mjs`)

4. Shared Helpers in scripts/

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
aem.js	scripts/aem.js	Edge Delivery core library: `decorateBlock(s)`, `loadSection(s)`, `loadBlock`, `loadCSS`, `loadScript`, `getMetadata`, `toClassName`, `createOptimizedPicture`, `sampleRUM`	everything	Library	RUM (`sampleRUM`)	None	Active; never modified
scripts.js exports	scripts/scripts.js	`moveInstrumentation` / `moveAttributes` (Universal Editor attributes to a new element), `decorateMain`, `decorateSectionsWithOptions`, `preloadBlockModules`, re-exported `decorateButtons`	`moveInstrumentation`: 18 blocks and 6 decorators; `decorateMain`: fragment, `editor-support.js`	DOM in, DOM out	Importing it runs `loadPage()` (eager, lazy, delayed phases); `autolinkModals` adds a document click listener for `/modals/` links	None	Active
scripts.js internals	scripts/scripts.js	`applySectionBlockOptions` (`container*` fields to classes), `buildColumnsSections`, `buildTabsSections`, `buildCarouselSections`, `loadTemplate`, `loadErrorPage`, `prefetchData` (query index, taxonomy, redirects in the lazy phase), `loadMainSections` (`loadSection(section, prepareMotion)`)	Page loading	Not exported	Fonts flag in `sessionStorage` (`fonts-loaded`)	None	Active
loadFragment	blocks/fragment/fragment.js	Fetches and decorates another page's sections; recursion stopped by the chain of including fragments (`from`)	header, footer, modal, tabs	`loadFragment(path, from)` returns a `main` element or null	Fetches `{path}.plain.html`	None	Active
editor-support.js	scripts/editor-support.js	Universal Editor live updates: sanitizes the patch with DOMPurify, re-decorates the changed block, section or main, falls back to a reload	Not referenced by `head.html` or `scripts.js`; loaded when the page is rendered for Universal Editor (loading mechanism requires validation)	`aue:content-*` events	Listeners on `main`; drops ids of the replaced element (re-audit R-C-9); a MutationObserver for rich text	None	Active
editor-support-rte.js	scripts/editor-support-rte.js	Groups rich-text instrumentation (`data-richtext-*`)	`editor-support.js`	`decorateRichtext(container)`	Moves instrumentation attributes	None	Active

Name	Path	Purpose	Consumers	Inputs / Outputs	Side effects	Tests	Status
module-graph.js	scripts/module-graph.js	Generated by `npm run build:modules`: per block, the modules its JavaScript imports beyond `scripts.js`, and the blocks with a stylesheet	`scripts.js` (`preloadBlockModules`)	`blockModules(name)`, `blockStylesheet(name)`	None	`test/modules/module-graph.test.mjs`, Passed	Active (generated; don't edit)
delayed.js	scripts/delayed.js	Delayed phase (3 s after load)	`scripts.js` (`loadDelayed`)	None	None: the file only has a comment; Adobe Tags / Web SDK are not configured (blocked: measurement plan)	None	Active (empty)

Block-local pure modules follow the same pattern and are unit tested: `blocks/list/list-data.js` (`test/list/`), `blocks/cards/cards-data.js` (`test/cards/`), `blocks/separator/separator-config.js` (`test/separator/`), `blocks/conversation-canvas/canvas-journey.js` (`test/conversation-canvas/`); `canvas-controller.js` applies the Canvas journey to the page.

5. Rules for Shared Code

Reuse before creating

Search this inventory, decorators/README.md and utils/README.md before writing a helper. Every button and link button goes through the button decorators; ids through `utils/ids.js`; UI text through `getPlaceholders()`; key-value properties through `readProperties`; item rows through `itemRows`. When two blocks need the same behaviour, move it into a decorator or util instead of copying it (the audit replaced per-block copies with `matchCells`, `availableId`, `getPages`, `buildImage` and `announceNewTab`).

Single responsibility and layering

One purpose per file; add a new module rather than growing an unrelated one. Decorators may import utils; **utils never import decorators** (verified). Utils build no page elements. `scripts/aem.js` is never modified.

No hidden global side effects

Importing a decorator or util does no work: their top level only declares constants, functions and caches (the one environment read at import is the masonry decorator's `CSS.supports` check for native masonry). The global effects that exist are deliberate and documented in the module header:

Effect	Module
Document-level capture listeners, registered once	`data-layer-decorator.js` (`trackInteractions`), `scripts.js` (`autolinkModals`)
`window.adobeDataLayer`	`utils/analytics.js`
One page-wide IntersectionObserver; `styles/motion.css` loaded when needed	`motion-decorator.js`
Browser storage	`utils/header-space.js` (`localStorage`), `utils/inherited-metadata.js` and `scripts.js` fonts flag (`sessionStorage`)
Module caches of fetched data	`placeholders`, `taxonomy`, `redirects`, `query-index`, `data`, `inherited-metadata`, `lottie`

Effect	Module
Per-page counters	`utils/ids.js`, `data-layer-decorator.js`

A new global effect needs the same: a reason, the module header, the README entry.

Idempotency

Decoration can run more than once (fragments, Universal Editor re-renders). The existing guards: `trackInteractions` registers once (`listening`); `trackBlocks` marks blocks `data-cmp-tracked` and `componentIdOf` keeps an existing `data-cmp-id`; `preload` in `scripts.js` preloads each path once; loaders memoize their promise; `availableId` keeps an id that is already the element's own; `editor-support.js` drops the ids of a replaced element so the new one gets them (R-C-9). New helpers must be safe to call twice on the same element.

Error handling

Shared code never breaks rendering: `pushToDataLayer` catches and returns `false`; sheet loaders resolve to empty values when a sheet is missing (placeholders: an empty map, data sheets: empty lists); `renderWhenLoaded` removes the skeleton on failure; a failed Lottie animation hides with its control (R-B-2); code always passes an English fallback to `t()`.

Async work

Loaders return one memoized promise per page; shared data (query index, taxonomy, redirects) is prefetched when the lazy phase starts, never in the eager phase where it would compete with the LCP image; data later than 150 ms shows a skeleton so a section isn't held back (`renderWhenLoaded`).

Multiple instances

Blocks appear several times per page: ids come from `uniqueId` / `createIdFactory` / `availableId`; data-layer ids are `{block}-{n}` unless the block has an id; scrollers, masonry grids and animations are per instance; the same fragment may be loaded twice side by side, but not inside itself (`loadFragment` chain, R-C-5).

Listener and observer cleanup

Helpers that observe return a way to stop: `createScroller` returns `destroy()`, `layoutMasonry` returns `disconnect()`, `onVisible` disconnects after its first call, the motion decorator unobserves each block and aborts its `animationend` listeners (`AbortController`, 4 s safety timeout). Known gap: editor-only listener leaks (audit deferral D-C-19).

Consent awareness

No consent mechanism is implemented. The data layer is filled without a consent check but stays in the page: no Adobe Tags, Web SDK, Analytics or Target library consumes it. Embeds use privacy-enhanced players (youtube-nocookie; Vimeo `dnt=1` in Embed and in Video with privacy mode). Any helper that would send data to a third party is blocked until a consent model exists (blocked: consent categories and CMP).

Data minimisation

`utils/privacy.js` `redactSearchTerm` replaces e-mail addresses and phone numbers and cuts search terms at 100 characters before `cmp:search` (unit tested); `cmp:submit` carries the form id and field names, never values; the data layer code writes no storage. Open items: `mailto:` / `tel:` link URLs are still pushed in full as `xdm:linkURL` (handover finding F-05, to be masked in Tags) and the Conversation Canvas Search page URL carries the raw query (re-audit R-D-12, blocked until Tags is set up).

Performance

- **No forced layout during decoration**: the scroller decorator measures all scrollers in one animation frame, reads before writes; the Canvas controller writes `<html>` custom properties only when they change and never measures while scrolling.
- **Module preloading**: there is no build step, so `head.html` preloads every module `scripts.js` imports and `preloadBlockModules` preloads each block's modules and stylesheet from `scripts/module-graph.js`. Run `npm run build:modules` after adding, removing or changing an import; `test/modules` fails when the lists are stale.
- **Load only when needed**: the Lottie player near the viewport, `styles/motion.css` only with animations, the placeholder hint only in the editor, shared data once per page.

Backward compatibility

Keep exported names and signatures stable (blocks import them directly); keep the readers of content authored before (`readMultifieldItems`, `addLegacyMotion`, the legacy Stats multifield in `buildStats`, the Video poster group); unknown option values are ignored rather than turned into classes (`utils/motion.js`).

Tests and documentation

Pure logic goes into utils or block-local data modules with a Node test in `test/**/*.test.mjs` (`npm test`); DOM helpers are validated on the block library pages. Update the module header, the table in `decorators/README.md` or `utils/README.md` and, when a block's dependencies change, `npm run build:docs` (the generated files section of the block READMEs). The "Used by" columns of the two READMEs have drifted from the imports (e.g. the heading, content, image, link and data-layer decorators, and `utils/site.js`); the consumers in this inventory come from the imports at commit 2d06316.

6. Related

- blocks.md: reuse strategy and anti-patterns for blocks
- universal-editor.md: models whose values these helpers read
- decorators/README.md (with the data layer contract) and utils/README.md
- templates/README.md: templates using the template, author-bio and contact-list decorators
- docs/animation/README.md: motion and skeleton decorators
- AGENTS.md: shared code rules, data layer, module preloading
- Handover Part 2.3 — Component Architecture and Part 11 — Analytics & Data Layer: TCS-CORP-HANDOVER.md
- Other knowledge base files: architecture.md, development-standards.md, performance.md, data-layer-and-analytics.md, security-privacy-consent.md, testing-and-quality.md

7. Module inventory (from the READMEs)

7.1 Decorators

Module	Purpose	Used by
`button-decorator.js`	`decorateButton` (option `newTabHint`: a link opening a new tab says so), `createButton`, `setButtonDisabled`, `setButtonAlignment`, `removeButtonDecoration`	most blocks (button-block: the Button), header, `scripts.js` (404 Go back)
`icon-button-decorator.js`	`createIconButton`	gen-carousel, article meta decorator, modal, embed, video, filter-list, conversation-canvas

Module	Purpose	Used by
content-buttons-decorator.js	decorateButtons (authored links in content become buttons)	scripts.js, editor-support.js
link-decorator.js	setLinkTarget (open in a new window), announceNewTab (tells screen reader users a link opens in a new tab: hidden text, or added to its aria-label)	button decorator, teaser, list, page-header
heading-decorator.js	buildHeading (Title + Title Type), HEADINGS	teaser, image-collage, quote, testimonial, filter-list, form-variations
content-decorator.js	fromCell (property content into an element, keeping editor instrumentation)	teaser, image-collage, quote, testimonial, page-header, form-variations
editor-item-decorator.js	buildEditorItems, editorItemHint (in Universal Editor, an item the block doesn't render yet, e.g. a CTA just added, as a dashed placeholder the author can select; styles in styles/lazy-styles.css; the hint loaded only in the editor)	hero, page-header, teaser, image-collage, testimonial, anchor-list, filter-list, footer, stats decorator
image-decorator.js	pictureOf (image with its alt text field)	teaser, image-collage, quote, testimonial, page-header
scroller-decorator.js	createScroller (previous / next, scroll indicator, mouse drag, keyboard, loop and initial item for a CSS horizontal scroller; active only while it scrolls)	card list decorator
card-list-decorator.js	buildCardListHeader, renderCardList, renderCard, renderRow, renderFeature, applyCardListLayout, buildCardListSkeleton (the List's skeleton in its own layout), renderCardListEmpty, cardListId, cardHeadingLevel, trackCardList (the TCS DS Figma List cards: image, panel, report, statistic, feature, rows; grid / scroll per breakpoint; data layer state; styles in styles/card-list.css)	list, cards
motion-decorator.js	prepareMotion (the Content animation and Background animation of the blocks of a section, before it is shown, or of a fragment, footer page or header Canvas once in the page: first frame paused, run by one IntersectionObserver (Animation Trigger On Scroll) or at once (On Page Load), at the Motion Speed of the block, every class and inline style removed at the end; nothing in Universal Editor or for reduced motion; styles in styles/motion.css, loaded only when needed), releaseMotion (a loaded block's content animation after its skeleton)	scripts.js, fragment, header, footer, skeleton decorator
skeleton-decorator.js	renderWhenLoaded (a block from data that may take time: directly within 150ms, else after a skeleton so its section is shown; removed on success and failure), skeletonWhile (a skeleton in a part that fills in by itself), showSkeleton, skeletonElement, bone (styles in styles/styles.css, .skeleton)	list, filter-list, conversation-canvas, card list decorator
masonry-decorator.js	layoutMasonry (a grid's items of different heights packed in columns, in reading order; native masonry where supported)	filter-list
responsive-image-decorator.js	buildImage (image from a URL; site media and EDAM delivery images optimized), buildDeliveryPicture (responsive picture of a Dynamic Media with OpenAPI delivery image), decorateDeliveryImages (standalone delivery image links become pictures before the blocks are decorated, KB-D-064)	scripts.js (decorateMain, section backgrounds), page-header, filter-list, conversation-canvas, cards

Module	Purpose	Used by
`date-decorator.js`	`buildTime` (<time> for a date)	article meta decorator, filter-list
`tag-decorator.js`	`buildTagList`	teaser, page-header
`media-decorator.js`	`createFrame`, `createYoutubeFrame`, `createVimeoFrame`, `createVideoElement`	embed, video
`animation-decorator.js`	`createAnimation`, `playAnimation` (Lottie animations of the Lotto image type, loaded when near the viewport; a pause / play button as `element.animationToggle`, WCAG 2.2.2; paused for reduced motion)	hero, teaser
`article-meta-decorator.js`	`buildPublished`, `buildReadingTime`, `buildPageActions` (listen / download / share), `buildAuthorLockup`	page-header, teaser, filter-list
`stats-decorator.js`	`buildStats` (Stat items, and the Stats composite multifield authored before them, as a description list)	testimonial, teaser
`seo-decorator.js`	`decorateHead` (bulk metadata SEO / social columns to the standard head tags, <html lang>)	`scripts.js`
`social-links-decorator.js`	`buildSocialLinks` (Social Links data sheet as an icon list)	footer
`template-decorator.js`	`ensurePageHeader`, `appendLoadedSection` (structure templates add before decoration)	article and blog templates (both), press-release template (`ensurePageHeader`)
`author-bio-decorator.js`	`buildAuthorBio` (About the author from the Authors sheet)	article / blog template
`contact-list-decorator.js`	`buildContactGroups` (groups of contacts, each with its heading and its mail / phone links one below the other; the links carry `data-cmp-interaction` email / phone)	media-contacts
`a11y-decorator.js`	`hideVisually`, `createHiddenText` (with `.visually-hidden` in `styles/styles.css`)	teaser, image-collage, testimonial, conversation-canvas, link decorator
`block-options-decorator.js`	`applyBlockOptions` (the Additional tab of every block dialog: an `anchor-…` class becomes the block's id, unique on the page; Hide block, padding and margin are classes styled in `styles/styles.css`)	`scripts.js` (`decorateMain`), `editor-support.js`

Module	Purpose	Used by
`data-layer-decorator.js`	`trackPage`, `trackBlocks`, `trackInteractions`, `componentIdOf` (Adobe Client Data Layer for every block, Core Components schema, read by Adobe Tags / Web SDK; see "Data layer" below)	`scripts.js`, list, modal

7.2 Utilities

Module	Purpose	Used by
`site.js`	`LOCALES`, `DEFAULT_LOCALE`, `getLocale`, `indexRoot`, `REVIEW_ROOTS` and `REVIEW_SHEETS` (the three review roots, KB-D-068: /language-master/en for testing, /language-masters/en the language master, /design-block-master/en the block library and curated design pages; each lists its own pages and reads its own placeholders, redirects, taxonomy and data sheets and its own nav and footer, `REVIEW_PAGES`), `getParentPages`, `localePagePath`, `resolvePagePath`, `pageLink` (on AEM author, a published page path as its AEM page link), `localeSheetPath`, `toPublishedPath`, `toAemPath`, `sheetRows` (the 13 locale sites, AEM ↔ published paths; references, configured nav / footer, error pages and sheets of a locale)	header, footer, `scripts.js`, placeholders, taxonomy, data, urls, inherited metadata, conversation-canvas
`inherited-metadata.js`	`getInheritedMetadata`, `INHERITED_METADATA` (nav, footer and error-page of the page, else of the nearest parent page up to Home; parent heads loaded in parallel and cached per session)	header, footer, `scripts.js` (404)
`metadata.js`	`getPageMetadata`, `getPageTags` (article:tag and the AEM page Tags published as `cq-tags`), `getPageLanguage` (page properties and bulk metadata columns: page-type, author-id, publish-date, reading-time, ...)	page-header, conversation-canvas, SEO decorator, templates
`placeholders.js`	`getPlaceholders`, `loadPlaceholders`, `fillPlaceholder` (UI text from the locale's placeholders, with English fallbacks)	all blocks with UI text, header, footer, `scripts.js` (404 Go back)
`taxonomy.js`	`loadTaxonomy`, `resolveTags` (tag labels from the locale's taxonomy)	teaser, page-header, filter-list, conversation-canvas
`data.js`	`loadDataSheet`, `getAuthor`, `getAuthors`, `getLocations`, `getPartners`, `getSocialLinks`, `getMediaContacts`, `inRegion` (the locale's data sheets)	page-header, filter-list, footer, media-contacts, article template
`privacy.js`	`redactSearchTerm` (data minimisation: e-mail addresses and phone numbers replaced, 100 characters), `contactLinkKind` (`mailto:` / `tel:` links as email / phone, so clicks are recorded without the address); unit tested	data layer decorator
`properties.js`	`readProperties`, `textOf`, `isOn`, `isOnByDefault` (a toggle that defaults to on), `optionOf`, `normalizeKey`, `textSource` (key-value block properties)	teaser, image-collage, quote, testimonial, page-header, anchor-list, form-variations, conversation-canvas, tab, tabs, table-list, decorators
`multifield.js`	`readMultifieldItems` (composite multifield items, authored before item children)	links util, stats decorator, anchor-list, filter-list
`links.js`	`NEW_WINDOW`, `linkOf`, `linkPathsOf` (the paths of a page property, one or many), `readLinkItems` (CTA lists with Link Type and the items' other select values), `readLinkRows` (the same from link item rows, e.g. Hero CTA items)	hero, teaser, image-collage, page-header, footer, list

Module	Purpose	Used by
buttons.js	BUTTON_VIEWS, BUTTON_TYPES, BUTTON_ALIGNMENTS, pick, readButtonOptions	button decorators, button-block, hero, list, conversation-canvas, tabs, table-list
dates.js	formatDate (DD-MM-YYYY, MM-DD-YYYY, MMM d, yyyy, DD/MM/YY, DD MMM YY, 4th September 2026; no time zone shift)	date decorator, list, cards
query-index.js	getPages (the current locale's pages from /query-index.json, or a review root's own pages from its own index, {root}/query-index.json, indexUrlOf; requests of 1,000 rows, in parallel after the first, remainingOffsets with their metadata columns: tags, page and content type, dates, images, author, featured, priority; loaded once per page and cached), tagsOf (a row's tags: the tags and cqTags columns), columnOf (a column by its metadata name, e.g. featured-image), listOf (a multi-value column), getPagesAt (pages the index doesn't have, e.g. on AEM author, read from their own HTML, once per page), rowOfDocument (an index row from a page's HTML)	filter-list, list, conversation-canvas
redirects.js	loadRedirects, resolveRedirect, normalizeRedirectPath (redirect targets from the locale's redirects sheet, loaded once)	list
analytics.js	pushToDataLayer, pushComponents, pushComponentEvent (Adobe Client Data Layer, Core Components schema; never throws)	data layer decorator, list, separator, tabs, gen-carousel, modal
cells.js	itemRows, KEY_VALUE_ITEM, FIELD_ROW_ITEM (the item rows of a block: by model in the editor, by their cells on the page), withoutTabCells (drops the empty cell AEM renders for each dialog tab of an item model), matchCells (block item cells to model fields when AEM leaves out unset fields)	list, table-list, cards, tabs, hero, page-header, teaser, image-collage, testimonial, anchor-list, filter-list, footer
card-list-config.js	normalizePresentation (the card list settings of List and Cards, with defaults and validation), PRESENTATION_PROPERTIES, kindOf and themeOf (the card's Card style and Card colour: item, list, page metadata, automatic), RESPONSIVE_DEFAULTS, CARD_STYLES / CARD_KINDS / CARD_THEMES and the other options, contentTypeLabel (a content type by its taxonomy label, e.g. Case Study), isContentTypeTag / visibleTags (a card's tag labels, without its content type tags for Show content type off), isSafeUrl, isExternal, toISODate, toSafeId, hash, normalizePath	list, cards, card list decorator
tags.js	readTagValues, parseTag, readTags (tag IDs; ID-derived labels as the taxonomy fallback)	teaser, filter-list, taxonomy util
urls.js	toSitePath (AEM content paths to published paths, e.g. /content/tcs-corp/de/de-de/x.html → /de-de/x)	table-list, conversation-canvas
ids.js	nextInstance, uniqueId, createIdFactory (unique ids per block instance), availableId (an authored id kept, suffixed -2, -3, ... only when taken)	tabs, gen-carousel, page-header, filter-list, form-variations, conversation-canvas, header, block options decorator
environment.js	prefersReducedMotion, scrollBehavior, isEditing, isInEditor, isAuthor (the page is rendered by AEM author, which has no query index)	gen-carousel, tabs, hero, table-list, anchor-list, video, form-variations, list, filter-list, motion decorator
header-space.js	reserveHeaderSpace, rememberHeaderSpace (the loaded header's height, the logo bar, remembered per breakpoint (header-height-v2-*) and reserved on the next pages before the header loads: no layout shift); setHeaderOverlay (the TCS Header's Header over the first block as <html data-header-overlay>, always / never, remembered as header-overlay-v1 and applied by reserveHeaderSpace on the next pages)	header, tcs-header, scripts.js

Module	Purpose	Used by
`viewport.js`	`onVisible` (once), `watchVisibility` (on / off screen), `currentBreakpoint` (mobile / tablet / desktop)	gen-carousel, hero, table-list, embed, video, data layer decorator, card list decorator, filter-list
`motion.js`	`CONTENT_MOTIONS`, `BACKGROUND_MOTIONS`, `contentMotionOf`, `backgroundMotionOf` (the Content animation / Background animation of the Additional tab, read from the block classes `motion-*` / `bg-motion-*`; unknown values give none), `staggerIndex` (capped at `STAGGER_MAX`), `isHorizontal`, `isVisibleColor`, `MOTION_SPEEDS`, `speedOf` (Motion Speed), `triggerOf` (Animation Trigger), `isSeen` (enough of a block in view to start), `addLegacyMotion` (Motion Type values authored before, mapped to the Additional tab animations)	motion decorator, hero, table-list
`media.js`	`isYoutube`, `isVimeo`, `isAnimation`, `youtubeEmbedUrl`, `vimeoEmbedUrl`, `videoType`, `deliveryAsset` (reads a Dynamic Media with OpenAPI delivery URL of a separate AEM Assets environment), `deliveryImageUrl` (its URL at a width, WebP preferred)	media decorator, embed, video, hero, teaser, responsive image decorator
`lottie.js`	`loadLottie` (the vendored Lottie player, loaded once per page)	animation decorator

8. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

9. Known limitations

- None recorded for this document.

10. Open items

- None.

11. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Local Development Setup

Attribute	Value
Document ID	DEV-001
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	Setting up and running the site locally
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	AGENTS.md, package.json
Related documents	—
Related AEM paths	—

1. Install and run

1. `git clone` the repository and `npm install`.
2. Start the dev server: `npx -y @adobe/aem-cli up --no-open --forward-browser-logs` (or `aem up` after `npm install -g @adobe/aem-cli`); it serves http://localhost:3000 with the local code and the previewed content.
3. Content without an authored page: static HTML in a `drafts/` folder and `--html-folder drafts`.
4. Inspect delivered markup: `curl http://localhost:3000/<path>`, `.md` and `.plain.html`.

2. Generated files

Command	Regenerates	When
npm run build:json	component-models.json, component-definition.json, component-filters.json	after any _<block>.json or models/ change
npm run build:modules	head.html modulepreload links, scripts/module-graph.js	after adding, removing or changing an import
npm run build:docs	generated sections of blocks/README.md and block READMEs	after a model change or a new block
npm run build:paths	paths.json	after a LOCALES or REVIEW_ROOTS change

3. Local development

Task	Command	Notes
Install dependencies	`npm install`	Also installs the Husky pre-commit hook (`prepare: husky`)

Task	Command	Notes
Start the dev server	`npx -y @adobe/aem-cli up --no-open --forward-browser-logs` (or `aem up` with the CLI installed globally)	`http://localhost:3000`: local code, previewed content
Serve static test content	Add `--html-folder drafts`	Files in `drafts/` (`.html` or `.plain.html`); 14 files are tracked
Lint	`npm run lint` (`lint:js`: ESLint on `.json`, `.js`, `.mjs`; `lint:css`: Stylelint on `blocks/**/*.css`, `styles/*.css`)	`npm run lint:fix` fixes what it can
Unit tests	`npm test` (`node --test 'test/**/*.test.mjs'`)	Pure modules, model layout, docs, module graph
Rebuild the Universal Editor files	`npm run build:json`	`component-models.json`, `component-definition.json`, `component-filters.json` from the partials
Regenerate the path mapping	`npm run build:paths`	After changing LOCALES in utils/site.js; commit both
Regenerate the module graph	`npm run build:modules`	tools/modules/generate-module-graph.mjs, used for module preloading
Regenerate block READMEs	`npm run build:docs`	Generated sections of the block READMEs (blocks/README.md)

Inspect delivered content with `curl http://localhost:3000/{path}`, `.md` and `.plain.html` (AGENTS.md). Node.js: README.md states 20 or newer; the workflows install Node 24. The local preview renderer used by the audits (`.home-work/render-lib.mjs`, named in AGENTS.md) is not tracked in Git.

4. Branches and feature previews

- `main` is production. Changes go through short-lived branches merged with pull requests (130 merge commits on `main` at 2d06316, latest #129).
- Branch prefixes in use: `feature/`, `feat/`, `fix/`, `perf/`, `docs/` (for example `fix/block-reaudit`, `perf/module-preload`, `docs/block-readmes`). The handover's `feature/tcs-<topic>` convention (12.2.2) is not what recent history uses; no naming rule is enforced.
- AEM Code Sync serves every pushed branch at `https://{branch}--tcs-corp--sradha-tcs.aem.page/` (slashes become hyphens; keep names short and lower case: branch, repository and owner form the host name).

5. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

6. Known limitations

- None recorded for this document.

7. Open items

- None.

8. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Performance Development Guide

Attribute	Value
Document ID	DEV-011
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	Performance rules for code changes
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/performance.md
Related documents	—
Related AEM paths	—

1. JavaScript loading

Phase	What runs (`scripts/scripts.js`)
Eager (`loadEager`)	`decorateHead` (SEO tags, `<html lang>`), `trackInteractions` (data-layer listeners and page entry), template / theme / page-type classes, the 404 content or the Page Type template (`templates/{name}`), `decorateMain`, `reserveHeaderSpace`, `body.appear`, `preloadBlockModules(first section)`, then the first section with `prepareMotion` and `waitForFirstImage` (sets the first image `loading="eager"` and waits for it); `fonts.css` only on wide viewports or when cached
Lazy (`loadLazy`)	Modal auto-links, `preloadBlockModules` for the rest of `main` and the chrome blocks (`header`, `footer`, `fragment`, `tcs-header`, `conversation-canvas`), `prefetchData`, `loadHeader`, `loadFooter` (hidden), `loadMainSections` (each section in order, animations prepared before it shows), then footer shown, `trackBlocks(main)`, Go back on 404 pages, hash scroll, `lazy-styles.css`, `loadFonts`
Delayed (`loadDelayed`)	After 3 s: `trackBlocks(document)` (blocks added later, such as header, footer, modals) and `import('./delayed.js')`, which is empty ("add delayed functionality here")

- **Per-block code splitting**: `loadBlock` (`scripts/aem.js`) loads a block's CSS and JS in parallel only when the block is on the page; blocks of a section load one after another, and sections load in order.
- **Module preloading**: there is no build step, so each import level is a network round trip. `head.html` has 20 `modulepreload` links (the modules `scripts.js` imports); `preloadBlockModules` adds `modulepreload` for each block's modules and a `preload as="style"` for its stylesheet from the generated `scripts/module-graph.js`. Stylesheet preloads go in the **body**, because `loadCSS` skips any stylesheet whose href is already linked in the head. Run `npm run build:modules` after any import change; `npm test` (`test/modules/module-graph.test.mjs`) fails when the graph or the head links are stale (AGENTS.md "Setup Commands").

- **Dynamic imports**: on-demand modules are imported only when needed: `utils/inherited-metadata.js` (404 pages), `utils/placeholders.js` (Go back), `utils/query-index.js`, `utils/taxonomy.js`, `utils/redirects.js` (`prefetchData`), `blocks/modal/modal.js` (modal links and the video pop-up), template modules; the Lottie player (`scripts/lib/lottie`, vendored) loads when an animation is near the viewport.

2. CSS loading

Stylesheet	When it loads
`styles/styles.css`	Render-blocking `<link>` in `head.html` (tokens, global styles, sections, buttons, theme contract, fallback font faces, skeletons, Additional tab classes)
`blocks/<name>/<name>.css`	With the block (`loadBlock`), preloaded by `preloadBlockModules` (`blockStylesheet`)
`styles/card-list.css`	`@import` from `blocks/list/list.css` and `blocks/cards/cards.css` (a chained request after the block stylesheet; not in the preload list)
`templates/<name>/<name>.css`	With the Page Type template in the eager phase (`templates/blog/blog.css` imports `article.css`)
`styles/motion.css`	Only on pages with an animated block (`prepareMotion`)
`styles/fonts.css`	Eager on viewports ≥ 900px or when the session flag is set, else end of the lazy phase
`styles/lazy-styles.css`	End of the lazy phase (currently only the Universal Editor item placeholder styles)

3. Images

- Authored images are optimised `/media_…` renditions served as `<picture>` with WebP and fallback sources (`createOptimizedPicture`: 2000px from 600px, 750px below; `format=webply&optimize=medium`).
- The first image of the first section is the LCP candidate: `waitForFirstImage` sets `loading="eager"` on it and waits for it before the lazy phase. The Hero sets `loading="eager"` and `fetchPriority="high"` on its image when it opens the page; the Page Header does the same for its featured image; List / Cards "Image loading" Eager gives only the first image a high fetch priority (`decorators/card-list-decorator.js`).
- `buildImage` (`decorators/responsive-image-decorator.js`) optimises same-origin media URLs and lazy-loads other URLs; section backgrounds use 2000px / 750px. Open: a featured image that isn't site media stays lazy (handover F-14).
- Icons are lazy `<img>` (16 × 16 attributes). Images committed to git must be optimised (AGENTS.md "Performance"); the only font file is 48,256 bytes.

4. Fonts

- One self-hosted variable font (`fonts/inter-variable.woff2`, `font-display: swap`, Latin `unicode-range`); no external font service.
- Eager only when `window.innerWidth >= 900` or `sessionStorage fonts-loaded` is set (set after the first load, except on localhost); otherwise loaded at the end of the lazy phase, so the font doesn't compete with the LCP image on small screens.
- Text first renders in a fallback face scaled to Inter: `inter-fallback` (Arial, Liberation Sans), `inter-fallback-roboto` (Android), `inter-fallback-dejavu` (Linux), in `styles/styles.css` (values in design-system.md). Keep the faces when the font changes, or text reflows when Inter arrives (2.36: CLS 0.33 → 0.003).

5. Layout shift prevention

Source of shift	Prevention
Header (lazy phase)	`header:not(:has(.nav-wrapper))` keeps `--header-reserved` (the logo bar `--nav-height`: 4rem, 7.75rem from 1200px) or the height remembered for the breakpoint (`utils/header-space.js`, `localStorage` key `header-height-v2-<breakpoint>`); from 1200px the header lies over a light first hero out of the flow; the header Canvas lies on the hero and takes no header space
Header / footer content	`header .header` and `footer .footer` stay `visibility: hidden` until `data-block-status="loaded"`
Footer	`display: none` until all main sections have loaded
Fonts	Metric-matched fallback faces
Data blocks (List, Filter List)	Skeletons shaped like the block (`renderWhenLoaded`) when data takes longer than 150 ms; shown at least 250 ms so they don't flash
Sticky Anchor List	Doesn't pin while its section is hidden (F-90)
Animations	Only `opacity`, `transform`, `clip-path`; first frames applied only on screen; Background animation on the first hero delays the LCP image paint (F-94: use it below the fold)

6. DOM size, listeners and observers

- DOM size: no measurement or limit is recorded. Skeletons add about 20–40 small spans per block, removed with the skeleton; animations add no wrapper elements (animation report, section I).
- Data-layer listeners are delegated: one capturing listener per event type on `document` (`click`, `toggle`, `close`, `play` / `pause` / `ended`, `submit`, `change`), registered once (`trackInteractions`).
- Observers: one IntersectionObserver for all animations (`decorators/motion-decorator.js`); `utils/viewport.js` `onVisible` / `watchVisibility` for deferred work (Embed, Video, Gen Carousel, Lottie in `decorators/animation-decorator.js`); own IntersectionObservers in the Separator (motion) and the Modal (inside the editor); ResizeObserver in scrollers, masonry, Table, Anchor List, Header and the Canvas controller (resizes measured once they settle).
- Scroll listeners are passive and scheduled (Anchor List, Canvas controller); the Canvas never measures while scrolling.
- Layout reads are batched: scrollers measure in one shared animation frame, reads before writes; the Canvas controller writes `<html>` custom properties only when they change (AGENTS.md module preloading paragraph).

7. Query and data requests

Data	Loading
Query index (`/query-index.json`)	`utils/query-index.js` `getPages()`: fetched once per page view in pages of 500 rows (`offset` / `limit`), kept as an in-memory promise, filtered to the locale (or the review root)
Taxonomy, redirects, placeholders, data sheets	Fetched once per page view, cached in memory
Inherited metadata (Navigation, Footer, 404 Page)	Parent pages' heads loaded in parallel, cached per browser session (`sessionStorage`)
Prefetch	`prefetchData` starts the query index and taxonomy (and, with a List, the redirects) when the lazy phase begins on pages with a List, Filter List or Conversation Canvas; not in the eager phase, so it doesn't compete with the LCP image

Data	Loading
Late data	`renderWhenLoaded`: data within 150 ms renders directly; else a skeleton, so the section is shown and the sections below aren't held back (a 1.5 s index showed the sections below a List at 0.5 s instead of after the index)

8. Budgets

No verified performance budget (bytes, requests, long tasks) is defined in the repository, CI or handover. The working targets are the NFRs above (PSI 100, LCP < 2.5 s, CLS ≤ 0.1, INP ≤ 200 ms at p75). Proposed budgets are a recommendation, not a decision.

9. Required performance validation

Validate before merging (AGENTS.md "Publishing Process"; handover 12.4 release checklist and 13.16):

Change	Validate
A block's JS or CSS	PageSpeed Insights 100 on the feature preview of a page using the block (mobile and desktop); `npm run build:modules` if imports changed, then `npm test`; no forced layout while decorating (reads before writes in one frame); no new eager work in the first section; the block README's "Performance Considerations" updated
A dialog option or combination	The block library page of the option, first section and below the fold; options that load media or data (Image loading Eager / Lazy, Background animation, Lotto, autoplay video, List sources) checked for LCP and CLS; the dialog options report "Responsive and performance" section is the model (CSS-only options: no new requests, listeners or blocking resources)
Block library pages	The harness run (0 errors, 0 failed requests, 0 overflow) and Lighthouse on a representative library page
Curated design pages and key templates (Home, Insights, What we do, article)	Lighthouse mobile / desktop and a throttled run against main on aem.live; CLS from the font swap on Android-like metrics (long intro paragraphs)
A new third-party script	Loaded from `scripts/delayed.js` or on interaction / visibility, never in `head.html`; CSP compatible
Fonts	Keep the fallback faces; measure CLS on mobile

Author guidance that affects performance: keep the LCP image in the first block (Hero or Page Header); set List / Cards "Image loading" to Eager only for lists in the first section; avoid page lists from the query index and Background animation as the first block (handover 13.16, F-94).

10. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

11. Known limitations

- None recorded for this document.

12. Open items

- None.

13. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Pull Request Review Checklist

Attribute	Value
Document ID	DEV-015
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	What every pull request must contain and what reviewers check
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	AGENTS.md, docs/handover/knowledge-base/development-standards.md
Related documents	—
Related AEM paths	—

1. Checklist

- ☐ Branch from `main`; PR into `main` with a feature preview URL (`https://<branch>--tcs-corp--sradha-tcs.aem.page/<path>`).
- ☐ `npm run lint` and `npm test` run; real results in the PR description.
- ☐ Generated files current: `build:json`, `build:modules`, `build:docs`, `build:paths` as applicable.
- ☐ Models follow the AEM delivery rules (tabs, Additional tab, item children, naming).
- ☐ Accessibility (WCAG 2.1 AA), performance (three-phase loading, no new runtime dependency), privacy (no personal data in the data layer).
- ☐ UI text from placeholders with an English fallback.
- ☐ Block README, decorators/utils READMEs, AGENTS.md, knowledge base and decision log updated.
- ☐ **docs/handover on migrated-page-content-edam updated** for the change (role documents, test cases, screenshots) or the PR says which follow-up does it.
- ☐ `gh pr checks` green; a human merges.

2. Code Review

- Template: .github/pull_request_template.md asks for the GitHub issue (`Fix #<id>`) and Before / After test URLs. It still contains the boilerplate `{repo}` / `{owner}` placeholders: write `tcs-corp` / `sradha-tcs` in the URLs.
- The PR description must link `https://{branch}--tcs-corp--sradha-tcs.aem.page/{path}` showing the change (AGENTS.md "Publishing Process"); the PageSpeed check reads the URLs from it.
- Reviewer checks: Build, Code Sync and PageSpeed green; `npm test` passed (not in CI); aggregated `component-*.json` included when partials changed; generated files current; scoped CSS in rem; placeholders for UI text; no clicks pushed by blocks; legacy content still renders; docs updated.
- Required reviewers and branch protection: Not recorded.

3. Definition of Done

A change is done when all of these hold:

1. `npm run lint` passes (ESLint with xwalk model rules, Stylelint).
2. `npm test` passes; new pure logic has unit tests.
3. Generators run and their output committed where relevant: `npm run build:json` (models), `npm run build:modules` (imports), `npm run build:docs` (models, new blocks), `npm run build:paths` (`LOCALES`).
4. Verified on the local preview and the feature preview at mobile, tablet and desktop widths; no console errors, no overflow; authored content from before still renders.
5. Keyboard, focus, names and contrast checked on changed blocks; reduced motion honoured.
6. Data layer: one `cmp:click` per click, no duplicate component ids, no personal data.
7. PageSpeed Insights on the feature preview (target 100); no new render-blocking resources.
8. Documentation updated in the same change (table above).
9. PR with the preview URL; checks green; reviewed and merged with a merge commit.
10. Items that need an author, a measurement plan or another environment are reported as **Blocked**, not as passed.

4. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

5. Known limitations

- None recorded for this document.

6. Open items

- None.

7. Change history

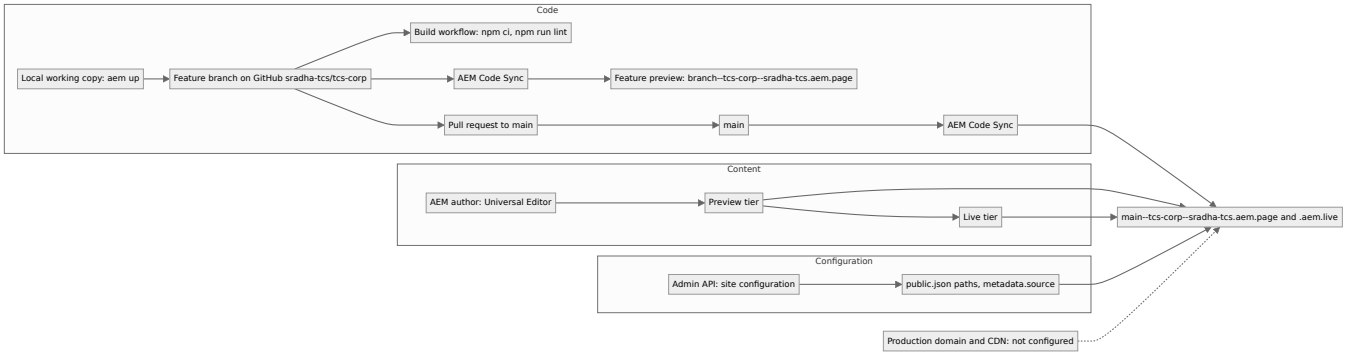
Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Release and Deployment Guide (developers)

Attribute	Value
Document ID	DEV-016
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How code reaches preview and live, and how it is rolled back
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/deployment-and-release.md, .github/workflows/main.yaml
Related documents	—
Related AEM paths	—

1. Delivery streams

Content, code and configuration are released independently: a code merge doesn't publish content and publishing content doesn't deploy code. A release that needs both goes **code first, then content** (handover Part 12, introduction).



2. Branches and feature previews

- `main` is production. Changes go through short-lived branches merged with pull requests (130 merge commits on `main` at 2d06316, latest #129).
- Branch prefixes in use: `feature/`, `feat/`, `fix/`, `perf/`, `docs/` (for example `fix/block-reaudit`, `perf/module-preload`, `docs/block-readmes`). The handover's `feature/tcs-<topic>` convention (12.2.2) is not what recent history uses; no naming rule is enforced.
- AEM Code Sync serves every pushed branch at `https://{branch}--tcs-corp--sradha-tcs.aem.page/` (slashes become hyphens; keep names short and lower case: branch, repository and owner form the host name).

3. Pull requests

The publishing process of AGENTS.md:

1. Push the feature branch; Code Sync makes it available on the feature preview.
2. Run PageSpeed Insights on the feature preview URL and fix issues; target a score of 100.
3. Open a pull request to `main`. The description **must** link `https://{branch}--tcs-corp--sradha-tcs.aem.page/{path}` showing the change, or the PR is rejected. Without a page, create test content in `drafts/` and ask an author to create a CMS page to link.
4. Check the status of code sync, linting and performance tests (`gh pr checks`).
5. A human reviewer reviews the code, opens the URL and merges; Code Sync updates `main`.

.github/pull_request_template.md asks for the GitHub issue (`Fix #<gh-issue-id>`) and Before / After test URLs. It still contains the boilerplate placeholders `{repo}` and `{owner}`: replace them with `tcs-corp` and `sradha-tcs` (the `cleanup-on-create` workflow that would replace them only runs on repository creation or manual dispatch).

Approval rules: the number of required reviews, required checks and branch protection on `main` are not recorded (handover 12.3; open item O-05).

4. Automated checks

Check	Defined in	When	What it does	Status at 2d06316
Husky pre-commit	.husky/pre-commit → .husky/pre-commit.mjs	Every local commit	If staged files include a model partial (`_*.json`), runs `npm run build:json` and stages the three aggregated files	Aggregated files match the partials: Passed
Build	.github/workflows/main.yaml	Every push, every branch (`on: [push]`)	`npm ci`, then `npm run lint` (ESLint with `airbnb-base`, `plugin:json`, `plugin:xwalk/recommended`; Stylelint `stylelint-config-standard`)	`npm run lint` locally: Passed
Unit tests	package.json `test`	Manual only: not in any workflow	311 tests in 53 suites (`test/`)	`npm test` locally: Passed (311 / 311)
cleanup-on-create	.github/workflows/cleanup-on-create.yaml	Repository creation or manual dispatch	Boilerplate clean-up; no role in releases	Not applicable
Renovate	.renovaterc.json	Renovate schedule	Dependency PRs; devDependency PRs labelled `ignore-psi-check` and auto-merged when tests pass; ESLint pinned to `^8.0.0`	Not applicable
AEM Code Sync	GitHub app (not a repository file)	Every push	Syncs the branch to the code bus and reports a check	Not verified here (no GitHub access)
PageSpeed (`aem-psi-check`)	AEM GitHub app (not a repository file)	Pull requests	PageSpeed Insights on the URLs in the PR description	Not verified here

The CI gate is lint only. Run `npm test` before every PR; adding it to `main.yaml` is a recommendation of the handover (14.9 #19, F-37), not a decision.

5. Environment promotion

Step	From → to	How	Owner
1	Local → feature preview	Push the branch; Code Sync	Developer

Step	From → to	How	Owner
2	Feature preview → `main` (preview and live code)	Reviewed PR merged; Code Sync updates `main--tcs-corp--sradha-tcs.aem.page` and `.aem.live`	Reviewer
3	Universal Editor	Authors reload the editor to get new definitions, models and filters	Content team
4	Configuration	Apply `public.json` paths, `metadata.source`, sitemap or CDN settings through the configuration Admin API, saving the current JSON first	Site Administrator
5	Content: author → preview → live	Publish sheets, then fragments, then pages	Content team

There is no separate staging environment in the repository: feature previews and the `main` preview tier play that role. The author environment reads the code of `main` (fstab.yaml mount point). The production domain and CDN are not configured (open item O-03).

6. Rollback

No rollback has been recorded; the procedures below are the handover's (Part 12, 12.6) and Require validation.

Stream	Procedure
Code	Revert the merged PR in GitHub (a revert PR, checks, merge) or `git revert -m 1 <merge-commit>` on a branch, newest first; Code Sync deploys the previous code. Never force-push or rewrite `main`
Models	The revert restores partials and aggregated files; authors reload the editor; values saved with removed fields stay stored but aren't shown
Content	Sites console **Timeline > Revert to this Version** (or Restore Version), check, publish again; sheets the same way; unpublish when no good version exists
Configuration	Re-apply the saved JSON (`public.json` paths from a previous `paths.json`: `git show <commit>:paths.json`; the previous `metadata.source` list); repository configuration files by reverting the PR, then re-index or regenerate the sitemap
Content package	Reinstall the previous package version from the `home-content-package` branch (installing replaces the package's pages)

Emergency change approval and the hotfix path are not recorded (handover 12.6.1, Part 17 R-15).

7. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

8. Known limitations

- None recorded for this document.

9. Open items

- None.

10. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Repository Structure

Attribute	Value
Document ID	DEV-002
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	The tracked tree, generated files and what is served
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/repository-structure.md
Related documents	—
Related AEM paths	—

1. Tracked Tree

Only paths tracked by Git (file counts from `git ls-files`):

```
.
├── .agents/settings.json          agent tooling: enabled plugins
├── .editorconfig                  indentation (js / json 2, css 4)
├── .eslintignore                  lint exclusions (*.min.js, dompurify)
├── .eslintrc.js                   ESLint: airbnb-base, json, xwalk model rules (max-cells)
├── .github/
│   ├── pull_request_template.md   boilerplate PR template ({repo} / {owner} not substituted)
│   └── workflows/
│       ├── main.yaml              "Build": npm ci + npm run lint on every push
│       └── cleanup-on-create.yaml boilerplate clean-up on repository creation
├── .gitignore                     .hlx, coverage, logs, node_modules, helix-importer-ui, ...
├── .hlxignore                     files not served by Edge Delivery (dot files, *.md, docs/*, test/*,
_*, ...)
├── .husky/pre-commit(.mjs)        rebuilds component-*.json when a _*.json partial is staged
├── .migration/project.json        migration tooling project metadata (site, assets folder)
├── .renovaterc.json               Renovate: dev dependencies auto-merged, ESLint pinned to v8
├── .stylelintrc.json              stylelint-config-standard
├── .well-known/adobe/             Cloud Manager challenge file
├── 404.html                       the 404 page (loads the authored, inherited error page)
├── AGENTS.md / CLAUDE.md          project rules for agents and developers (CLAUDE.md points to
AGENTS.md)
├── CODE_OF_CONDUCT.md, CONTRIBUTING.md, LICENSE, README.md   boilerplate files (Apache 2.0)
├── blocks/                        30 block folders + README.md (125 files)
├── component-definition.json      generated: Universal Editor palette
├── component-filters.json         generated: what each container accepts
├── component-models.json          generated: dialog fields
├── decorators/                    28 decorator modules + README.md
├── docs/                          not served: handover/knowledge-base/ (this knowledge base); the
handover, audit,
│                                  blocks report, animation, proposals and EDAM set-up are on the branch
```

```
│                                  migrated-page-content-edam under docs/ (KB-D-066)
├── drafts/                        14 tracked local-preview drafts (header / Canvas journey pages, 2
logos)
├── favicon.ico
├── fonts/inter-variable.woff2     self-hosted Inter variable font
├── fstab.yaml                     content source: AEM author via the xwalk delivery bridge
├── head.html                      CSP, aem.js / scripts.css, generated modulepreload links
├── helix-query.yaml               query index definition (/query-index.json)
├── helix-sitemap.yaml             sitemap (/sitemap.xml) from the query index
├── icons/                         22 SVG icons
├── models/                        default content, page, section and shared models (_*.json)
├── package.json / package-lock.json   scripts and dev dependencies (no runtime dependencies)
├── paths.json                     generated: AEM path -> published path mappings, includes, excludes
├── scripts/                       aem.js, scripts.js, delayed.js, editor support, module graph,
vendored libs
├── styles/                        styles.css, lazy-styles.css, fonts.css, motion.css, card-list.css
├── templates/                     home, article, blog, press-release + README.md
├── test/                          unit tests (node --test) and helpers
├── tools/                         docs, modules and paths generators; sidekick config
├── utils/                         26 util modules + README.md
├── well-known/adobe/              Cloud Manager challenge file (second copy, without the dot)
└── xwalk.json                     xwalk site option: multi-field enabled
```

2. Local Folders, Not in Git

These exist in the delivery workspace but are not tracked. Except `node_modules/` (in .gitignore), they are ignored only by the workspace's local `.git/info/exclude`, so a fresh clone would not ignore them: don't add them to commits.

Path	What it holds (observed)	Note
`.home-work/`	Delivery workspace scripts: the local library preview renderer (`render-lib.mjs`, referenced by AGENTS.md), package builders, reports, Figma crops, fixtures	Local, not in git. `tools/docs/library-snapshot.json` keeps the block library data the docs generator needs from it
`.handover-work/`	Handover build scripts (`build-volumes.mjs`, `build-index.mjs`, `compile-pdf.sh`, `typst-watch.sh`) and working notes	Local, not in git; described in docs/handover/README.md
`content/`	Local content and guide workspace: earlier guides (Markdown and PDF), the whitepaper template, local content (`content`, en, `language-master`, `home-implementation`)	Local, not in git
`en/`	Local `/en` preview content (`data`, `media`, `references`, `shots`, `taxonomy.json`)	Local, not in git
`migration-work/`	Content package workspace (`jcr-content`, `packages`, `profile.json`)	Local, not in git
`drafts/dam/`, `drafts/en/`, `drafts/language-master/`	Local preview content beside the 14 tracked drafts	Local, not in git
`query-index.json`	A local copy of the query index for the dev server	Local, not in git
`node_modules/`	`npm install` output	Ignored by `.gitignore`
`.claude/`, `.claude-plugin/`	Agent runtime and plugin configuration	Local, not in git

3. Ownership

No ownership is recorded in the repository: there is no `CODEOWNERS` file, and the handover's ownership matrix (1.11) names roles only (code: Development Lead accountable and responsible; component models:

Development Lead; site configuration: Site Administrator), with every name "To be confirmed" (open item O-01). The tables below therefore give no owner per folder.

4. Folders

blocks/

Aspect	Rule
Purpose	One folder per block, section container, child item or compatibility block (30 folders; inventory and status in blocks/README.md)
Belongs	`{block}.js` (default export `decorate(block)`), `{block}.css`, `_{block}.json` (definitions, models, filters for Universal Editor), `README.md`; pure, unit-tested helpers of that block only (`list/list-data.js`, `cards/cards-data.js`, `separator/separator-config.js`, `conversation-canvas/canvas-journey.js`, `conversation-canvas/canvas-controller.js`). `header/` has no model (built from the nav page)
Must not	Shared behaviour (goes to `decorators/` or `utils/`); unscoped selectors or `{block}-container` / `{block}-wrapper` classes; px / em sizes (rem only, breakpoints in px); hard-coded UI text (placeholders); own click pushes to the data layer (describe clicks with `data-cmp-*`); sample content in templates or model defaults; Canvas logic in `header` or `footer`
Consumers	`loadBlock` in scripts/aem.js (by block class), `npm run build:json` (aggregation into `component-*.json`), tools/modules/generate-module-graph.mjs, tools/docs/generate-block-docs.mjs
Dependencies	`decorators/`, `utils/`, `styles/` tokens (and `styles/card-list.css` for List and Cards), `models/_card-list.json` (shared List / Cards fields), placeholders, query index and sheets at runtime
Lifecycle	Status in tools/docs/blocks.json: Active, Compatibility-only (Columns), Experimental (Interactive Dashboard). A replaced component keeps its definition so content authored before stays editable (Columns, Button (basic)); model changes must respect the AEM delivery rules in AGENTS.md
Testing and documentation	`npm run lint` (ESLint incl. xwalk model rules, Stylelint); `npm test` for pure modules, block names (test/models/block-names.test.mjs), item rows and docs; block library pages; compare aem.page with the local preview after model changes. After a model change: `npm run build:json`, `npm run build:docs`; after an import change: `npm run build:modules`. Update the block README (written sections) and, for significant changes, the handover

models/

Aspect	Rule
Purpose	Models that are not a block's: page properties (`_page.json`: Title, Description, Keywords, Page Type, Navigation, Footer, 404 Page), section (`_section.json`), built-in default content (`_text.json`, `_title.json`, `_image.json`, `_button.json` with "Button (basic)"), shared field groups (`_card-list.json`), and the three aggregators `_component-definition.json`, `_component-models.json`, `_component-filters.json`
Belongs	Universal Editor JSON partials (file names start with `_`, so `.hlxignore` keeps them off the site)
Must not	Block models (they live in `blocks/{block}/_{block}.json`); hand edits to the generated root `component-*.json` instead of the partials
Consumers	`npm run build:json` (`merge-json-cli`), the Husky pre-commit hook, Universal Editor (through the generated files), templates/README.md (adding a Page Type)
Dependencies	The block partials (`../blocks/*/_*.json`), referenced by the aggregators
Lifecycle	Page Type options must match TEMPLATES in scripts/scripts.js; the main filter lists `section`, `gen-carousel`, `tabs`, `columns-container`
Testing and documentation	`npm run lint` (xwalk rules, `max-cells` per model in .eslintrc.js); handover 15.4 Content Modeling Standards

scripts/

Aspect	Rule
Purpose	Page decoration and loading: aem.js (core library, `sampleRUM`, `loadBlock`, `loadCSS`), scripts.js (eager / lazy / delayed phases, templates, section containers, error page, preloading, data prefetch), delayed.js (empty; the place for martech), editor-support.js and editor-support-rte.js (Universal Editor live updates), module-graph.js (generated), vendored `dompurify.min.js` and `lib/lottie/`
Belongs	Global page lifecycle code and vendored third-party libraries
Must not	Changes to `aem.js` (never modified); block-specific logic; edits to the generated `module-graph.js`
Consumers	`head.html` (loads `aem.js` and `scripts.js`), `404.html`, Universal Editor (editor support), every block
Dependencies	`decorators/`, `utils/`, `templates/`, `styles/`
Lifecycle	`module-graph.js` regenerated by `npm run build:modules`; vendored libraries updated by hand (not by Renovate)
Testing and documentation	test/modules/module-graph.test.mjs checks the generated preload lists; AGENTS.md (three-phase loading, module preloading)

styles/

Aspect	Rule
Purpose	styles.css (tokens, LCP-critical global styles, fallback font faces, Additional tab classes, section backgrounds and the dark-background theme contract), lazy-styles.css (post-LCP), fonts.css (Inter), motion.css (loaded only when a block animates), card-list.css (shared by List and Cards)
Belongs	Global styles and design tokens
Must not	Block-specific styles (in the block's CSS); colours defined for animations; primitives instead of tokens for text colours in new rules
Consumers	`head.html`, `scripts.js` (loadCSS of fonts and lazy styles), `decorators/motion-decorator.js`, the List and Cards CSS
Dependencies	`fonts/inter-variable.woff2`
Lifecycle	Keep the fallback font faces when changing the font (CLS)
Testing and documentation	`npm run lint:css` lints `blocks/**/*.css` and `styles/*.css` (not `templates/`); check the theme on the block library page "Sections: colours with blocks"

decorators/ and utils/

Aspect	Rule
Purpose	Shared single-purpose modules: `decorators/` create or decorate elements (28 modules: buttons, links, headings, images, media, data layer, motion, skeletons, SEO, ...); `utils/` build no UI (26 modules: site and locales, metadata, placeholders, taxonomy, data sheets, query index, properties, cells, ids, environment, privacy, ...)
Belongs	One purpose per file; exports listed in decorators/README.md and utils/README.md
Must not	Imports from `decorators/` in `utils/` (utils never import decorators); block re-implementations of shared behaviour
Consumers	Blocks, templates, `scripts.js`, `editor-support.js` (the "Used by" column of each README)
Dependencies	`scripts/aem.js`; site data at runtime (placeholders, taxonomy, sheets, query index)
Lifecycle	New shared behaviour goes here, not into a block; update the README's table with the export and its users

Aspect	Rule
Testing and documentation	Unit tests for pure modules (test/privacy/privacy.test.mjs, test/motion/motion.test.mjs, test/site/index-root.test.mjs, test/items/item-rows.test.mjs, test/media/vimeo-url.test.mjs); the folder README

templates/

Aspect	Rule
Purpose	Code of the Site Global Templates, loaded by Page Type: `home`, `article`, `blog`, `press-release` (templates/README.md)
Belongs	`templates/{name}/{name}.js` (default export gets `main` before decoration) and `{name}.css`
Must not	LCP content built by code (keep it authored); blocking sheet loads (use `appendLoadedSection`)
Consumers	`loadTemplate` in scripts/scripts.js (names in `TEMPLATES`)
Dependencies	`decorators/template-decorator.js`, `author-bio-decorator.js`; the Authors sheet (the media contacts are the Media Contacts block, KB-D-054)
Lifecycle	A new template needs its AEM template page, the code, a `TEMPLATES` entry and a Page Type option in `models/_page.json`
Testing and documentation	`npm run lint:js` (CSS here is not in the Stylelint globs); templates/README.md

test/

Aspect	Rule
Purpose	Unit tests run by `npm test` (`node --test 'test/**/*.test.mjs'`): cards, conversation-canvas (journey), docs, items, list (data, display precedence, query-index tags), media, models, modules, motion, privacy, separator, site; helpers `test/model-layout.mjs` and `test/list/fixtures.mjs`
Belongs	Tests of pure modules (no browser) and repository consistency checks
Must not	Browser or network tests; tests that need AEM author access
Consumers	Developers; not run by CI (.github/workflows/main.yaml runs lint only)
Dependencies	The modules under test, the models, `tools/docs/`
Lifecycle	Keep item layouts (`ITEM_LAYOUT`, `CARD_LAYOUT`) and their tests in step with model tabs
Testing and documentation	ESLint override for `test/**/*.mjs` (Node globals); not served (`test/*` in `.hlxignore`)

tools/

Path	Purpose	Run
tools/paths/generate-paths.mjs	Generates `paths.json` from `LOCALES` (plus the temporary `REVIEW` block for the language master)	`npm run build:paths` after changing `LOCALES`; then apply the mappings to the site configuration
tools/modules/generate-module-graph.mjs	Generates the `head.html` modulepreload links and `scripts/module-graph.js`	`npm run build:modules` after an import change
tools/docs/generate-block-docs.mjs	Rewrites the generated sections of `blocks/README.md` and the block READMEs	`npm run build:docs` after a model change
tools/docs/validate-block-docs.mjs	Validates the block READMEs (structure, links, anchors, no environment ids)	Used by `test/docs`

Path	Purpose	Run
tools/docs/report-block-docs.mjs	Builds docs/blocks/BLOCK-DOCUMENTATION-REPORT.md	`node tools/docs/report-block-docs.mjs [base-ref]`
tools/docs/blocks.json, tools/docs/library-snapshot.json	Hand-maintained block metadata (status, type, purpose); block library snapshot (package 1.0.13) for examples and usage	Inputs of the docs generator
tools/sidekick/config.json	AEM Sidekick configuration ("AEM Editor" edit URL pattern)	Read by Sidekick

Generated output must not be edited by hand; every generator is checked by `npm test` except `generate-paths.mjs`.

docs/

Aspect	Rule
Purpose	Project documentation, not served (`docs/*` in `.hlxignore`): handover/ (handover Markdown and PDF, earlier guides, 629 screenshot files, this knowledge base), audit/ (block ecosystem audit, re-audit, dialog options report, CSV data, evidence tools), blocks/ (documentation report and per-block review notes), animation/, proposals/
Belongs	Reports, proposals, handover sources, evidence
Must not	Code the site loads; secrets or environment identifiers
Consumers	Project and support teams; `tools/docs/` (screenshots)
Lifecycle	Update the handover's component chapter, Part 16 and revision history for significant changes, then regenerate the PDF (docs/handover/README.md)
Testing and documentation	`docs/audit/tools` is excluded from ESLint (.eslintignore)

drafts/, fonts/, icons/

Path	Purpose	Rule
`drafts/`	14 tracked static pages for the local dev server (`aem up --html-folder drafts`): Canvas journey, header and nav drafts, two logo images	Test content only; not authored content. Whether these files are reachable on the code bus is not verified (`drafts/` is not in `.hlxignore`)
`fonts/`	`inter-variable.woff2`	Optimised assets only
`icons/`	22 SVG icons (arrows, chevrons, social, media controls) rendered by `decorateIcons` (`scripts/aem.js`, called from `decorateMain`) and used by the button and icon decorators	Optimised SVGs only

5. Root Configuration Files

File	Purpose	Consumers	Change rule
fstab.yaml	Content source: one mount point `/` to the AEM author's xwalk delivery bridge (`bin/franklin.delivery/sradha-tcs/tcs-corp/main`), type `markup`, suffix `.html`	Edge Delivery Services	Only with the platform administrator; assumption: no other mount points (handover 1.8)

File	Purpose	Consumers	Change rule
paths.json	42 mappings (13 locale folders, 13 locale Home pages, 13 taxonomy pages, 3 for the language master review; sheets covered by the folder mappings), 15 includes (13 locale pages, `/content/dam/tcs-corp/`, the language master root), 28 excludes (13 block libraries and Site Global Templates, each as page and subtree)	AEM uses the site configuration's `public.json` → `paths`, not this file	Generated: `npm run build:paths`; never edit by hand; apply to the site configuration after a change
helix-query.yaml	Query index `/query-index.json`: includes `/**`; excludes `.json`, `references/`, `error-pages/`, `data/`, nav / footer, block libraries, templates, fragments, modals; indexes the metadata columns	List, Filter List, Conversation Canvas menus (utils/query-index.js); the sitemap	Add a property when a block needs a new metadata column; reindex after changes
helix-sitemap.yaml	`/sitemap.xml` from `/query-index.json`, `lastmod` YYYY-MM-DD, origin the `main--tcs-corp--sradha-tcs.aem.live` host until a production domain exists	Search engines	Replace the origin when the production domain is configured
head.html	Content Security Policy (`script-src 'nonce-aem' 'strict-dynamic' ...`, moved to an HTTP header), viewport, `aem.js`, `scripts.js`, `styles.css`, the generated modulepreload block	Every page	The block between `<!-- modulepreload -->` markers is generated by `npm run build:modules`; no external scripts
404.html	The 404 page: same CSP, `sampleRUM('404')` with the referrer, built-in fallback content replaced by the inherited authored 404 page (`loadErrorPage` in `scripts.js`)	Edge Delivery (served for missing pages)	Keep the built-in content as the last fallback
xwalk.json	xwalk option `multi-field` enabled	Universal Editor / xwalk delivery	—
component-definition.json, component-models.json, component-filters.json	The Universal Editor palette, dialogs and container filters	Universal Editor	Generated by `npm run build:json` from `models/_component-*.json` and `blocks/*/_*.json` (also by the pre-commit hook); never edit by hand
package.json	Scripts (`lint`, `lint:js`, `lint:css`, `lint:fix`, `test`, `build:json`, `build:paths`, `build:modules`, `build:docs`, `prepare`) and dev dependencies only	Developers, CI, Renovate	Name, version (`1.3.0`) and repository URL are still the boilerplate's
.eslintrc.js	Airbnb base, JSON and xwalk recommended rules; `.js` extensions in imports; Unix line breaks; `xwalk/max-cells` limits per model (default 4; e.g. `columns-container` 22, `tabs` 21, `list-item` 19, `card` 16)	`npm run lint:js` (`.json`, `.js`, `.mjs`)	Raise a `max-cells` limit only with the model change that needs it
.stylelintrc.json	`stylelint-config-standard`	`npm run lint:css`	—
.eslintignore	`helix-importer-ui`, `dompurify.min.js`, `*.min.js`, `docs/audit/tools`	ESLint	—
.hlxignore	Not served: `.*`, `*.md`, `karma.config.js`, `LICENSE`, `package.json`, `package-lock.json`, `test/*`, `_*`, `docs/*`	Edge Delivery code bus	Add any file that must not be public
.editorconfig	Indent 2 for `.js` / `.json`, 4 for `.css`	Editors	—
.gitignore	`.hlx/*`, `coverage/*`, `logs/*`, `node_modules/*`, `helix-importer-ui`, `.DS_Store`, `*.bak`, `.idea`; keeps `CLAUDE.md`, `AGENTS.md` tracked	Git	Workspace folders are excluded only locally (see above)
.github/workflows/main.yaml	"Build": checkout, Node 24 (the step is labelled "Use Node.js 20"), `npm ci`, `npm run lint` on every push	GitHub Actions	Unit tests are not run in CI
.github/workflows/cleanup-on-create.yaml	Boilerplate workflow that removes template helper files when a repository is created from the template (or on manual dispatch)	GitHub Actions	Boilerplate leftover; Unused candidate

File	Purpose	Consumers	Change rule
.github/pull_request_template.md	Issue and before / after test URLs; {repo} / {owner} were never substituted	Pull requests	—
.husky/pre-commit, .husky/pre-commit.mjs	When a _*.json partial is staged, runs npm run build:json and stages the three component-*.json	Git commits (installed by npm install through prepare)	—
.renovaterc.json	config:recommended; dev dependencies labelled ignore-psi-check and auto-merged; ESLint allowed ^8.0.0	Renovate	—
tools/sidekick/config.json	Sidekick project "AEM XWalk Boilerplate", edit URL label "AEM Editor"	AEM Sidekick	—
.migration/project.json, .agents/settings.json	Migration and agent tooling metadata (site names, assets folder, enabled agent plugins)	Delivery tooling, not the site	Not served (dot folders)
.well-known/adobe/, well-known/adobe/	Cloud Manager challenge files (added on 10 October 2026); contents not reproduced here	Adobe Cloud Manager	Requires validation: .hlxignore (.*) keeps the dot folder off the code bus
AGENTS.md, CLAUDE.md	Verified project rules; CLAUDE.md contains only see @AGENTS.md	Developers and AI agents	Update with significant changes
README.md, CONTRIBUTING.md, CODE_OF_CONDUCT.md, LICENSE	Boilerplate texts (README still "Your Project's Title..."; CONTRIBUTING refers to Project Helix and an npm run commit script that doesn't exist); Apache License 2.0	—	Requires validation: project-specific README not written

6. Generated Files

Output	Generator	Command	Checked by
component-definition.json, component-models.json, component-filters.json	merge-json-cli over models/_component-*.json	npm run build:json (and the pre-commit hook)	npm run lint (xwalk rules)
paths.json	tools/paths/generate-paths.mjs	npm run build:paths	Not tested
head.html modulepreload block, scripts/module-graph.js	tools/modules/generate-module-graph.mjs	npm run build:modules	test/modules/module-graph.test.mjs
Generated sections of blocks/README.md and the block READMEs	tools/docs/generate-block-docs.mjs	npm run build:docs	test/docs/block-docs.test.mjs
docs/blocks/BLOCK-DOCUMENTATION-REPORT.md	tools/docs/report-block-docs.mjs	node tools/docs/report-block-docs.mjs	—
Handover PDF	Local scripts in .handover-work/ (pandoc, typst)	See docs/handover/README.md	—

7. What Is Served

Edge Delivery serves the code bus of each branch minus .hlxignore: blocks/, scripts/, styles/, decorators/, utils/, templates/, fonts/, icons/, drafts/, the root HTML / JSON / YAML files and favicon.ico. Not served: Markdown (all READMEs and AGENTS.md), docs/, test/, _* model partials, package*.json, LICENSE and dot files

and folders. Everything served is public client-side code: no secrets, keys or environment identifiers belong anywhere in the repository.

8. Related

- project-overview.md: project scope, locales, delivery model and stack
- glossary.md: terms used on this page
- references.md: every repository document and cited Adobe page
- architecture.md, blocks.md, content-architecture.md, data-layer-and-analytics.md, design-system.md, personalization.md: the other knowledge base topics
- AGENTS.md: project structure, setup commands, code style
- blocks/README.md, decorators/README.md, utils/README.md, templates/README.md
- Handover Part 2 — Solution Architecture and Part 12 — Publishing, Deployment & Release: TCS-CORP-HANDOVER.md

9. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

10. Known limitations

- None recorded for this document.

11. Open items

- None.

12. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

SEO Development Guide

Attribute	Value
Document ID	DEV-012
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	Metadata, SEO decorator, sitemap and indexing rules
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	decorators/seo-decorator.js, helix-query.yaml, helix-sitemap.yaml
Related documents	—
Related AEM paths	—

1. Metadata model

Purpose. Page-level values (SEO, social, card content, page type, author, dates, flags) that code, the query index and the data layer read. Every value is a `<meta>` in the server-delivered head.

Sources and precedence.

Source	Where authored	Published as
Page properties (`models/_page.json`)	Universal Editor page properties	`<title>`, `description`, `keywords`, `page-type`, `nav`, `footer`, `error-page` metas
Standard AEM page properties	Sites console (e.g. Tags `cq:tags`)	`cq-tags` (one comma-separated meta)
Bulk metadata sheet	`/content/tcs-corp/{country}/{language}/metadata`	`/{locale}/metadata.json`; each column becomes `<meta name="{column}">` on pages matching the row's `url` pattern

Page-level values win over bulk metadata; bulk rows apply top to bottom (handover Part 8.1.2). The bulk sheets apply only once all 13 are listed in the site configuration's `metadata.source`, and each sheet's URL patterns must start with its locale (`/de-de/**`) (AGENTS.md "Notes").

Bulk metadata columns (AGENTS.md, 41 columns): `url`, `page-type`, `meta-title`, `meta-description`, `meta-keywords`, `author-id`, `publish-date`, `modified-date`, `reading-time`, `og-title`, `og-description`, `og-image`, `og-type`, `twitter-card`, `twitter-title`, `twitter-description`, `twitter-image`, `canonical-url`, `robots`, `region`, `country`, `language`, `persona`, `thumbnail-image`, `featured-image`, `campaign-id`, `content-type`, `featured`, `priority`, `listen-link`, `author`, `card-title`, `card-description`, `card-image`, `card-image-alt`, `image-alt`, `eyebrow`, `category`, `hide-in-nav`, `card-style`, `card-theme`.

Consumers.

Consumer	Reads	Source
SEO decorator (eager)	`meta-title` → <title>; `meta-description`, `meta-keywords`; `og-*` (falling back to `meta-*`); `twitter-*` (falling back to `og-*`, `meta-*`); `canonical-url` → <link rel="canonical"> and `og:url`; `language` → <html lang>	`decorators/seo-decorator.js`
`getPageMetadata()`	title, description, image (`featured-image`, else `og:image`), `page-type` (else `template`), `content-type`, `category`, tags, `author-id`, `author`, `publish-date` (else `publication-date`), `modified-date`, `reading-time`, `listen-link`, `thumbnail-image`, `featured-image`, `image-alt`, `region`, `country` (else the locale's), language, `persona`, `campaign-id`, `featured` (yes / true / 1 / on / y), `priority` (number)	`utils/metadata.js`
`getPageTags()`	`article:tag` metas and `cq-tags`, else a comma-separated `tags` meta	`utils/metadata.js`
`getPageLanguage()`	`language` when it is a language code, else the locale's `lang`	`utils/metadata.js`
Page Header, templates, data layer page entry	`getPageMetadata()`	page-header, `decorators/data-layer-decorator.js`
Query index	the server head (see indexing-and-search.md)	`helix-query.yaml`

Rules. The query index reads the server-delivered head, so values set by JavaScript (the SEO decorator's) aren't indexed (handover Part 8.1.3). `robots` is a standard name and works server side. Use `card-*`, `eyebrow`, `hide-in-nav`, `card-style`, `card-theme` for List cards (List README "Page metadata contract").

Example.

```
url                   | page-type      | hide-in-nav
/de-de/newsroom/**    | press-release  |
/de-de/insights/**    | article        |
/de-de/legal/imprint  |                | yes
```

Common mistakes. A sheet row pattern without the locale prefix; adding columns to a sheet that isn't listed in `metadata.source` (no effect); expecting a metadata change to reach lists before the page is republished and indexed. Field reference per column: handover Part 8 — Metadata Documentation (8.2, 8.4).

2. Sitemap

`helix-sitemap.yaml` defines one sitemap, `default`: source `/query-index.json`, destination `/sitemap.xml`, `lastmod: YYYY-MM-DD`, origin `https://main--tcs-corp--sradha-tcs.aem.live` (to be replaced by the production domain once `cdn.prod.host` is configured, per the file's comment).

- It covers every indexed page of all locales; there are no per-locale sitemaps and no `hreflang` alternates (handover Part 4.9.6).
- Handover Part 2.10.7 states that pages with `robots noindex` are left out of the sitemap while they stay in the index and in List / Filter List (no code checks `robots`); the sitemap behaviour is platform behaviour (Requires validation).
- The sitemap is regenerated when the index changes; the Admin API `sitemap` operation regenerates it on demand (handover Part 12.1.9).

3. Query index definition

`helix-query.yaml` defines **four indexes** (8 November 2026): `pages`, published as `/query-index.json`, for the 13 locales, and one per review root (KB-D-068): `language_masters` (`/language-masters/en/query-index.json`), `language_master` (`/language-master/en/query-index.json`) and `design_block_master` (`/design-block-master/en/query-index.json`), with the same exclusions and columns. A page reads only its root's index (`indexUrlOf` in `utils/query-index.js`); the sitemap (`helix-sitemap.yaml`) is built from `/query-index.json`, so it lists the locales only.

Included: `/**`.

Excluded:

Pattern	Why (from the file's comments)
`/**.json`	Sheets and other JSON resources
`/*/references/**`	Fragments, nav and footer pages, modals of each locale
`/*/error-pages/**`	Error pages
`/*/data/**`	Data sheets
`/*/nav`, `/*/footer`	Nav and footer pages at the locale root
`/*/block-library`, `/*/block-library/**`	Block libraries (also excluded from publishing in `paths.json`; safeguard)
`/site-global-templates/**`	Template pages (not published; safeguard)
`/language-masters/*/references/**`, `/language-masters/*/error-pages/**`, `/language-masters/*/data/**`, `/language-masters/*/nav`, `/language-masters/*/footer`, `/language-masters/*/block-library`, `/language-masters/*/block-library/**`	The same for the language master under review, a folder level deeper (#131, F-108; to remove with the REVIEW settings after the rollout)
`/**/fragments/**`, `/**/modals/**`	Fragment and modal folders anywhere

The `/*/…` patterns name one path segment before the folder (e.g. `/en/block-library`); the language master (`/language-masters/en/…`) is a level deeper and has its own patterns. Until 21 October 2026 it had none: 166 block library pages, 6 reference pages and the 404 page of the language master were in the live index (F-108). After the fix (#131) a reindex reports "requested path does not match index configuration" for these pages but doesn't remove their rows: they were removed with the Admin API index DELETE (173 / 173; the pages stay published). The index went from 354 to 181 rows (137 language-master content pages, 44 `/en` pages), and the Lists and Filter Lists of the library and the curated Insights page still filled.

Columns (37 properties, all read from the server-delivered `<head>`):

Group	Index columns (meta source)
Platform / standard	`lastModified` (`last-modified` header, parsed timestamp), `robots`, `title` (`og:title`), `description`, `image` (`og:image`, path only), `tags` (all `article:tag`, multi-value), `cqTags` (`cq-tags`, AEM page Tags as one comma-separated meta)
SEO	`metaTitle`, `metaDescription`
Classification	`pageType`, `contentType`, `category`, `eyebrow`
Author and dates	`author`, `authorId`, `publishDate`, `modifiedDate`, `readingTime`
Images	`thumbnailImage`, `featuredImage`, `imageAlt`
Audience	`region`, `country`, `language`, `persona`, `campaignId`
Featuring	`featured`, `priority`, `featureCard` and `featuredPriority` (earlier names, read as fallbacks)
List cards	`cardTitle`, `cardDescription`, `cardImage`, `cardImageAlt`, `cardStyle`, `cardTheme`, `hideInNav`

Rules:

- Values set by JavaScript (e.g. the SEO decorator) aren't indexed: only the delivered head counts (handover Part 8.1.3).
- `/query-index.json` is public: don't put internal-only values in metadata (handover Part 2.10.7).
- A new metadata column used by a block needs a property here, then a reindex (List README "Extending the Block", Filter List README step 3).

4. SEO decorator

Module	Purpose
`seo-decorator.js`	`decorateHead` (bulk metadata SEO / social columns to the standard head tags, `<html lang>`)

5. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

6. Known limitations

- None recorded for this document.

7. Open items

- None.

8. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Unit Testing Guide

Attribute	Value
Document ID	DEV-013
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	The unit tests, how to run them and how to add tests
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	test/, package.json
Related documents	—
Related AEM paths	—

1. Unit Tests

`npm test` runs `node --test 'test/**/*.test.mjs'` (Node's built-in runner, no dependencies). Tests cover pure modules (no DOM) and generated files. At commit 2d06316: 14 files, 53 suites, 311 tests, all passing.

File	Module(s) under test	What it covers	Tests
test/list/list-data.test.mjs	`blocks/list/list-data.js`, `utils/cells.js`, `utils/redirects.js`	List configuration defaults and clamping, author hints, child pages and depth, page metadata precedence, redirects (chains, loops), fixed lists, unsafe links, ordering, List Item rows (including rows rendered before the Display toggles)	57
test/list/display-precedence.test.mjs	`blocks/list/list-data.js`	Card style and colour precedence (List Item > List > page > automatic), field-by-field overrides from the linked page	16
test/list/query-index-tags.test.mjs	`utils/query-index.js`	Tags from `article:tag` and `cq-tags`, merged without duplicates	4
test/cards/cards-data.test.mjs	`blocks/cards/cards-data.js`, `utils/dates.js`	Card item cells in model order, field mapping, dates, new tab, external links, unsafe links, shared card-list settings	15
test/items/item-rows.test.mjs	`utils/cells.js` (`itemRows`)	Item rows told apart by model in the editor and by cells on the page	24
test/separator/separator-config.test.mjs	`blocks/separator/separator-config.js`	Defaults, ids, accessibility mode, variants, widths (legacy spellings), responsive values, decorations, animation, print, analytics	76
test/conversation-canvas/canvas-journey.test.mjs	`blocks/conversation-canvas/canvas-journey.js`	Floating journey roles, states, allowed transitions, hidden state, page variations	32

File	Module(s) under test	What it covers	Tests
test/motion/motion.test.mjs	utils/motion.js, block models, styles/motion.css	Animation values, speed and trigger, stagger, background colour detection, the authoring fields offered per block (container sections use container*)	15
test/site/index-root.test.mjs	utils/site.js	indexRoot, localeSheetPath, localePagePath, author paths, pageLink, language master review, getParentPages	21
test/media/vimeo-url.test.mjs	utils/media.js	Vimeo embed URLs (ids, channels, unlisted hashes, parameters)	5
test/privacy/privacy.test.mjs	utils/privacy.js	Search term redaction (e-mail addresses, phone numbers, length)	5
test/models/block-names.test.mjs	component-definition.json	Each block component's template name renders as an existing block folder's class	1
test/modules/module-graph.test.mjs	scripts/module-graph.js, head.html	The module graph is current (npm run build:modules), head.html preloads every scripts.js import, only existing modules listed	5
test/docs/block-docs.test.mjs	tools/docs/	Every block folder has a README listed in blocks/README.md; generated sections current; links and anchors resolve	35

Helper: test/model-layout.mjs derives the cells AEM renders for an item model (tabs as empty cells, collapsed suffix fields folded, classes left out), so cell tests follow model changes.

Not unit tested (no DOM test environment): block decoration, the data layer decorator, the header, editor support. These are covered by the browser harness and manual checks.

2. Lint

npm run lint = lint:js (ESLint on .json, .js, .mjs) + lint:css (Stylelint on blocks/**/*.css, styles/*.css). It is the only check in the Build workflow. Rules:

- airbnb-base, .js extensions required, Unix line endings.
- plugin:json/recommended for JSON files.
- plugin:xwalk/recommended for the models: xwalk/max-cells (default 4, per-model limits in .eslintrc.js), xwalk/no-duplicate-fields, xwalk/invalid-field-name, xwalk/no-orphan-collapsible-fields (a field such as hideTitle without its base field), xwalk/no-custom-resource-types.
- stylelint-config-standard.

Ignored: *.min.js, dompurify.min.js, docs/audit/tools (.eslintignore). Not linted: templates/*/*.css. npm run lint:fix fixes what it can.

3. Test inventory (19 files, 257 test cases)

File	Suites	Tests
test/cards/cards-data.test.mjs	card items; shared card list settings; content type labels and long dates; cards call to action	16
test/conversation-canvas/canvas-journey.test.mjs	roles from placement; the journey down the page; transitions; HIDDEN; page variations; the header Canvas at the bottom of the window; a row in place that holds focus (focused)	36
test/docs/block-docs.test.mjs	block documentation (npm run build:docs); block documentation helpers	5

File	Suites	Tests
test/items/item-rows.test.mjs	item models; itemRows	6
test/list/display-precedence.test.mjs	Card style of List Items (manual); Card style of source cards (child pages, tags); Card colour; List Item content from the linked page (field by field)	16
test/list/list-data.test.mjs	configuration; feature card author (Show author on feature card); content type on image cards (Show content type); parent / child pages; page metadata; fixed list; fixed list: Pages and List Items; tag-based pages; deduplicate, sort and limit; card kinds; list item cells; list item display; list call to action	69
test/list/query-index-tags.test.mjs	index tags	4
test/media-contacts/media-contacts-data.test.mjs	groupHeading; groupContacts; telHref	8
test/media/delivery-url.test.mjs	deliveryAsset; deliveryImageUrl	8
test/media/vimeo-url.test.mjs	vimeoEmbedUrl	5
test/models/block-names.test.mjs	block component names	1
test/modules/module-graph.test.mjs	module graph (npm run build:modules)	4
test/motion/motion.test.mjs	content animation; background animation; motion speed and trigger; stagger; background colour; authoring fields	15
test/privacy/privacy.test.mjs	redactSearchTerm; contactLinkKind	7
test/separator/separator-config.test.mjs	defaults; component id; accessibility mode; variants and line settings; width and alignment; responsive; decorations; animation, print and analytics	21
test/site/index-root.test.mjs	indexRoot; localeSheetPath; localePagePath; on AEM author; pageLink; the language master root page; getParentPages (inherited Navigation, Footer, 404 Page)	23
test/site/page-tree.test.mjs	pageTree	5
test/site/paths.test.mjs	paths.json	5
test/site/query-index-url.test.mjs	indexUrlOf; remainingOffsets	3

4. Result in this program

- `npm run lint`: Passed (exit 0)
- `npm test`: 387 of 389 passed, 2 failed: the 2 Form Variations motion tests (known before this program)

5. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

6. Known limitations

- None recorded for this document.

7. Open items

- None.

8. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

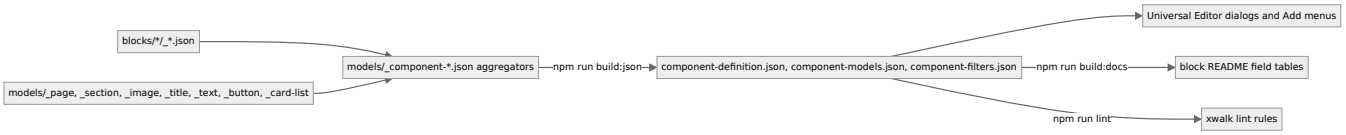
Universal Editor Model Guide

Attribute	Value
Document ID	DEV-007
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	Definitions, filters, field types, tabs, naming, the Additional tab, defaults, conditions, validation and the procedure for model changes
Scope	—
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/universal-editor.md, models/, component-models.json
Related documents	—
Related AEM paths	—

1. How the Configuration Is Built

File	Role	Edit?
`blocks/<block>/_<block>.json`	The block's `definitions`, `models` and `filters` (partial)	Yes
`models/_page.json`, `_section.json`, `_image.json`, `_title.json`, `_text.json`, `_button.json`	Page properties (`page-metadata`), Section, and the default content components	Yes
`models/_card-list.json`	Shared field groups (`header`, `cards`, `layout`, `scroll`, `advanced`) included by List and Cards with `"...": "../../models/_card-list.json#/fields/<group>"`	Yes
`models/_component-models.json`, `_component-definition.json`, `_component-filters.json`	Aggregators: include the `models/` partials and `../blocks/*/_*.json`; the filters aggregator also defines the `main` filter	Rarely
`component-models.json`, `component-definition.json`, `component-filters.json`	Generated by `npm run build:json` (`merge-json-cli`), read by Universal Editor	Never by hand
`xwalk.json`	`public.xwalk.multi-field.enabled: true`	No
`.husky/pre-commit.mjs`	When a staged file matches `_*.json`, runs `npm run build:json` and stages the three generated files	No

Partials start with _ and aren't served (`.hlxignore`).



2. Definitions

Definitions are grouped in `component-definition.json`: Default Content (Text, Title, Image, Button (basic)), Sections (Section) and Blocks (57 components: blocks, items and section containers).

Kind	resourceType	Template
Block	`core/franklin/components/block/v1/block`	name (becomes the block class), `model`, optional `filter`, `key-value: true` for key-value blocks, default values
Item	`core/franklin/components/block/v1/block/item`	name, `model`, default values
Section container	`core/franklin/components/section/v1/section`	name, `model`, `filter`, `style` (e.g. `columns-container-section`), default values
Built-in	`core/franklin/components/{text,title,image,button,columns}/v1/...`	Fixed fields only

- **The template name is the block class**: AEM renders a block with the class-name form of its template name, not the component id ("Button Block" gives `button-block`; "Column Break" gives `column-break`). `test/models/block-names.test.mjs` checks that every block component's template name names a block folder.
- **Key-value blocks** (17: Anchor List, Button, Cards, Column Break, Conversation Canvas, Filter List, List, Page Header, Quote, Separator, Tab, Table List, Tabs block, TCS Header, Teaser, Testimonial, Video) render one `name | value` row per field, read with `readProperties` (`utils/properties.js`). The other blocks render a row per field (or group) in model order.
- **Only standard resource types** are allowed (`xwalk/no-custom-resource-types`).

3. Filters

`component-filters.json` holds 26 filters. `main` (page level) allows Section, Gen Carousel, Tabs Container and Columns Container; `section` allows the blocks authors add to a section; `columns-container`, `tabs` and `gen-carousel` restrict the section containers; each container block has a filter of its items (e.g. `hero: hero-cta`). The compatibility-only Columns and Button (basic) and the experimental Interactive Dashboard are in no filter; the built-in Columns keeps its implicit `columns > column > text, image, title` filters so Columns authored before stay editable. The full table is in blocks.md.

4. Field Types in Use

Counted in `component-models.json` at commit 2d06316: 63 models, 991 top-level field entries (82 of them tabs), 1027 including the sub-fields of the 14 `container` fields.

Component	Count	Typical use
`select`	465	Options, Title Type, Additional tab spacing and animation
`text`	179	Titles, labels, ids, alt text
`boolean`	128	Show / hide toggles, Hide block
`tab`	82	Dialog tabs (also an empty cell in item rows)

Component	Count	Typical use
`aem-content`	45	Links, pages, fragments, transcripts, captions
`richtext`	41	Descriptions, captions, card text
`number`	38	Counts, columns, intervals, times
`reference`	18	Images, posters, logos, icons
`container`	14	Legacy composite multifields (12, `multi: true`) and the Conversation Canvas `menu` / `searchBar` groups (collapsible)
`aem-tag`	9	Tags (`multi: true` on Card, List, List Item, Teaser)
`multiselect`	6	`classes` of Cards, List and Table; Columns Container `style`; Filter List Page types and Content types
`date-time`	2	Teaser and Card date

Other properties: `description` on 418 fields, `condition` on 155, `options` on 471 (4292 options), `validation` on 35, `required` on 23, `min` / `max` on 34, `readOnly` on 6 (the form items' Field type), `placeholder` on 4. The audit counts (926 dialog fields after the first audit, 924 after the re-audit) used another method (`docs/audit/data/dialog-fields.csv`).

5. Dialog Tabs

- **At most three short tabs per model, the last one Additional.** Universal Editor collapses the tabs into a dropdown when they don't fit on one line. No model has more than three tabs; 28 models end with Additional (e.g. Content / Display / Additional for List and Cards, Content / Style / Additional for Hero and Teaser, Video / Playback / Additional for Video). Header and Footer have no Additional tab.
- **Key-value blocks**: tabs only group fields in the dialog; regrouping them changes nothing in the content.
- **Other blocks**: AEM adds an empty row for each dialog tab (Form Variations, Table, Accordion, Hero, Embed, Fragment, Modal): read rows by `data-aue-prop` in the editor, else skip empty tab rows, never "the first row".
- **Item models** (List Item, Card, Tab item, Anchor Link, Stats, Filter Category / Page): each tab is an empty cell of the row, so a tab change shifts the cells. Keep the layout constants and their tests in step: `ITEM_LAYOUT` (`blocks/list/list-data.js`, `blocks/anchor-list/anchor-list.js`, `blocks/tabs/tabs.js`), `CARD_LAYOUT` (`blocks/cards/cards-data.js`), `CATEGORY_LAYOUT` / `PAGE_LAYOUT` (`blocks/filter-list/filter-list.js`); `withoutTabCells` and `matchCells` in `utils/cells.js` drop tab cells and match cells to fields; `test/model-layout.mjs` derives the expected cells from the model.

6. Field and Property Naming

The naming rules follow what AEM delivers, verified on aem.page in the block audit of 19 October 2026 (AGENTS.md, "AEM delivery rules"):

Rule	Effect	Convention	Example
Field collapse	Fields ending in `Text`, `Title`, `Type`, `Alt` or `MimeType` fold into their base field (`linkText` is the link's text, `imageAlt` the image's alt, `linkType` becomes bold / italic around the link)	Use the suffix only to fold; for a value that must be read, use another name	`buttonVariant` (not `buttonType`); `posterAlt` folds into `poster`

Rule	Effect	Convention	Example
Element grouping	`<group>_<field>` fields render in one cell in row-per-field blocks; **in key-value blocks grouped fields aren't delivered**	Plain names in key-value blocks	Video `poster` / `posterAlt` (was `placeholder_image`, lost: audit L-2); Hero `content_*` (row-per-field block)
Block options	A field named `classes` or `classes_<name>` arrives as block classes, never as a content row	Presentation options and the Additional tab	`classes_anchor`, `classes_paddingtop`, Teaser `classes`
Section fields	**Sections don't get their `classes_*` fields delivered**: only `style` (as classes) and other fields as data attributes	Section containers use `container*` names, turned into classes by `applySectionBlockOptions` (`scripts/scripts.js`)	`containerAnchor`, `containerHide`, `containerAnimation`, `containerPaddingTop` (re-audit R-L-1)
Item node name	An item's `name` property is delivered in class-name form, and an item model with `classes` starts with a cell `<node name>, <classes>`	A value whose case matters gets its own field	Form Variations `fieldName` (re-audit R-D-1); first cell `email, field-text`
Non-image references	A Lottie `.json` or PDF reference is delivered as a link whose text is its Alt	Read it with `animationLabel` (`utils/media.js`)	Hero / Teaser Lottie alt (re-audit R-B-3)
Built-in components	Built-in Button, Title, Image and Text deliver only their fixed fields; built-in Columns no `classes_*`	Don't add fields to built-in models	Button (basic), Columns Additional tab removed
Field name syntax	Letters, digits, `_`, `-`, `:`; starts with a letter; at most one `_` and one `:`	`xwalk/invalid-field-name`	`classes_animationspeed`, `jcr:title`

Property naming habits in the models: camelCase names (`showEyebrow`, `desktopColumns`); `show*` booleans for optional parts; `*Alt` next to each image; `titleType` next to `title`; Additional tab names are lower case after the prefix (`classes_paddingleft`), container names camelCase (`containerPaddingLeft`).

7. The Additional Tab

Field	Blocks (`classes_*`)	Section containers (`container*`)
Anchor ID	`classes_anchor` (24 models), validated `^anchor-[a-z0-9][a-z0-9-]*$`	`containerAnchor` (same validation)
Hide block	`classes_hideblock` (25 models)	`containerHide`
Animation Type, Motion Speed, Animation Trigger	`classes_animation`, `classes_animationspeed`, `classes_animationtrigger` (17 models)	`containerAnimation`, `containerAnimationSpeed`, `containerAnimationTrigger` (Columns Container, Gen Carousel, Tabs Container)
Background animation	`classes_bganimation` (Hero, Teaser, Quote, Testimonial, Form Variations, Conversation Canvas)	none
Padding / Margin Left, Right, Top, Bottom	`classes_padding*`, `classes_margin*` (`--space-*` tokens: `pt-400` = `var(--space-400)`)	`containerPadding*`, `containerMargin*`

`decorators/block-options-decorator.js` makes the anchor class the block's id (de-duplicated with `-2`, `-3`); `styles/styles.css` applies Hide block and spacing; `utils/motion.js` and `decorators/motion-decorator.js` run the animations. Motion Speed and Animation Trigger show only once an animation is chosen (condition on `classes_animation`). Copy the tab from an existing `_<block>.json` into a new block model; the Tab item's Additional tab has only Hide block and the Column Break's has no animation fields.

8. Defaults and the Template-Value Rule

- **Template values are stored.** New components get their definition template values written to the content, so a sample text in a template appears on every new item: no sample content in templates or model defaults. Use them only for settings (`buttonView: button`, `titleType: h2`, `desktopMode: grid`).
- Corrected cases (re-audit): Table List "Explore" (R-A-6), Tab title "Tab" (R-C-10), Accordion Item Lorem Ipsum (R-C-11), Table rows "Column1" (R-D-9). Retained exceptions: the Submit Button `label` "Send" (template value; localised by translation only, re-audit review D, low) and the TCS Header logo alt defaults ("Tata Consultancy Services", "TATA Group").
- **Model `value`** (672 fields) is the dialog default; keep it equal to the code's default so an empty value and the default render the same (Hero CTA Arrow default corrected to Auto in the audit).
- UI text that isn't authored content comes from placeholders (`utils/placeholders.js`), never from defaults.

9. Enumerations

- `select` and `multiselect` options are `{ name, value }`: `name` is the dialog label, `value` the stored value (lower case, hyphenated, usually the class or keyword the code reads, e.g. `align-top`, `motion-fade-up`, `per-row-2`).
- An option with the empty value `''` is the default / none choice in 39 models (e.g. Conversation Canvas `classes` Default, Teaser Layout Image & Text); code treats `''` and a missing value alike.
- Shared vocabularies live in code and must match the options: `BUTTON_VIEWS`, `BUTTON_TYPES`, `BUTTON_ALIGNMENTS` (`utils/buttons.js`), `CARD_STYLES`, `CARD_THEMES`, `IMAGE_RATIOS` (`utils/card-list-config.js`), `CONTENT_MOTIONS`, `BACKGROUND_MOTIONS`, `MOTION_SPEEDS`, `MOTION_TRIGGERS` (`utils/motion.js`); unknown values are ignored.
- When an option is removed, keep reading its stored value (Separator Width "Auto" = 100 %, Variant "Custom" = Default).

10. Conditional Visibility

155 fields have a `condition` in JsonLogic, built from `var`, `===`, `!==`, `!`, `and` and `or` (`var` 277 times, `and` 59, `or` 40, `!` 16). Examples:

```
{ "===": [{ "var": "buttonView" }, "icon"] }
```

```
{ "and": [{ "var": "classes_animation" }, { "!==": [{ "var": "classes_animation" }, ""] }] }
```

Uses: Button Icon only for the Icon view; Motion Speed and Trigger only with an animation; List source-specific fields; List / Cards layout fields per breakpoint mode (`desktopMode` grid vs horizontal-scroll); Teaser single vs multiple links. Conditions only control the dialog; the code reads whatever AEM delivers, so it applies the same rule when it reads a value (whether a hidden field's stored value is delivered was not verified in the repository). Legacy-only fields (Separator Component ID, Background contrast, Stretch to container width) are shown only where a value already exists (re-audit D-D-17).

11. Validation

Kind	Fields	Rule
regExp Anchor ID	`classes_anchor`, `containerAnchor`	`^anchor-[a-z0-9][a-z0-9-]*$`

Kind	Fields	Rule
regExp anchor name / tab id	Section, Columns Container, Tabs `anchorName`; Tab and Tabs item `tabId`	`^#?[a-z0-9][a-z0-9-]*$`
regExp Component ID	Separator `componentId`	`^[A-Za-z][A-Za-z0-9_-]*$`
`numberMin` / `numberMax`	Table List `maxCards`	1 to 10
`maxLength`	Card `analyticsLabel`	120
`min` / `max`	34 `number` fields	Ranges of counts and columns
`required`	23 fields: item links (`hero-cta`, `page-header-cta`, `teaser-link`, `list-item`, `footer-link`), form field `fieldName` and `label`, Tab / Tabs item `title`, List `source`, Video `uri`, Filter List item fields, Contact Card region and name	Enforced in the dialog

Every `validation` has a `customErrorMsg`. The code still validates what it reads (ids through `availableId`, URLs through `isSafeUrl`), because content authored before a rule isn't revalidated.

12. Help Text

418 fields have a `description` (the help text under the field) and 4 a `placeholder`. Descriptions say what the field does on the page and its default ("Optional tooltip (most buttons need none)."), label legacy fields ("Links authored before Anchor Link items. Add links to the list with Add (Anchor Link) instead.") and name the AEM delivery consequence where an author could be surprised. Outdated guidance is a defect (re-audit R-D-16, R-A-4). Labels and help text are written in English in the models; no localisation mechanism for dialog text exists in the repository.

13. Multifields and Item Children

Approach	Status	Rule
Item children (components added with Add, own model, in the container's filter)	Standard	All repeating content: List Items, Cards, CTAs, Teaser Links and Stats, Testimonial Stats, Anchor Links, Filter List tags / categories / pages, Footer Links, form fields, Table rows, Tabs items, Canvas buttons
Composite multifields (`container` with `multi: true`)	Compatibility-only	They didn't render from content packages; the 12 remaining ones are labelled "authored before" and still read (`readMultifieldItems` in `utils/multifield.js`, `readLinkItems` in `utils/links.js`, `buildStats` in `decorators/stats-decorator.js`)
`multi: true` on simple fields	Used	`aem-tag` (tags) and Page Keywords; Universal Editor doesn't save `multi: true` on `aem-content` fields
Non-multi `container`	Used	Collapsible groups in the Conversation Canvas dialog (`menu`, `searchBar`)

The audit recommends removing the legacy multifields after confirming that no production content uses them (deferral D-B-14, blocked: production content check).

14. Container-Child Models

- An item model is a `core/franklin/components/block/v1/block/item` definition with its own model, listed in the container's filter; it has no folder and is documented in its container's README.
- AEM renders each item as a row after the block's own rows, one cell per field (empty and tab cells included); an item model with a `classes` field starts with a `<node name>`, `<classes>` cell (form items: `classes` is a read-only Field type).

- `itemRows(block, models, isItem)` in `utils/cells.js` finds the item rows: by `data-aue-model` in Universal Editor, else by the cell count (`KEY_VALUE_ITEM` for key-value blocks: 1 or 3+ cells, so a two-field item starts with a tab; `FIELD_ROW_ITEM` for row-per-field blocks: 2+ cells), tested in `test/items/item-rows.test.mjs`.
- An item without content yet (just added) is a placeholder in the editor (`decorators/editor-item-decorator.js`, placeholder key `itemEditorHint`) and is left out on the page.
- `xwalk/max-cells` limits the cells of non key-value models; the item models that need more cells have their own limits in `.eslintrc.js`.

15. Accessibility and Analytics Fields

Purpose	Fields
Alt text (collapses into its image)	16 *`Alt` fields: Image, Section and Columns Container background, Card, List Item, Hero image and icon, Quote, Teaser and author images, Testimonial author, TCS Header logos, Embed placeholder, Canvas icon, Video poster
Accessible names	Card and List Item `ariaLabel`, Gen Carousel `carouselAriaLabel`, Tabs / Tabs block `tabsAriaLabel`, Separator `accessibleLabel` and `accessibilityMode` (decorative / semantic)
Headings	`titleType` and similar Title Type selects (h1 to h6) next to titles
Media	Video `transcript`, `captions`, `captionsLanguage`, `captionsLabel`, `privacyMode`
Analytics	`analyticsEnabled` (Cards, List, Separator), `analyticsLabel` (Card, List Item, Separator)

Blocks don't push clicks; the central data layer reads `data-cmp-*` attributes (decorators-and-utilities.md). Adobe Tags and the Web SDK are not configured (blocked: measurement plan), so analytics fields only shape the Adobe Client Data Layer.

16. Legacy Properties and Migrations

The code keeps reading values authored before a model change; content is migrated separately (the library by `.home-work/bl/bl-migrate.mjs`, not in git; AEM content by authors or a script).

Former property	Current	Code compatibility	Content action
Video `placeholder_image` / `placeholder_imageAlt` (grouped, not delivered)	`poster` / `posterAlt`	Rows of the former group are read (`blocks/video/video.js`)	Copy the property in AEM (re-audit R-D-7)
Hero `classes_motion`, Table List `motion` (Motion Type)	Additional tab Animation Type / Speed	`addLegacyMotion` (`utils/motion.js`)	Set the Additional tab (R-A-7, R-B-6)
Form item `name` as the key	`fieldName`	Older items keep working with lower-case keys	Set `fieldName` where the case matters (R-D-1)
Canvas Button `linkType`	`buttonVariant` (own `conversation-canvas-button` model)	Old items keep model `button` and render as before	Copy `linkType` to `buttonVariant`
Built-in Button View / Icon / Alignment / Tertiary	Button block	Primary / Secondary still render	Replace by a Button block
Built-in Columns	Columns Container / Column Breaks	Columns still render	Rebuild as Columns Container or Column Breaks
Composite multifields (CTA List, Links, Stats, Pages, Tags, Categories)	Item children	Read when present	Re-author as items
List / Cards Component ID, Touch swipe; Reduced motion	Anchor ID; removed	Component ID and Touch swipe read; Reduced motion ignored (always honoured)	None required

Former property	Current	Code compatibility	Content action
Separator Width Auto, Variant Custom	Widths, Default	Still read	None required
Page Header CTA style `''`	`tertiary`	`''` still a text link	None required

Sources: BLOCK-ECOSYSTEM-AUDIT.md section 11, READUIT-2026-10-20.md section 5, DIALOG-OPTIONS-2026-10-20.md section 5.

17. Procedure: Adding or Changing a Field

1. **Check what exists**: the block README ("Authorable Fields", "Extending the Block"), the shared groups in `models/_card-list.json` and the Additional tab; don't duplicate an existing control.
2. **Choose the name** against the delivery rules (no collapse suffix unless folding, no grouping in key-value blocks, `container*` on section containers, a dedicated field for case-sensitive values).
3. **Edit the partial** (`blocks/<block>/_<block>.json` or `models/_*.json`): field, label, description, `value`, options, condition, validation; no sample content in the template.
4. **Rebuild**: `npm run build:json` (the pre-commit hook also does it); never edit `component-*.json` by hand.
5. **Read it in code** through `readProperties` / `itemRows` / `matchCells`; update `ITEM_LAYOUT` / `CARD_LAYOUT` and their tests when an item model's fields or tabs change; keep reading the old property when renaming.
6. **Lint and test**: `npm run lint`, `npm test`; `npm run build:modules` if imports changed.
7. **Document**: `npm run build:docs` (field tables), the written sections of the README, `tools/docs/blocks.json` if status or relationships change, the handover for significant changes.
8. **Validate** in the editor and on aem.page (table below); migrate the library and record the content action for AEM content.

18. Lint Rules

`.eslintrc.js` extends `plugin:xwalk/recommended` (`eslint-plugin-xwalk` 0.1.1); `npm run lint` passed at commit 2d06316.

Rule	Checks
`xwalk/max-cells`	Cells a model renders (collapse suffixes fold into their base field, `prefix_*` groups count once, tabs don't count, a multi container counts once); key-value models and `*-metadata` models are exempt. Limits: * 4, `section` 6, `tabs` 21, `gen-carousel` 18, `columns-container` 22, `table-list` 10, `table-list-card` 6, `list-item` 19, `card` 16, `table-col-5` 5, `field-text` / `field-select` / `field-textarea` 6, `field-checkbox` / `contact-card` 5
`xwalk/invalid-field-name`	Name syntax (see Naming)
`xwalk/no-duplicate-fields`	No duplicate names in a model
`xwalk/no-orphan-collapsible-fields`	A suffixed field needs its base field
`xwalk/no-custom-resource-types`	Only standard resource types

19. Common Errors

Symptom	Cause	Fix
Dialog field missing or outdated in Universal Editor	`component-*.json` not rebuilt or not deployed	`npm run build:json`, commit, wait for code sync (handover 13.5)
Value saved but not on the page	Collapse suffix, grouped field in a key-value block, or `classes_*` on a section	Rename per the delivery rules; keep reading the old name
Block renders empty or shifted on aem.page but fine locally	An empty tab row or tab cell read as content; item first cell `<name>, <classes>`	Skip tab rows / cells (`withoutTabCells`), read by model
Every new item shows the same text	Sample text in the template or model default	Empty the template; use placeholders for UI text
`max-cells` lint error	Too many cells in a non key-value model	Make the block key-value, group or fold fields, or raise the limit with a reason in `.eslintrc.js`
Block class wrong or block code not loaded	Template name doesn't match the folder	Fix the template name (`test/models/block-names.test.mjs`)
Key lower-cased	Item `name` delivered in class-name form	A dedicated field (`fieldName`)

20. Validation Required for Every Model Change

Area	How	Recorded state
Model	`npm run build:json`, `npm run lint`	Passed at commit 2d06316
Persistence	Save, reload and re-open the dialog in Universal Editor; check the stored property	Blocked: AEM author access (audit section 9; D-A-26 date storage)
Rendering	Library page in the local preview and on aem.page, block by block	Executed in the audits (package 1.0.9 / 1.0.10)
Responsive	1440, 1024, 768, 600 and 390 px	Executed in the audits (940 loads)
Accessibility	axe at 1440 and 390 px, keyboard, names, contrast; screen reader	axe executed (0 violations after fixes); screen reader Blocked
Data layer	One component entry, `cmp:*` events, no duplicate ids	Executed in the audits (758 of 758 clicks)
Tests	`npm test` (layouts, item rows, model names, docs)	311 passed at commit 2d06316
Docs	`npm run build:docs`, README written sections, handover	Required per change

21. Related

- blocks.md: block governance, statuses, variation strategy
- decorators-and-utilities.md: `properties.js`, `cells.js`, `multifield.js`, `links.js` and the other shared modules
- AGENTS.md: "AEM delivery rules", "Dialog tabs", "Item children", "Additional tab", "Content Modeling"
- blocks/README.md and the generated field tables in each block README
- docs/audit/DIALOG-OPTIONS-2026-10-20.md, docs/audit/BLOCK-ECOSYSTEM-AUDIT.md, docs/audit/READUIT-2026-10-20.md
- Handover Part 15.4 — Content Modeling Standards and 13.5 — Component Dialog Fields Missing or Outdated in Universal Editor: TCS-CORP-HANDOVER.md
- aem.live: component model definitions
- Other knowledge base files: authoring.md, content-architecture.md, development-standards.md, testing-and-quality.md, troubleshooting.md, decision-log.md

22. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

23. Known limitations

- None recorded for this document.

24. Open items

- None.

25. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages